SECTION **5.0**

THORACIC SURGERY

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INTRODUCTION: SURGEON'S PERSPECTIVE

AIRWAY AND LUNG ACCESS CONFLICTS

Induction and maintenance of anesthesia for thoracic surgery require close cooperation between the surgeon and anesthesiologist. For example, during periods of one-lung ventilation (OLV), significant hypoxia and hypotension may occur, and surgery may need to be stopped temporarily while the hypoxia is corrected by reinflation of the unventilated lung. Hypotension in the absence of bleeding can sometimes be corrected by less vigorous retraction of the lung and heart by the surgeon. Quick and timely communication between the anesthesiologist and the surgeon can be lifesaving. Occasionally, during critical parts of the dissection, cessation of all respiration for short periods of time can make the surgeon's job much easier.

TUBES AND TUBE SIZES

The size of the endotracheal tube (ETT) is important for many thoracic surgical procedures. Fiberoptic bronchoscopy (FOB) through the ETT is a common event. The standard operating FOB just fits through an 8.0 single-lumen endotracheal tube (SLT). The fiberoptic laryngoscope (FOL) used for difficult intubations will fit smaller ETTs and double-lumen tubes (DLTs). Often the patient is intubated with a SLT initially so that bronchoscopy can be performed, and then either the SLT is switched to a DLT or a bronchial blocker is placed. The bronchoscope should be lubricated with a polyethylene glycol–based (e.g., Carbowax®) or silicone-based (e.g., EndoLube®) ointment rather than an aqueous jelly, which will dry out quickly making manipulation quite difficult. The FOL can be used for correct positioning of a DLT. If the bronchial portion of the DLT cannot be advanced into the left main bronchus, the FOL can be advanced through the bronchial side of the DLT into the left main bronchus. Then, the DLT can be advanced over the bronchoscope into the left bronchus. The depth of the tube can be determined by FOL observation of the right main bronchus through the tracheal side of the DLT.

PATIENT POSITIONING AND SURGICAL INCISIONS

The most common open incisions used by thoracic surgeons are the **posterolateral thoracotomy** (and its variations; Fig. 5.1A), the **median sternotomy** (Fig. 5.1B), and the anterior thoracotomy (often bilateral; Fig. 5.1C). The standard posterolateral thoracotomy involves division of the latissimus dorsi and sparing of the serratus anterior muscles. For procedures where exposure of both lungs is mandatory (e.g., bilateral lung transplantation), bilateral anterior thoracotomy with a transverse sternotomy, or the **clamshell incision** (Fig. 5.2), is useful. Generally, patients are in the **supine position** for anterior incisions (sternotomy, cervical incisions, anterior thoracotomy, and the clamshell incision) and in the **lateral position** for lateral and posterolateral thoracotomies. Minimally invasive techniques include **video-assisted thoracic surgery (VATS)** and **robot-assisted** approaches, which are quickly becoming the standard of care for many thoracic operations. They are performed using several (often three for VATS and five for robot-assisted) small incisions without spreading the ribs with a retractor. (NB: A preincision review of recent imaging in the OR will help ensure that the operation is performed on the correct side.)

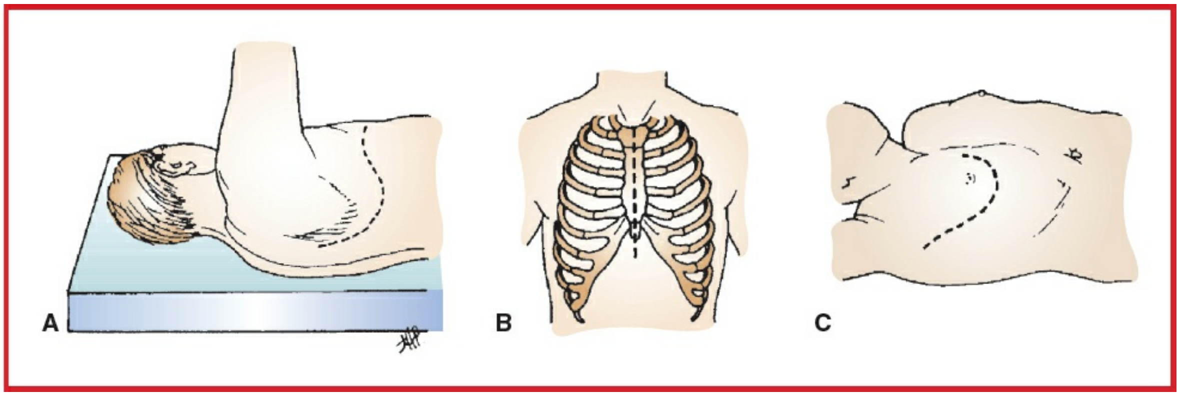

Figure 5.1 Primary incisions for thoracic surgery. **A:** Posterolateral thoracotomy, in lateral position. The incision curves in an S shape, passing under the tip of the scapula over in the fifth interspace anteriorly. **B:** Median sternotomy, in supine position, arms at side: the incision is made from the suprasternal notch to a point between the xiphoid process and the umbilicus. **C:** Anterior thoracotomy in supine position. (Reproduced with permission from Fry WA: Thoracic incisions. In: Shields TW, LoCicero J III, Ponn RB, eds. General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

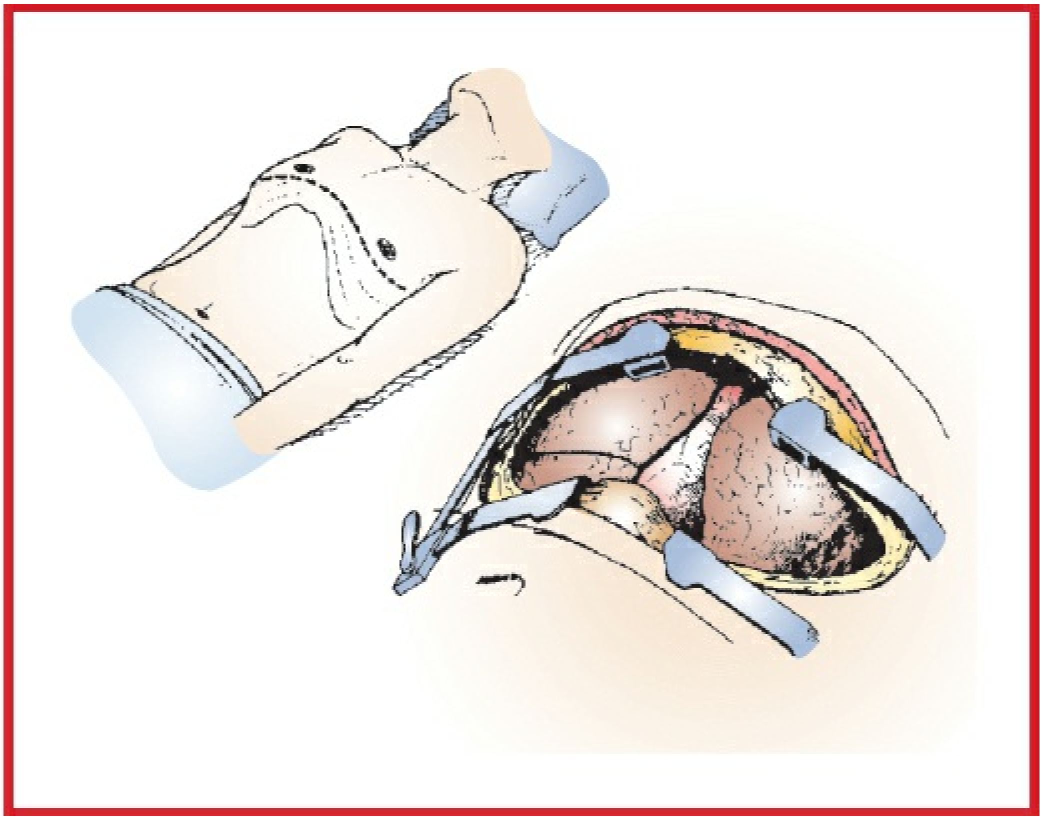

Figure 5.2 The “clamshell” incision, in classic supine position, affords excellent exposure, especially for bilateral lung procedures. (Reproduced with permission from Fry WA: Thoracic incisions. In: Shields TW, LoCicero J III, Ponn RB, eds. General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Most thoracic operations are performed with the patient in the **lateral decubitus** position. A patient in the right lateral decubitus position has the left side up and vice versa. After general anesthesia is induced and the patient is intubated, the patient is positioned relative to the break in the bed, usually at the level of the xiphoid or umbilicus. The patient is then rotated onto their side, using either bolsters or a beanbag to stabilize them. An axillary roll may be used to prevent axillary compression. The lower leg is flexed, and the upper leg is extended and supported by pillows. Then the table is flexed to assist in spreading the ribs and sometimes placed in slight Trendelenburg or reverse Trendelenburg position to level out the chest. The head and neck require extra support to remain in a neutral position to avoid brachial plexus injuries. The lower arm can be either extended on an arm board or flexed and placed next to the patient’s head (Fig. 5.3A). The upper arm is extended and held in position with either an airplane holder or an arm board with several pillows (Fig. 5.3B). Wide adhesive tape is placed across the hips to further secure the patient. A lower-body warming blanket (e.g., Bair Hugger™) should be used to avoid hypothermia. At the end of the operation, when the surgical incisions are being closed, the table can be returned to the level position. The neck and arms should be checked after the position is changed.

The anesthesiologist may be asked to exert posterior pressure on the patient's shoulder to align and diminish tension on the latissimus dorsi closure.

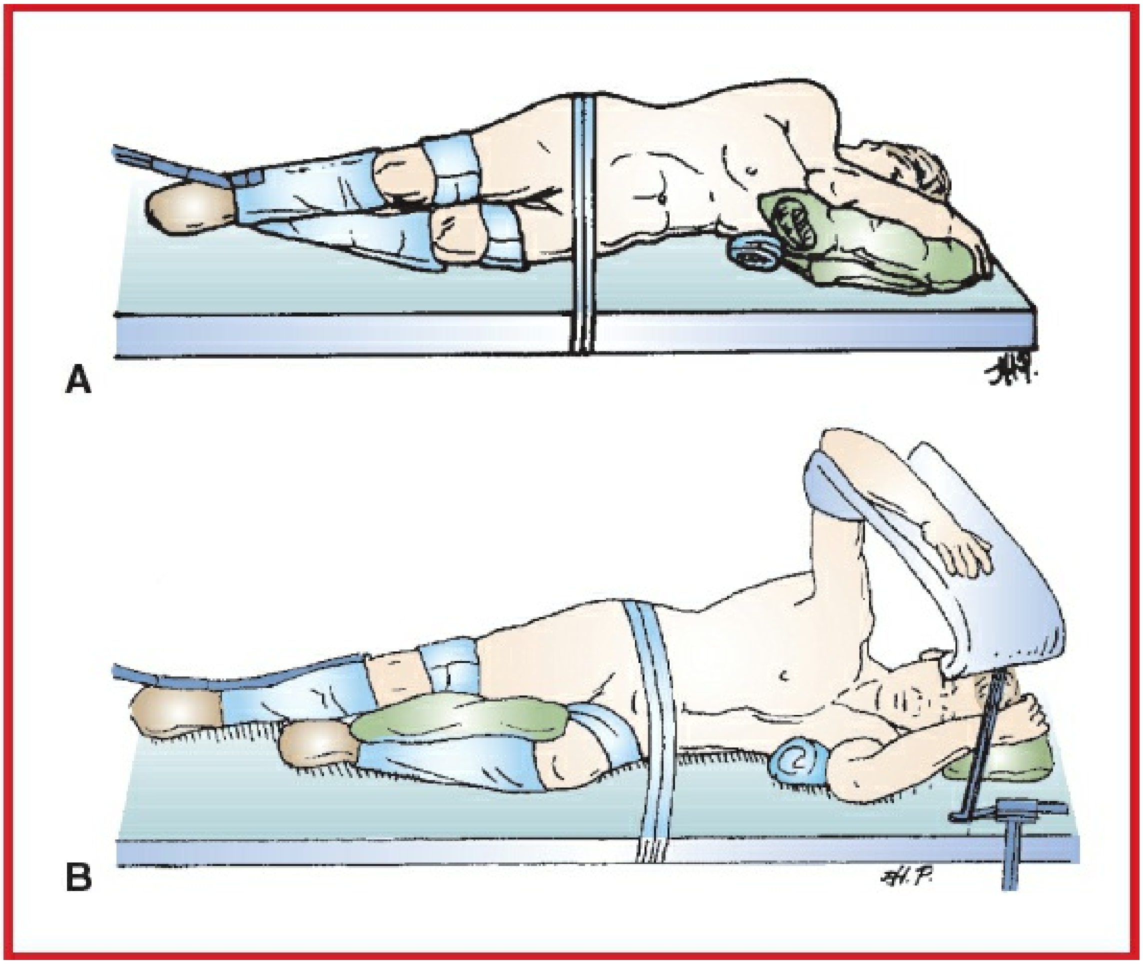

Figure 5.3 Lateral positioning for thoracic lateral and posterolateral procedures. **A:** Patient on his side, with kidney rest, axillary roll, pillows between knees, and padding under elbows. Wide adhesive tape secures the position. **B:** The upper arm abducted 90% on arm board. (Reproduced with permission from Fry WA: Thoracic incisions. In: Shields TW, LoCicero J III, Ponn RB, eds. General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Thoracic operations using an anterior incision are performed with the patient in the **supine** position. A small roll is placed under the shoulder blades to extend the neck and facilitate access to the upper mediastinum. This is particularly important for mediastinoscopy and proximal tracheal operations. In general, the arms should be tucked at the patient's sides. Having an arm extended during a sternotomy can place undue stretch on the brachial plexus.

The probability of deep venous thrombosis (DVT) in general thoracic cases is controversial, whereas the risks of prophylaxis are minimal. Given that patients often have protracted periods of immobility, both during and after surgery, and often have a malignancy, the use of DVT prophylaxis is good practice. We use sequential compression devices (SCD) for all patients, except those undergoing short video-assisted procedures, and reserve subcutaneous heparin for those at higher than normal risk.

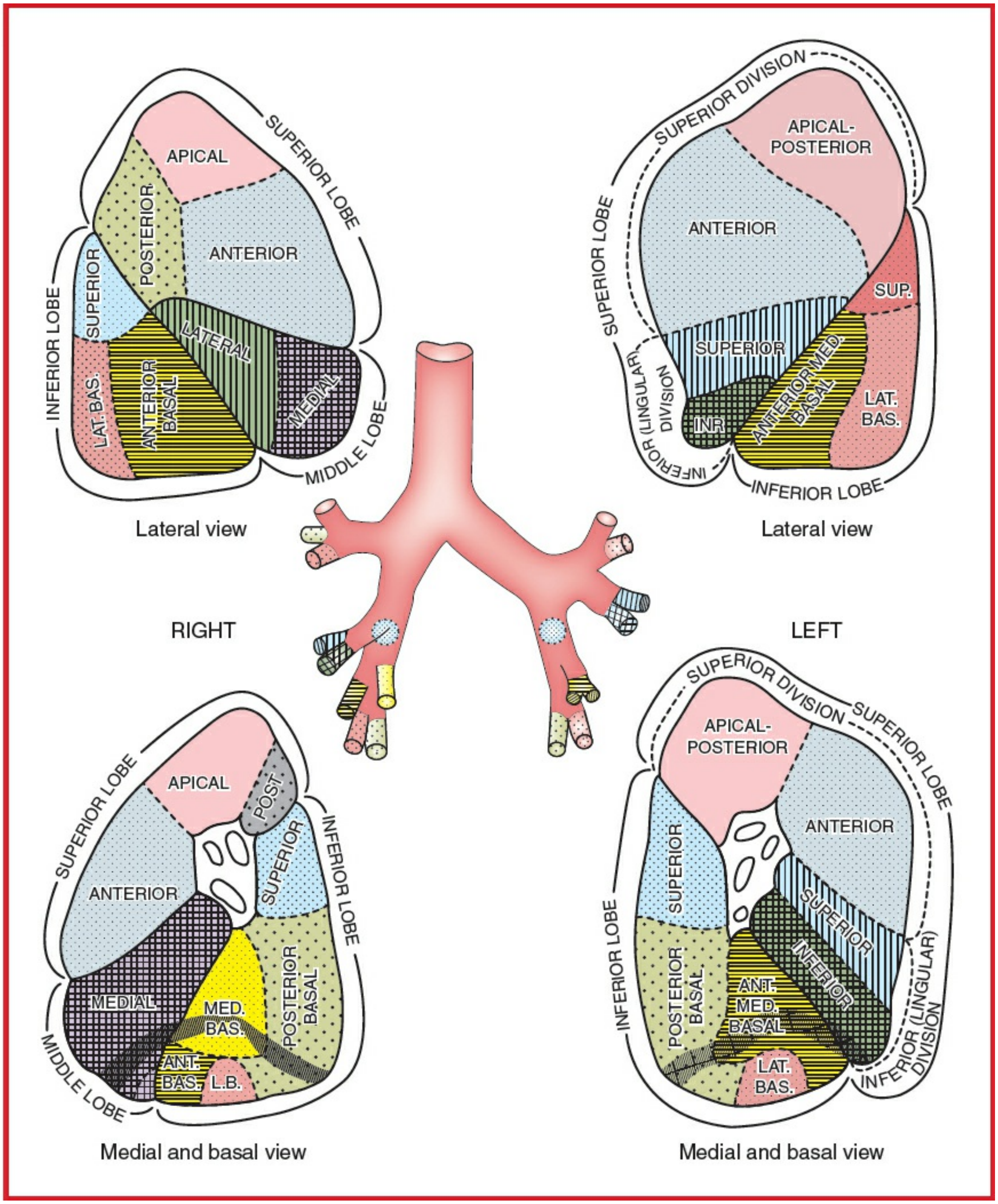

Figure 5.4 Segmental anatomy of the lungs. (Reproduced with permission from Clemente CD:Gray's Anatomy, 30th American edition. Lea & Febiger, Philadelphia: 1985.)

POSTOPERATIVE ISSUES

The most common postop issues relevant to anesthesiologists are:

- The need for postop mechanical ventilation
- Airway management
- Hemodynamic instability
- Pain control

The majority of patients can be extubated immediately after the case in the OR. This includes patients undergoing shorter or minimally invasive procedures and even those undergoing lung volume reduction surgery (LVRS). The most common exceptions are patients requiring preop mechanical ventilation, lung transplant patients, and those with difficult airways. Patients undergoing prolonged surgery may require postop mechanical ventilation. Because DLTs are larger than standard SLTs, there is greater potential for laryngeal trauma and airway edema and thus loss of airway following extubation. By exchanging the DLT for a SLT over a tube changer, this potentially catastrophic complication can be avoided.

Hemodynamic instability following surgery may be due to several causes, the most important being ongoing blood loss, inadequate intraop fluid replacement, and cardiac dysfunction. Epidural anesthesia may cause intraop and postop hypotension.

With the use of epidural anesthesia and systemic analgesics (e.g., acetaminophen and ketorolac), postop pain can be managed effectively. It is important for the anesthesiologist to communicate to the surgeon (and postop care team) which agents have been used, how the patient responded to them, and what types of hemodynamic, pulmonary, and neurologic effects can be expected in the postop period. Most patients will go to the ICU or a step-down unit with telemetry, depending on the institution policy and procedure.

Suggested Readings

1. D'Ercole F, Arora H, Kumar PA: Paravertebral block for thoracic surgery. J Cardiothorac Vasc Anesth 2018; 32(2):915–27.
2. Fischer GW, Cohen E: An update on anesthesia for thoracoscopic surgery. Curr Opin Anaesthesiol 2010; 23(1):7–11.
3. Gerner P: Postthoracotomy pain management problems. Anesthesiol Clin 2008; 26:355–67.
4. Harris RJ, Benveniste G, Pfitzner J: Cardiovascular collapse caused by carbon dioxide insufflation during one-lung anaesthesia for thoracoscopic dorsal sympathectomy. Anaesth Intensive Care 2002; 30(1):86–9.
5. Hutter J, Miller K, Moritz E: Chronic sequels after thoracoscopic procedures for benign diseases. Eur J Cardithorac Surg 2000; 17(6):687–90.

6. Louie BE: Catastrophes and complicated intraoperative events during robotic lun resection. J Vis Surg 2017; 3:52.
7. Refai M, Andolfi M, Gentili P, et al: Enhanced recovery after thoracic surgery patient information and care-plans. J Thorac Dis 2018; 10(Suppl 4):S512–6.
8. Steenwyk B, Lyerly R III: Advances in robotic assisted thoracic surgery.Anesthesiol Clin 2012; 30:699–708.
9. Taylor R, Massey S, Stuart-Smith K: Postoperative analgesia in video-assiste thoracoscopy: the role of intercostal blockade. J Cardiothorac Vasc Anesth 2004; 18(3):317–21.
0. Tobias JD: Anaesthesia for minimally invasive surgery in children. Best Pract Res Clin Anaesthesiol 2002; 16(1):115–30.

VIDEO-ASSISTED THORACIC SURGERY

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

Video-assisted thoracic surgery refers to endoscopically assisted direct, robotic, and other minimally invasive techniques for thoracic surgery. Although initially used for the assessment of pleural processes of unknown etiology (e.g., pleural effusion that has defied diagnosis), VATS has largely replaced traditional thoracotomy. VATS is now accepted for treatment of spontaneous pneumothorax due to apical blebs, biopsy of peripheral infiltrates or nodules, talc pleurodesis, drainage of pleural effusions and other fluid collections, decortication of early empyemas, evaluation and evacuation of traumatic hemothorax, and standard pulmonary resections including wedge resection and lobectomy. Although not appropriate for all non–small-cell lung cancers, small, peripheral tumors without significant hilar or mediastinal lymph node involvement are often appropriate for VATS lobectomy. VATS also has been used for **lung volume reduction surgery**, **esophageal (Heller) myotomy**, and **upper dorsal sympathectomy**. Less well-accepted procedures include **pneumonectomy** and **esophagectomy.**

In most VATS cases, the use of a DLT or other means (e.g., bronchial blocker) to provide collapse of the ipsilateral lung is necessary because satisfactory visualization of the pleural cavity is impossible without collapse of the lung. The patient is usually in the lateral position. Several small incisions are used—usually three, sometimes four, or more. The video thoracoscope is placed through the first incision, and the pleural cavity is inspected. Other small incisions are then made for insertion of instruments. The position of the video thoracoscope and instruments may be interchanged, depending on the location of the problem.

More recently, **nonintubated VATS** (NIVATS) under local/regional anesthesia is being considered because of the perceived advantages of avoiding GA, reduced hospital costs, and faster discharge. Contraindications to NIVATS include BMI > 29, CAD, E < 50, asthma, and sleep apnea. In the spontaneously breathing patient, the lung on the operative side will collapse with pleural opening, and spontaneous OLV will begin. Face mask oxygen is usually sufficient. Permissive hypercapnia may occur and is usually tolerated. Epidural anesthesia or nerve blocks should be considered. Dexmedetomidine is useful for procedural sedation. Conversion to GETA may become necessary.

Usual preop diagnosis

Pleural disease (e.g., effusions); recurrent empyema; recurrent pneumothorax; localized lung masses; achalasia; pulmonary infiltrates; hyperhidrosis; reflex sympathetic dystrophy (RSD)

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. See Appendix K: Enhanced Recovery After Surgery, p. K-1.**

■ SUMMARY OF PROCEDURES	
Position	Lateral
Incision	Usually 3–5 small incisions (portals; Fig. 5.5)
Special instrumentation	Video thoracoscope with thoracoscopy instruments, DLT required
Antibiotics	Cefazolin 2 g iv
Surgical time	1–3 h
Closing considerations	Chest tube placed
EBL	Minimal, although there is a risk for major bleeding
Postop care	Chest tube

Mortality	Minimal
Morbidity	Major vascular injury: rare Conversion to open thoracotomy: 4% Pneumothorax/persistent air leak
Pain score	2–3

PATIENT POPULATION CHARACTERISTICS	
Age range	All age groups
Male:Female	1:1
Associated conditions	Pleural effusions, lung mass, pneumothorax

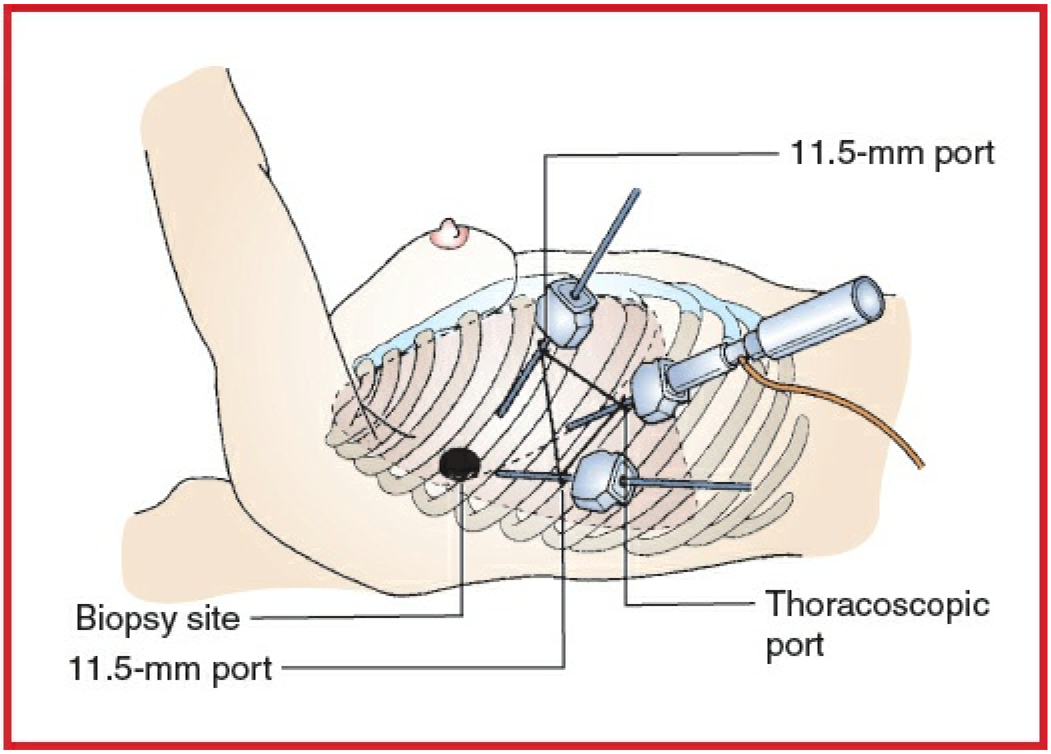

Figure 5.5 Example of thoracoscopic port placement. Positioning varies for individual need, but the principle of triangulation used for laparoscopic surgery is equally applicable in the thorax. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds:Surgery: Scientific Principles and Practice, 2nd edition. Lippincott-Raven Publishers, Philadelphia: 1997, 741.)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

As VATS is used for both diagnostic and therapeutic purposes, this patient population is quite diverse: patients may be of any age group and may present with an asymptomatic mass or be in respiratory distress due to undiagnosed interstitial disease. VATS is also used for nonthoracic procedures such as sympathectomy, pericardial window, and minimally invasive cardiac surgery. For robotic-assisted thoracic surgery (RATS), the requirements are similar to those for a VATS procedure except for (a) lateral decubitus position for prolonged periods of time with the table turned 90° and (b) limited access to the patient, requiring extensions to vascular lines and anesthetic circuit tubing, and increased difficulty ensuring OLV with a properly positioned DLT.

Respiratory	The preop evaluation should focus on the patient's ability to tolerate OLV, as well as the postop effects of the planned surgery. PFTs and ABG are useful prognostic tests. Question patient about dyspnea, productive cough, and cigarette smoking; examine for cyanosis, clubbing, RR, and pattern. Listen to the chest for wheezes, rhonchi, and rales. Preop lung function may be improved by treating respiratory tract infections, stopping smoking for several weeks, and treatment with bronchodilators and steroids, as indicated. **Tests:** PFT; CXR; if chest CT is available, look for airway anomalies that could interfere with DLT placement; ABG.
Cardiovascular	Directed at any underlying disease process
Laboratory	As indicated from H&P
Premedication	Standard premedication. Avoid heavy sedation that might impair postop respiratory function.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, typically with OLV, using DLT or BB. OLV may be difficult to achieve in pediatric patients, so chest insufflation with CO_2 has been described but is fraught with risks. Thoracoscopy has also been described in the awake, sedated patient under local or regional anesthetic technique. Although tissue trauma is significantly less than with the open thoracotomy, pain scores are not insignificant, and chronic pain syndromes are common. Referred pain due to lung dissection, chest wall pain due to decortication, and intercostal nerve impingements secondary to trocar insertion may cause marked postop pain. Multimodal therapy with opioid and nonsteroid anti-inflammatory drug (NSAID) is sufficient in many, but patient-controlled analgesia (PCA) may on occasion be required. Regional techniques (epidural or paravertebral) should be considered in compromised patients, in procedures with high conversion risk, or in procedures with significant tissue trauma (e.g., decortication).

Regional anesthesia

The incision site is infiltrated with local anesthetics, and intercostal nerve blocks are performed at the level of incision and at several levels above and below incision. The same effect can be accomplished by multiple-level single-shot paravertebral nerve blocks, which can be performed awake or asleep in the lateral position. Alternatively, continuous paravertebral or epidural catheters can be placed.

General anesthesia

The patient is intubated with a DLT or BB to selectively collapse the operative lung. Lung collapse is slower than with thoracotomy because the chest cavity is not completely opened to atmospheric pressure. To accelerate lung collapse, (a) start OLV early, use an FiO_2 of 1.0, and apply gentle suction to the unventilated lung; (b) make sure trocars are opened to atmosphere for 30–60 sec following insertion; and (c) CO_2 insufflation can be used to accelerate lung collapse. CO_2 insufflation is often well tolerated if begun slowly. There is a risk of venous gas embolism. Sudden insufflation can result in cardiovascular collapse as a result of ↑ intrathoracic pressure → ↓ BP, ↓ HR, and hypoxemia. Insufflated gas will be absorbed and may → hypercarbia.

Induction	Standard induction. Placement of DLT is discussed on p. 320.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and

	normothermia. Consider preincision placement of nerve blocks.	
Maintenance	O_2 (60–100%) and isoflurane, desflurane, or sevoflurane (0.6–1 MAC). No N_2O. Short-acting muscle relaxant and opioids as required. Consider remifentanil infusion (0.1–0.2 mcg/kg/min).	
OLV		
Emergence	Extubation in OR. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 2 mL/kg/h	See Restrictive Fluid Management, Appendix K, p. K-4
Monitoring	Standard monitors ± Arterial line	Arterial catheter generally not required, unless indicated by patient's medical condition
Positioning	✓ and pad pressure points ✓ eyes, ears, and genitals	See p. 306 for proper positioning
Complications	Air leak from the lung Hemorrhage Injury to intrathoracic structures Air embolism	Air leak observed on reexpansion of the lung Excessive blood drainage via chest tube, falling Hct Repair may require open thoracotomy

POSTOPERATIVE

Complications	Tension pneumothorax	In the absence of chest tube, air leak can → pneumothorax that can progress to tension

		pneumothorax if not treated. This may manifest as hyperresonance, ↓ chest wall movement, dyspnea, subcutaneous emphysema, tracheal shift, dysrhythmias, cardiovascular collapse, ↓ PO_2, and ↓ SaO_2. CXR diagnostic, but need to diagnose clinically if hemodynamically unstable. Requires immediate decompression of tension pneumothorax with 14-ga iv catheter through 2nd intercostal space at midclavicular line, followed by chest tube and continuous suction.
Multimodal pain management	IV opioids, ketorolac (30 mg)	Analgesic requirements less than for lateral thoracotomy, but moderate to severe pain may occur and chronic pain is not uncommon. Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
	Intrapleural anesthesia	Intrapleural local anesthetics (0.25% bupivacaine + 1:200,000 epinephrine 0.5 mL/kg) via thoracostomy drainage tube after the lung is reinflated but before chest tube suction applied.

	Epidural or paravertebral	Indicated in patients with significant comorbidities and/or for procedures with significant tissue trauma (decortication, lobectomy).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	CXR postop	

Suggested Readings

1. Harris RJ, Benveniste G, Pfitzner J: Cardiovascular collapse caused by carbo dioxide insufflation during one-lung anaesthesia for thoracoscopic dorsal sympathectomy. Anaesth Intensive Care 2002; 30(1):86–9.
2. Mineo TC: Epidural anesthesia in awake thoracic surgery. Eur J Cardiothorac Surg 2007; 32(1):13–9.
3. Steenwyk B, Lyerly R III: Advances in robotic assisted thoracic surgery. Anesthesiol Clin 2012; 30:699–708.
4. Tacconi F, Pompeo E: Non-intubated video-assisted thoracic surgery: where does evidence stand? J Thorac Dis 2016; 8(Suppl 4):S364–75.
5. Taylor R, Massey S, Stuart-Smith K: Postoperative analgesia in video-assisted thoracoscopy: the role of intercostals blockade. J Cardiothorac Vasc Anesth 2004; 18(3):317–21.
6. Umari M, Falini S, Segat M, et al: Anesthesia and fast-track in video- assisted thoracic surgery (VATS): from evidence to practice. J Thorac Dis 2018; 10(Suppl 4):S542–54.
7. Zhao ZR, Lau RWH, Ng CSH: Anaesthesiology for uniportal VATS: double lumen single lumen and tubeless. J Vis Surg 2017; 3:108.

LUNG RESECTIONS

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON’S PERSPECTIVE ABOVE)

Description

The most common indication for lung resection is lung cancer. Patients with Stage I or I non–small-cell lung cancer (disease confined to the lung and intrapulmonary or hilar lymph nodes) are generally offered surgery, unless their pulmonary function is prohibitively poor or their comorbidities pose an unacceptable risk. Patients with lymph node involvement often undergo adjuvant chemotherapy. Patients with Stage IIIA disease (including those with mediastinal lymph node involvement) often receive neoadjuvant chemotherapy and sometimes radiation, with restaging scans before surgery. Patients with Stage IIIB or IV disease are rarely offered an operation. Othe indications for lung resection include infection (particularly mycobacterial and fungal disease and bronchiectasis), congenital abnormalities such as pulmonary sequestrations, and trauma.

For lung cancer, the standard of care is lobectomy. Bilobectomy or pneumonectomy may be required for complete resection and segmentectomy, or wedge resection (nonanatomic lung resection) may be performed to spare lung parenchyma in patients with poor pulmonary function and patients with multiple indolent tumors. Patients without a preop diagnosis may undergo diagnostic wedge resection, followed by completion lobectomy if frozen section shows malignancy. Wedge resections are also performed for metastatic disease. In certain cases, sleeve lobectomy may be performed instead of pneumonectomy.

Lung resections can be performed through open or minimally invasive techniques. Minimally invasive techniques are becoming the standard of care for wedge resections and early-stage lung cancer without lymph node involvement. Large or central tumors or cases with lymph node involvement often require an open operation. Multiple wedge resections for metastases may be performed through open incisions in order to palpate the lung thoroughly.

Regardless of the underlying disease, the preop evaluation should include an assessment of pulmonary function. Spirometry with diffusing capacity is standard for any anatomic lung resection (Table 5.1). Predicted postop FEV_1 and D_LCO should be >40%. If the results are borderline, quantitative ventilation/perfusion scans or formal exercise testing may be necessary (Fig. 5.6). Preop smoking cessation is strongly encouraged.

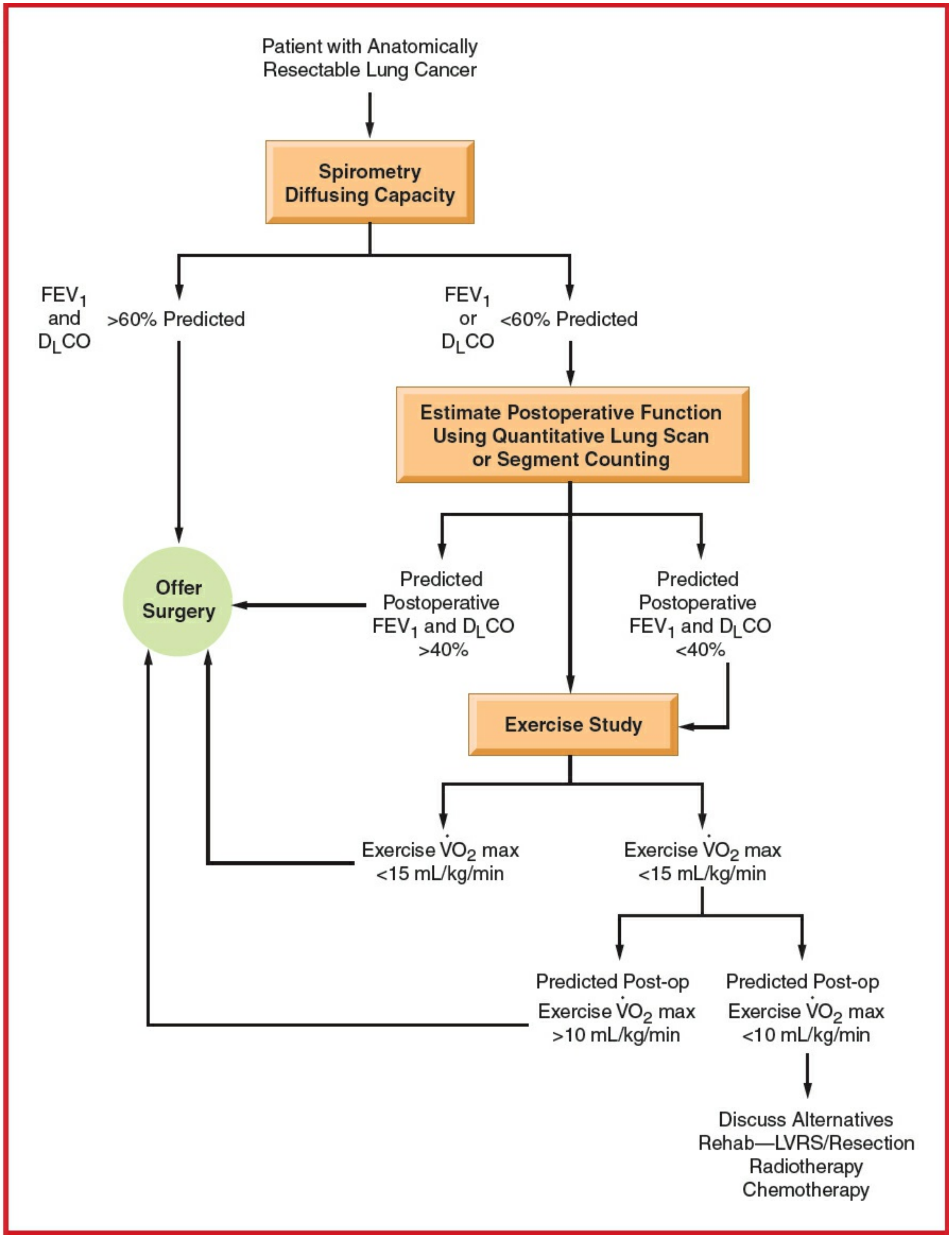

Figure 5.6 Physiologic assessment of patients with compromised lung function. (Reproduced with permission from Olson GN: Pulmonary physiologic assessment of operative risk. In: Shields TW, LoCicero J III, Ponn RB, edsGeneral Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Table 5.1 Spirometric Criteria for Pulmonary Resection		

Spirometry	**Operable**	**Further Study Suggested**
Forced vital capacity (FVC)	>60% predicted	<60% predicted
Forced expired volume in 1 sec (FEV_1)	>60% predicted	<60% predicted
FEV_1 FVC ratio	>50%	<50%
Maximum voluntary ventilation	>50% predicted	<50% predicted
Gas exchange		
Diffusing capacity for carbon monoxide	>60% predicted	<60% predicted
Arterial carbon dioxide tension	<45 mm Hg	>45 mm Hg

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Following induction of GA and intubation with a SLT, many surgeons perform a flexible bronchoscopy to confirm normal airway anatomy and check for secretions. In patients with tumors in the trachea or mainstem bronchi, bronchoscopy may be important in determining whether the patient should undergo a lobectomy, sleeve lobectomy, or pneumonectomy. After the bronchoscopy, the SLT is switched to a DLT, or a bronchial blocker is placed. The patient is then positioned in the lateral decubitus position (Fig. 5.3). The lung should be isolated at this time to make entry into the chest safer. Following entry into the chest, the lung on the operative side is allowed to deflate, and sometimes carbon dioxide insufflation is used. If the lung remains ventilated, a flexible bronchoscope should be used to verify correct positioning of the DLT or bronchial blocker (Fig. 5.7). After stable OLV has been obtained, the lung is mobilized and the bronchovascular structures are identified. Generally, the vascular structures are divided first and then the bronchus. Surgical staplers are used to divide vascular structures, the bronchus, and the lung parenchyma. Hypotension and arrhythmias may occur when the hilar structures or pericardium is retracted vigorously but generally resolve quickly on restoration of normal anatomic relationships. Inadvertent entry into the pulmonary artery or its branches during dissection can result in rapid blood loss. Because these vessels are usually under low pressure, bleeding generally can be temporarily controlled with direct pressure while the anesthesiologist resuscitates the patient. If the operation is performed through VATS, conversion to thoracotomy may be necessary for the surgeon to obtain more definitive vascular control. Before division of the bronchus, the surgeon may ask the anesthesiologist to ventilate the lung while the bronchus leading to the lobe that will be removed is occluded. This will ensure that the remaining lobes inflate appropriately. Thorough suctioning immediately before the lobectomy eliminates secretions as a cause of continued atelectasis. After the lobe has been resected, the surgeon may ask the anesthesiologist to ventilate the lung again to check for an air leak from the bronchial stump. Large air leaks are best addressed at the time of surgery rather than waiting for them to resolve postop.

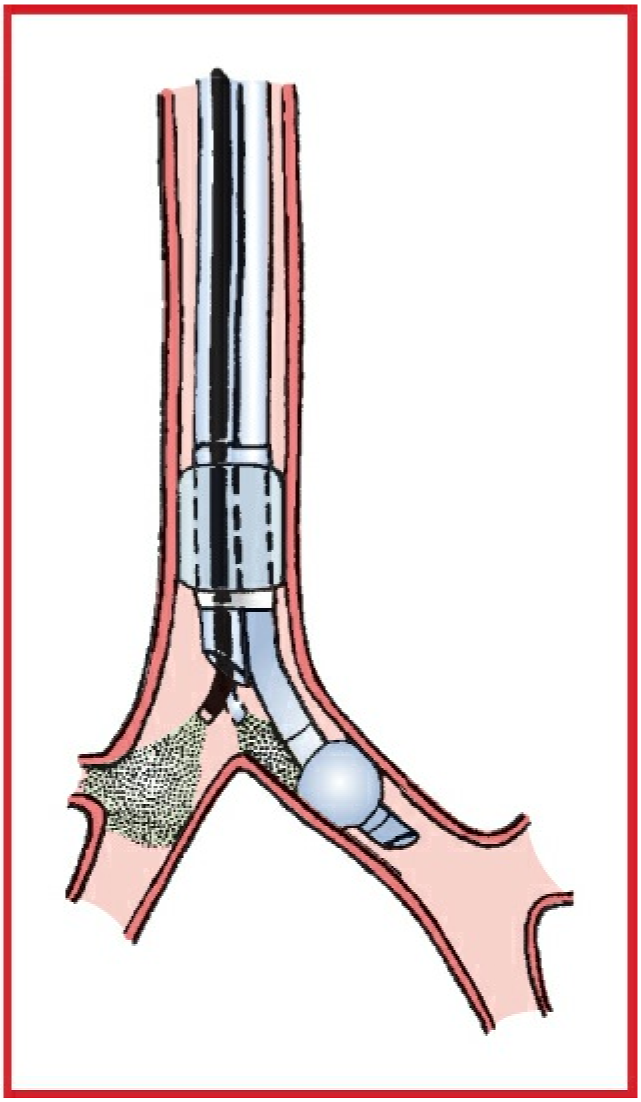


Figure 5.7 The use of a fiberoptic bronchoscope to ensure correct positioning of a left-side DLT. (Reproduced with permission from Fry WA: Thoracic incisions. In: Shields TW, LoCicero J III, Ponn RB, eds. General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Chest drainage is standard following lung resection and involves placement of a chest tube (usually about 28 Fr in size) attached to water seal or suction. Following pneumonectomy, chest drainage is not always performed; if a chest tube is placed, a balanced drainage system or water seal must be used. Otherwise, the mediastinum will shift to the operative side, creating adverse hemodynamic consequences. An alternative to drainage (after the patient is placed supine) is to aspirate air from the operative pleural space until a slight negative pressure is obtained.

Usual preop diagnosis

Carcinoma of the lung; infection; congenital abnormalities; trauma

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. **See Appendix K, p. K-1.**

Summary of Procedure

	Lung resection—anatomic (segmentectomy, lobectomy, bilobectomy, pneumonectomy, sleeve resections) and nonanatomic (wedge resection)
Position	Lateral/supine
Incision	Posterolateral thoracotomy or VATS
Special instrumentation	DLT or bronchial blocker
Antibiotics	Cefazolin 2 g iv
Surgical time	2–4 h
EBL	<500 mL (more in redo or inflammatory cases)
Postop care	PACU or ICU; careful attention to pulmonary toilet; chest tube output; chest tubes should not be placed to suction after pneumonectomy
Mortality	±1%
Morbidity	Dysrhythmias: 10% after lobectomy, 30% after pneumonectomy DVT: 5% ARDS PE

	MI Bronchopleural fistula Chylothorax Subcutaneous emphysema Phrenic nerve injury Recurrent laryngeal nerve injury
Pain score	7–8

PATIENT POPULATION CHARACTERISTICS

Age range	60–80 yr
Male:Female	15:1
Incidence	Common thoracic procedure, increasing in females
Etiology	Smoking
Associated conditions	Cardiopulmonary disease, PVD

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The main indication for lung resection is neoplasm. Other indications include infection and congenital malformation. Many patients have a history of cigarette smoking with associated emphysema and/or chronic bronchitis. As most patients are older, other comorbidities are common (CAD, DM, HTN). Morbidity and mortality following thoracotomy are increased with preexisting pulmonary, cardiovascular, and neurologic disease. Lung resections are commonly being performed via thoracoscopy, which decreases patient morbidity. Lung isolation (DLT or BB) and OLV are mandatory for surgical exposure.

Respiratory	Question patient about exercise tolerance, dyspnea, productive cough, and cigarette smoking. Examine patient for cyanosis, clubbing, RR, and pattern. Listen to the chest for wheezes, rhonchi, and rales. Timely cessation of smoking (>8 wk), adequate management of bronchospasm with bronchodilator treatment ± steroids, and prompt treatment of

	preexisting lung infections are important to reduce postop pulmonary complications. **Tests:** PFT (see below and Table 5.2); CXR; chest CT (if available), always examine chest imaging for ease of lung isolation; ABG (only if indicated from H&P).
Pulmonary function	Nonsmoker with normal lungs may not require any studies for simple lobectomy. Any preexisting respiratory disease (or possibility of pneumonectomy) should trigger lung function studies. Whole-lung tests (ABG, spirometry) are sufficient in most cases. Split-function lung tests (V/Q scan) should be considered if the patient has heterogeneous disease or a planned pneumonectomy or borderline lung resection is planned. Many variables have been shown to correlate with poor outcome. The most important are FEV_1, D_LCO, and VO_2 max, which focus on different aspects (mechanics, parenchymal function, and cardiopulmonary reserve, respectively). Postop predictive values of FEV_1 < 40%, D_LCO < 40%, or baseline VO_2 max < 15 mL/kg/min predict ↑ risk for postop complications following pulmonary resection. Preop hypercapnia and PHTN/RV dysfunction are relative contraindications to lung resection, except for specific circumstances (e.g., combined with LVRS in severe emphysema). Trial pneumonectomy with PA balloon occlusion is frequently discussed, but not used in clinical practice. Preop optimization of respiratory function with bronchodilator therapy should be routine.
Cardiovascular	RV dysfunction is a relative contraindication to lung resection (particularly pneumonectomy). Prophylactic digoxin or amiodarone to ↓ risk of postop atrial fibrillation has limited efficacy; however, drugs like diltiazem are often used in the postop period. **Tests:** ECG—look for evidence of RV hypertrophy,

	conduction problems, ischemia, and prior MI. Echo/stress echo to evaluate ventricular function, others as indicated from H&P.
Neurologic	✓ Hx of previous back surgery and peripheral neuropathy. Examine thoracolumbar area for skin lesions, infection, and deformities.
Musculoskeletal	Patients with lung cancer may have myasthenic (Eaton-Lambert) syndrome with resistance to depolarizing muscle relaxants and ↑ sensitivity to NDMRs. Monitor relaxation with peripheral nerve stimulator.
Hematologic	Adequate O_2-carrying capacity is important. Optimize Hb preop if possible (iron, EPO); consider transfusion with Hb < 7 g/dL (<10 g/dL with CAD), depending on vital signs. Although rare, bleeding can be profuse, so blood should be immediately available. Coagulopathy may preclude neuraxial anesthesia. **Tests:** Hct, PT, PTT (if epidural anesthesia is planned).
Laboratory	Other tests as indicated from H&P.
Premedication	Midazolam 0.5–2 mg iv if patient is anxious (unless respiratory compromise). When epidural opioids are planned, avoid additional systemic opioid or sedative premedication that can potentiate postop respiratory effects of neuraxial opioids.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

Table 5.2 Assessment of Risk of Postop Pulmonary Complications Following Thoracic and Abdominal Procedures

Category	Point
I. Expiratory Spirogram	
a. Normal (%FVC + %FEV_1/FVC > 150)	0
b. %FVC + %FEV_1/FVC = 100–150	1
c. %FVC + %FEV_1/FVC < 100	2
d. Preop FVC < 20 mL/kg	3
e. Postbronchodilator FEV_1/FVC < 50%	3
II. Cardiovascular System	
a. Normal	0
b. Controlled HTN, MI sequelae for more than 2 yr	0
c. Dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, dependent edema. CHF, angina	1
III. ABGs	
a. Acceptable	0
b. $PaCO_2$ > 50 mm Hg or PaO_2 < 60 mm Hg on room air	1
c. Metabolic pH abnormality >7.50 or <7.30	1
IV. Nervous System	
a. Normal	0
b. Confusion, obtundation, agitation, spasticity, discoordination, bulbar malfunction	1
c. Significant muscular weakness	1
V. Postop Ambulation	
a. Expected ambulation (minimum, sitting at bedside) within 36 h	0
b. Expected complete bed confinement for at least 36 h	1

0 points = low risk, 1–2 points = moderate risk, 3 points = high risk.

INTRAOPERATIVE

Anesthetic technique

Combined GA with continuous regional technique (epidural or paravertebral). Anesthesia for lobectomy/pneumonectomy relies on OLV techniques to improve surgical exposure and minimize damage to the operative lung in the case of lobectomy or bilobectomy. The challenges to the anesthesiologist include maintaining adequate oxygenation in patients with poor pulmonary reserve and ensuring that the patient is comfortable, warm, and awake at the end of surgery.

Preinduction	In patients undergoing open operations, placement of a regional analgesia catheter is important for postop pain control. Continuous epidural (lumbar or thoracic) and paravertebral blocks have been shown to be effective. Epidural catheters should be placed in the awake patient, whereas paravertebral catheters can be sited asleep or intraop under direct vision. The catheter tip should be as close as possible to the level of incision to minimize the sympathectomy. Intraop use of regional analgesia reduces the amount of systemic anesthetics/analgesics required and, therefore, facilitates rapid emergence. An adequate block can be established with lidocaine or bupivacaine; however, higher concentrations will be required for intraop anesthesia than postop analgesia. In patients undergoing minimally invasive operations, paravertebral blocks or intraop nerve blocks placed by the surgeon are often used.
Induction	Standard induction. If flexible bronchoscopy is planned prior to lung resection, intubate with an ETT (≥8 mm), which will be replaced with DLT after bronchoscopy (see below). Otherwise, proceed to intubation with an appropriate-sized DLT (check imaging, rough guideline: adult male, 39–41 Fr; adult female, 35–37 Fr) after induction.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks
Maintenance	Air/O_2 and isoflurane/desflurane/sevoflurane (around 0.5–0.7 MAC if neuraxial block is established). Avoid N_2O. Use high FiO_2 = 0.8–1.0 at onset of OLV; titrate to lowest possible after HPV is well established and

	operative lung collapsed. A local anesthetic (e.g., 2% lidocaine or 0.25–0.5% bupivacaine) can be infused or injected hourly into a thoracic (1–3 mL) or lumbar (5–10 mL) epidural catheter. Continuous infusion of local anesthetic (0.125% bupivacaine) generally provides better hemodynamic stability than hourly bolus injection. To enhance the effect of epidural analgesia, a loading dose of epidural opiate (e.g., hydromorphone 200–500 mcg [thoracic] or 1–1.5 mg [lumbar]) can be administered prior to incision. Epidural hydromorphone has a superior side-effect profile over morphine at equipotent doses. IV (compared to inhalational) anesthetics have a clinically insignificant benefit on OLV oxygenation (particularly if volatile agents are limited to <1 MAC); therefore, TIVA is not necessary in the majority of cases.
Emergence	Before chest closure, lungs are inflated gradually to 20–30 cm H_2O pressure to reinflate atelectatic areas and to ✓ for significant air leaks. Surgeon inserts chest tubes to drain pleural cavity and aid lung reexpansion. Patient is extubated in OR. If postop ventilation is required (rare), DLT is exchanged for single-lumen ETT. Patient is transferred in head-elevated position to PACU or ICU, breathing mask O_2. If hemodynamically unstable, monitor ECG, pulse oximetry, and arterial pressure during transfer. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.
Blood and fluid requirements	IV: 18 ga × 1 + 14 or 16 ga × 1 Maintain stable hypovolemia; limit BSS to 10–15 mL/kg if — Postop, PVR is increased in proportion to the amount of lung tissue removed. An overhydrated patient is at risk of RV failure and pulmonary

	possible Blood: available, but rarely required; use vasopressor (ephedrine 5–10 mg iv bolus or phenylephrine 50–100 mcg iv bolus) if hypotensive	edema. Replace blood loss with colloid (1:1) to minimize volume load. Third-space loss is negligible and need not be replaced. The use of epidural local anesthetics can cause ↓ BP in a volume-restricted patient; vasopressor may be needed. See Goal Directed Fluid Management, Appendix K, p. K-4
Monitoring	Standard monitors Arterial line Urinary catheter ± CVP line ± PA line or TEE (rare)	It is mandatory to follow oxygenation continuously during OLV. Typically, this can be done with pulse oximetry, although continuous intra-arterial PO_2 monitoring is now commercially available. CVP and/or PA line is optional for pneumonectomy and for patients with coexisting cardiac disease. CVP monitoring may be inaccurate intraop and is mostly placed for postop care. PA lines are rarely necessary and may interfere with PA stapling or endanger the PA stump. TEE may be of benefit in the borderline pneumonectomy to check for RV tolerance of PA cross-clamp.
Positioning	Axillary roll, "airplane" for the upper arm Avoid hyperextending arms	✓ radial pulses to ensure correct placement of axillary roll (if misplaced, it will compromise distal pulses).

	✓ and pad pressure points ✓ eyes, ears, and genitals	Placing the oximeter probe on the down arm may assist in monitoring arm perfusion.
Fiberoptic bronchoscopy	FOB performed immediately before thoracotomy to evaluate resectability of lesion. Patient intubated with large ETT (≥8 mm), replaced with DLT or BB following bronchoscopy (see Bronchoscopy)	Use the largest DLT that atraumatically passes through the glottis (typically, 39–41 Fr for men, 35–37 Fr for women). DLT can be placed accurately by careful auscultation ±confirmation by FOB. For a DLT, fiberoptic confirmation is done both through the tracheal lumen to confirm that the blue bronchial cuff is not herniating above the carina and through the bronchial lumen to confirm that the end of the tube is not abutting the secondary carina (L-DLT) or through the bronchial lumen to confirm that the end of the tube is in the bronchus intermedius and the opening in the blue bronchial cuff is facing the (R)UL orifice (R-DLT). For small children, the balloon of a Fogarty embolectomy catheter is used as a BB; for adults, either a BB or a Univent tube may be used if the proper size DLT cannot be placed. BB not ideal as FOB is always needed to confirm placement, lung collapse is slow, suction and CPAP application is more difficult, and repeated

		inflation and collapse may be difficult through the small lumen.
Lung isolation	Separate lung ventilation	Separate lungs to prevent contralateral contamination (infection, pus, blood, tumor), allow selective ventilation, and facilitate operation
OLV	Two-lung vent: Vt = 6–8 mL/kg Normocapnia PEEP 3–5 cm H_2O OLV: Vt = 4–6 mL/kg Permissive hypercapnia ($PaCO_2$ 50–60 mm Hg) except in PHTN PEEP 3–8 cm H_2O (unless BPF) FiO_2: 0.6–1.0 PIP <35 cm H_2O and plateau pressures <25 cm H_2O, consider PCV	Issues during OLV are oxygenation, ventilation, and lung injury. Oxygenation is rarely an issue if the DLT is adequately placed and derecruitment is avoided in the nonoperative lung. Ventilation is impaired by the smaller lumen of the DLT and the fact that only one lung is ventilated, resulting in higher ventilatory pressures. However, permissive hypoventilation allows for limiting the ventilatory stress. Acute lung injury may result in postpneumonectomy pulmonary edema, which may occur even after lesser resections. Limiting Vt, peak and plateau pressures, FiO_2, duration of OLV, and atelectasis formation help to minimize the risk.
Complications	Hypoxemia	Hypoxemia is now relatively infrequent (<5% of cases) due to better lung isolation techniques and anesthetic agents with less suppression

		of HPV. If hypoxemia occurs, tube position should immediately be confirmed and FiO_2 increased toward 1.0. Suctioning of secretions and lung recruitment maneuvers are often all that is required. If derecruitment has occurred, higher levels of PEEP should be employed; however, this may potentially worsen oxygenation. CPAP to the (recruited) operative lung is always helpful in open thoracotomy cases, as is clamping of the PA or one of its branches to reduce shunt flow. Return to two-lung ventilation (if possible) will always improve oxygenation (even if used intermittently only) and should be considered with refractory hypoxemia.
	Hypercarbia	Mild hypercarbia is well tolerated except in the setting of severe PHTN. CO_2 levels above 70 mm Hg may be associated with tachycardia, dysrhythmias, and cardiac depression. Treat with higher minute ventilation.
	Arrhythmia	✓ for mechanical compression of the heart or great vessels.
	Hypotension	✓ volume status and cardiac function. Avoid fluid overload.

		Consider phenylephrine if ↓ BP is 2° epidural anesthetic.
	Airway rupture	✓ integrity of the intubated bronchus after reexpanding the lung
	DVT	Preventive measures: TED hose or SCD
	Airway trauma from intubation, tracheobronchial rupture	Force should NEVER be used during insertion of a DLT, as it may result in catastrophic airway disruptions. Do not overdistend bronchial balloon or DLT cuffs. DLT bronchial cuff usually requires <3 mL air for airtight seal, if an appropriate (large) DLT is used.

POSTOPERATIVE

Complications	Injuries related to lateral positioning	Pressure damage to ear, eye, nose, deltoid muscle, iliac crest, brachial plexus, and radial, ulnar, common peroneal, and sciatic nerves has all been reported.
	Structural injuries related to thoracotomy	Neurologic (phrenic and recurrent laryngeal nerves), and spinal cord injury; bronchopleural fistula; tracheobronchial and thoracic duct disruption
	Surgical complications	Cardiac herniation, tension pneumothorax, bleeding, torsion of the residual lobe, acute lung injury/ARDS

	Cardiopulmonary complications	Supraventricular dysrhythmias, SVT, acute RV failure, atelectasis, BPF, pneumonia, PE. For SVT, treat underlying cause and correct electrolyte abnormalities. Most postop SVTs are 2° atrial fibrillation or 2° catecholamine surge and may resolve spontaneously. Hemodynamically unstable patients will require cardioversion. Beta-blockers, amiodarone, Ca^{++} channel blockers, and overdrive cardiac pacing are effective in patients with unstable AF.
Multimodal pain management	Neuraxial opioids—epidural or intrathecal Parenteral opioids (iv, im, continuous iv, PCA) Intercostal blocks Interpleural analgesia Epidural local anesthetics Cryoanalgesia NSAID (ketorolac)	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6). Effective analgesia via epidural or paravertebral route is essential for patient to cough, breathe deeply, and ambulate early. Epidural infusions consist of a local anesthetic + opioid mixture (smaller volume and higher concentration with thoracic placement). Paravertebral infusions are local anesthetic only, thereby avoiding opioid side effects. Paravertebral analgesia interferes less with postop lung function than

		epidural analgesia. ESP (erector spinae plane) block is a newer technique for pain management. Ketorolac (15–30 mg iv) is helpful as adjunct analgesic, particularly with referred shoulder pain.
Tests	Hct, CXR, ABG, and others as indicated	
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Alam N, Flores RM: Video-assisted thoracic surgery (VATS) lobectomy: the evidence base. JSLS 2007; 11(3):368–74.
2. Bernstein WK, Deshpande S: Preoperative evaluation for thoracic surgery. Semin Cardiothorac Vasc Anesth 2008; 12(2):109–21.
3. Brodsky JB, Macario A, Mark JBD: Tracheal diameter predicts double-lumen tube size: a method for selecting left double-lumen tubes. Anesth Analg 1996; 82:861–4.
4. Brodsky JB: Lung separation and the difficult airway. Br J Anaesth 2009; 103(Suppl 1):i66–75.
5. Davies RG, Myles PS, Graham JM: A comparison of the analgesic efficacy and side effects of paravertebral vs epidural blockade for thoracotomy—a systematic review and meta-analysis of randomized trials. Br J Anaesth 2006; 96(4):418–26.
6. Forero M, Adhikary SD, Lopez H, et al: The erector spinae plane block: a novel analgesic technique in thoracic neuropathic pain. Reg Anesth Pain Med 2016; 41(5):621–7.
7. Karzai W, Schwarzkopf K: Hypoxemia during one lung ventilation. Anesthesiology 2009; 110:1402–11.
8. Neustein M: The use of bronchial blockers for providing one-lung ventilation. J Cardiothorac Vasc Anesth 2009; 23(6):860–8.
9. Refai M, Andolfi M, Gentili P, et al: Enhanced recovery after thoracic surgery: patient information and care-plans. J Thorac Dis 2018; 10(Suppl 4):S512–6.
10. Slinger P: Update on anesthetic management for pneumonectomy. Curr Opin Anaesthesiol 2009; 22(1):31–7.

CHEST WALL RESECTION

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

The two most common indications for chest wall resection are (a) lung cancer that has invaded the chest wall and (b) primary chest wall tumors (the notable exceptions being Ewing sarcoma and rhabdomyosarcoma). If the operation is performed for lung cancer, the chest wall resection is often performed en bloc with lung resection. This operation may be done open or VATS for the lung resection but requires a thoracotomy for the chest wall resection. If the operation is performed for sarcoma, some patients may have received Adriamycin, which is associated with cardiotoxicity at high doses. If the tumor involves the skin, an appropriate area of the skin—typically, 4 cm around the tumor—must be resected along with the specimen. Underlying subcutaneous tissue and muscle should always be resected en bloc; however, the tumor itself must not be exposed. Wide skin flaps are frequently necessary as well. Limited resection (<5-cm segments of one or two ribs) generally requires no specific reconstructive measures, but larger resections may require reconstruction using polytetrafluoroethylene (PTFE) mesh or myocutaneous flaps. Removal of anterolateral or anterior portions of the chest wall, particularly resections that include the sternum, is associated with greater postop instability than are resections of posterior portions of the chest wall, which are protected by the back muscles and scapula. Larger defects can be tolerated posteriorly without reconstruction, as the scapula provides chest wall stabilization and prevents lung herniation. If a prosthesis is required, it must be covered by viable muscle to avoid erosion through the skin. Extensive reconstruction of the chest wall is often carried out in conjunction with plastic surgeons.

Usual preop diagnosis

Lung cancer with chest wall invasion; primary tumor of the chest wall (bone, cartilage, or soft tissue); radiation necrosis

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

Position	Supine or lateral
Incision	Over mass to be resected, sometimes en bloc lung resection is performed VATS
Special instrumentation	Bone instruments, Marlex® (or other) mesh, methyl methacrylate
Antibiotics	Cefazolin 2 g iv
Surgical time	1–6 h
Closing considerations	May require help of plastic surgeon in extensive cases
EBL	<500 mL
Postop care	PACU or ICU; some patients require temporary ventilator support
Mortality	<5%
Morbidity	Paradoxical chest wall motion (less in posterior resections) Pneumothorax Wound complications
Pain score	3–8

PATIENT POPULATION CHARACTERISTICS

Age range	Adults of all ages; children, rarely
Male:Female	1:1
Incidence	Relatively rare
Etiology	Unknown
Associated conditions	Lung cancer, metastatic disease, smoking-related diseases, cardiovascular disease

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Repair of Pectus Excavatum or Carinatum.

Suggested Reading

1. Hazel K, Weyant MJ: Chest wall resection and reconstruction: management of complications. Thorac Surg Clin 2015; 25(4):517–21.

REPAIR OF PECTUS EXCAVATUM OR CARINATUM

SURGICAL CONSIDERATIONS

Description

Standard bony and cartilaginous repair of a pectus excavatum (funnel chest) or carinatum (pigeon breast) deformities is usually elective surgery with the aim of improving contour and body image. Evidence that these repairs have any positive effect on cardiopulmonary function is controversial, although some surgeons feel that it can be more than a cosmetic procedure—particularly in patients with prominent deformities. Recent evidence suggests that, although resting cardiopulmonary function tests do not improve after pectus repairs, maximal exercise capacity may improve.

Pectus excavatum repair is performed with the Ravitch or Nuss procedures. The Ravitch procedure involves removing enough pairs of costal cartilages (usually four to six) to be able to mobilize and elevate the sternum. Depending on the severity of the defect and patient's age, fixation of the sternum in the corrected position may be necessary. The Nuss procedure is much less invasive and involves placing a metal bar across the chest wall under the sternum. VATS is often used for safe placement. **Pectus carinatum repair** is somewhat more complicated because the defects are more varied —often with a rotational component as well as anterior/posterior displacement. Removal of cartilages and correction of the position of the sternum are still the mainstays of treatment.

A midline incision provides the most satisfactory access to the cartilages and sternum. For cosmetic reasons, however, it may be important to use a curvilinear transverse incision, particularly in women. This incision requires extensive mobilization of subcutaneous and muscle flaps. The wound complication rate is somewhat greater after transverse incisions. The costal cartilages are moved by subperichondrial dissection. This may be tedious and time consuming, especially because several pairs of cartilages need to be removed. The elevation of the sternum is usually fairly straightforward and usually is accompanied by a transverse sternal osteotomy (Fig. 5.8). Intercostal muscle bundles may be left attached to the sternum or may be detached and reattached for better positioning of the sternum. Sternal support normally is not used in infants but may be used in older children. One common method of support is the use of a temporary transverse metal strut resting on the ribs but beneath the sternum. The final position of the sternum is easier to predict following repair of the pectus carinatum than following repair of pectus excavatum. Because of the negative intrathoracic pressure, it is easier to hold the sternum down than up. Ideally, patients for repair of pectus excavatum are just under school age. Satisfactory repair, however, may be carried out at almost any time during childhood. As full growth is attained, results tend to be less favorable. Pectus carinatum generally has its onset during adolescence, and it is well to let the patient complete his or her growth spurt prior to undertaking repair. (Also see Repair of Pectus Excavatum/Carinatum in Pediatric General Surgery.)

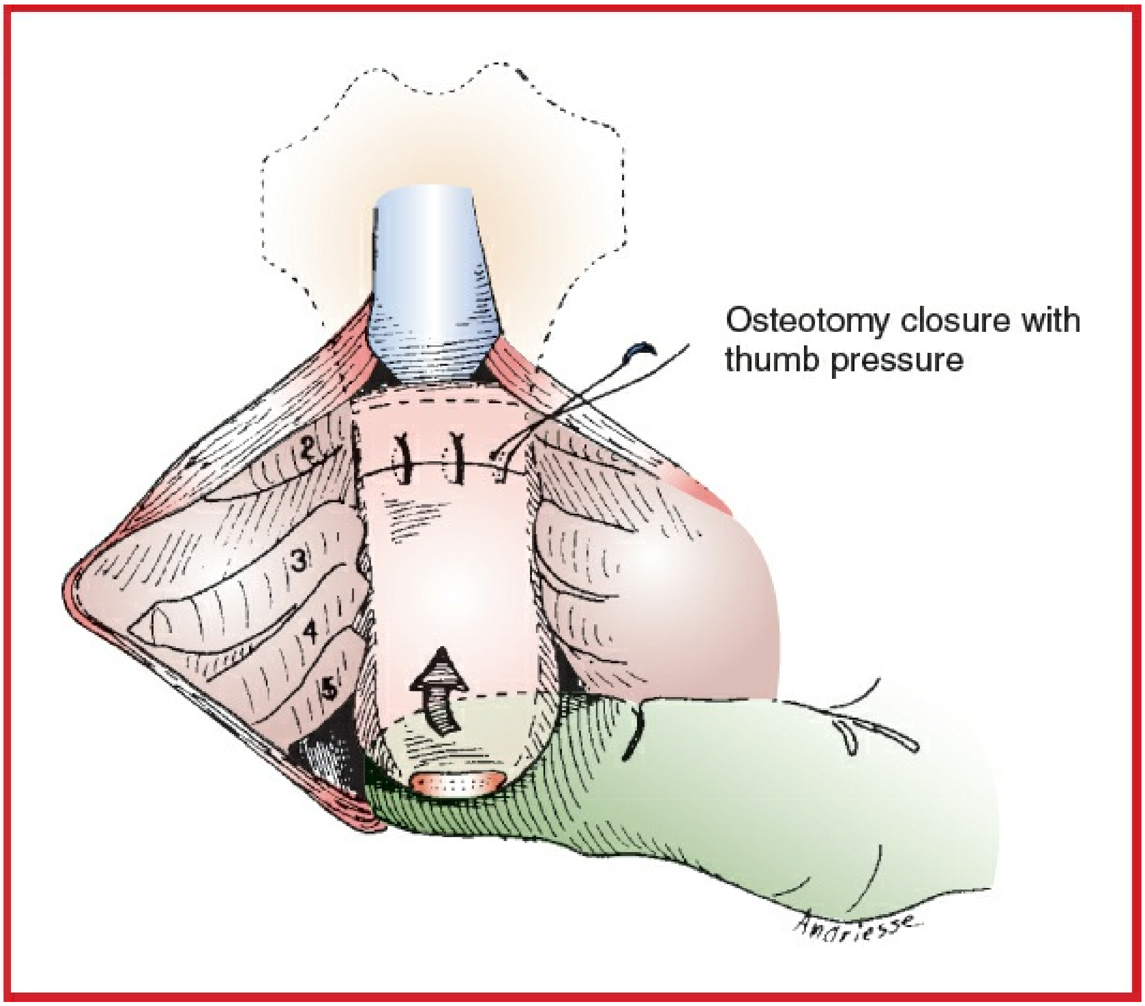


Figure 5.8 Correction of a pectus excavatum defect in a child. After subperichondrial resection of the involved costal cartilages, a wedge osteotomy permits anterior mobilization of the lower portion of the sternum. (Reproduced with permission from Shamberger RC: Chest wall deformities. In: Shields TW, LoCicero J III, Ponn RB, edsGeneral Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

Variant procedure or approaches

In certain circumstances, particularly in teenage girls and patients who do not engage in strenuous sports, subcutaneous, custom-made implants may be placed to improve body contour without necessitating major bony and cartilaginous repairs. These are usually carried out by plastic surgeons.

Usual preop diagnosis

Pectus excavatum or carinatum

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

Position	Supine
Incision	Transverse or vertical or VATS for Nuss procedure
Special instrumentation	Bone instruments, sometimes metal struts or wires for reconstruction
Antibiotics	Cefazolin 2 g iv
Surgical time	2–3 h
Closing considerations	Pleural and wound drainage common
EBL	100–500 mL
Postop care	ICU
Mortality	Minimal
Morbidity	Pneumothorax 5% Wound infection Sternum necrosis Immigration of strut Paradoxical chest wall motion → hypoventilation/atelectasis
Pain score	4–5

PATIENT POPULATION CHARACTERISTICS

Age range	Usually children, 5–10 yr; teenagers, sometimes; adults, rarely
Male:Female	1:1
Incidence	Unusual
Etiology	Unknown
Associated conditions	Marfan syndrome; MVP

ANESTHETIC CONSIDERATIONS

(Procedures covered: chest wall resection, repair of pectus excavatum/carinatum)

PREOPERATIVE

Patients for chest wall resection often have extensive cancer and may be weak and debilitated. A very large resection may create a “flail chest” situation, compromising postop ventilation.

Respiratory	Mild pectus seldom interferes with ventilation; no special studies are indicated. Severe pectus deformity can be associated with restrictive lung defects. **Tests:** CXR, PFT, ABG, if indicated from H&P
Cardiovascular	With severe pectus, the heart is displaced to the left and compressed; arrhythmias and RVOTO can occur 2° impaired filling, especially during exercise or in upright position. ECG may show right axis deviation, atrial and ventricular arrhythmias. A functional murmur may be detected. Echo may reveal ↓ SV with MVP. **Tests:** ECG, cardiac catheterization if indicated. Echocardiogram if symptoms or signs suggest MVP or RVOTO.
Hematologic	**Test:** Hct
Musculoskeletal	Chest wall resection is performed for invasive or metastatic cancer; patient may be markedly

	debilitated: pectus repair of chest wall deformity for cosmetic, orthopedic, or cardiopulmonary indications—pectus deformity usually asymptomatic.
Laboratory	Other tests as indicated from H&P
Premedication	Consider anxiolysis with short-acting benzodiazepine. When epidural opioids are planned, minimize systemic opioid or sedative premedication to avoid potentiating postop respiratory depression from central neuraxial opioids.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA +/– epidural anesthesia for minimal chest wall resection; however, epidural anesthesia is a recommended adjunct for extensive chest wall resections or repair of pectus deformities.

Induction	Standard induction. In the setting of RVOTO, avoid myocardial depressants, hypovolemia, and a short diastolic filling time (e.g., tachycardia). Lung isolation (DLT or BB) is usually required for chest wall resections. Sequential OLV can often be accomplished with a single-lumen ETT and an EZ-Blocker™.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.

Maintenance	Standard maintenance or high-dose opioid technique (fentanyl 10–25 mcg/kg) for patient with severe RVOTO. Patients with MVP will require prophylactic antibiotics for bacterial endocarditis. OLV, if necessary, as outlined for lobectomy/pneumonectomy.	
Emergence	Extubate in OR, or if high-dose opioid →ICU for later extubation. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18–16 ga × 1 BSS @ 1–2 mL/kg/h	Usually minimal blood loss. Fluid restriction is unnecessary as this is extrapulmonary operation. See Goal Directed Fluid Strategy, Appendix K, p. K-4
Monitoring	Standard monitors ± Arterial line	Close monitoring with arterial and central venous catheters may be required in patients with significant cardiopulmonary compromise.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Pneumothorax	Unintentional pleural tear can cause pneumothorax. Intraop deterioration characterized by ↑ peak inspiratory pressure, hypotension, and hypoxemia suggests pneumothorax. Increase FiO_2; d/c N_2O. Ensure pleural space is decompressed appropriately with needle or tube thoracostomy.

	Cardiac perforation	Rare, catastrophic event reported with aberrant bar placement during minimally invasive approach. Heralded by signs of hemorrhagic or obstructive shock. Supporting bar may interfere with external cardiac compressions.

POSTOPERATIVE

Complications	Hypoventilation Flail chest Atelectasis/pneumonia Pericarditis Bar displacement	Although most patients do not require postop ventilator support, with extensive chest wall resection, patient may hypoventilate. Respiratory stimulants, such as doxapram, should be avoided as they may → deep inspirations that can → severe sternal retractions. Paradoxical chest wall movement may occur during spontaneous ventilation with flail chest, postop atelectasis 2° splinting. Obtain postop CXR. Similar to postpericardiotomy syndrome, usually responsive to NSAIDs though occasionally corticosteroids or percutaneous drainage is required. Following minimally invasive repair (Nuss), the bar may become displaced and require

		repositioning. Inadequate analgesia and absence of a stabilizing bar are often cited as contributing factors.
Multimodal pain management	Depends on site and extent of chest wall resected Parenteral or epidural opioids with local anesthetic	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6). Epidural opioids and local anesthetics are particularly useful if flail chest is present—which reduces need for ventilator support.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Frantz FW: Indications and guidelines for pectus excavatum repair. Curr Opin Pediatr 2011; 23(4):486–91.
2. Frawley G, Frawley J, Crameri J: A review of anesthetic techniques and outcomes following minimally invasive repair of pectus excavatum (Nuss procedure). Paediatr Anaesth 2016; 26(11):1082–90.
3. Iida H: Surgical repair of pectus excavatum. Gen Thorac Cardiovasc Surg 2010; 58(2):55–61.
4. Jacobs JP, Quintessenza JA, Morell VO, et al: Minimally invasive endoscopic repai of pectus excavatum. Eur J Cardiothorac Surg 2002; 21(5):869–73.
5. Mavi J, Moore DL: Anesthesia and analgesia for pectus excavatum surgery Anesthesiol Clin 2014; 32(1):175–84.
6. Nuss D, Croitoru DP, Kelly RE, et al: Review and discussion of the complications o minimally invasive pectus excavatum repair. Eur J Pediatr Surg 2002; 12(4):230–4.

THORACOPLASTY

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

The objective of a **thoracoplasty** (removal of several ribs) is to permanently obliterate an existing pleural space, when the remaining lung cannot expand to fill it. Formerly, this operation was used in the treatment of tuberculosis (TB); however, because of better drug therapy, appropriate pulmonary resection, and the decrease in incidence of TB, thoracoplasty is now very rare. The procedure also was used for obliterating empyema spaces and helping to close bronchopleural fistulas (BPFs). The use of **pedicled muscle flaps** (serratus anterior, pectoralis major, and latissimus dorsi are the most common) or an **omental transposition** has largely replaced thoracoplasty for filling empyema spaces and encouraging closure of BPFs. These operations are less deforming and better tolerated physiologically because they do not result in paradoxical motion of the chest wall.

For patients whose lung will never expand to fill the space—such as those who have had a pneumonectomy or who have a permanently noncompliant lung—resection of multiple overlying ribs may be necessary (Fig. 5.9). Thoracoplasty is accomplished by removing several ribs in a subperiosteal fashion, allowing the underlying chest wall to collapse. This collapse is aided by the normally negative intrapleural pressure. Because the periosteum is left intact, the ribs will regenerate, resulting in a permanent, bony collapse of the chest wall. If the objective of the thoracoplasty is to obliterate a relatively small space (meaning that segments of only two to three ribs need be removed), the procedure may be done in a single stage, with little postop physiologic impairment of respiration. If extensive thoracoplasty is necessary, however, the procedure may be done in stages to minimize postop chest wall instability and resultant respiratory problems.

Usual preop diagnosis

Pulmonary TB; BPF; empyema

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. See Appendix K, p. K-1.**

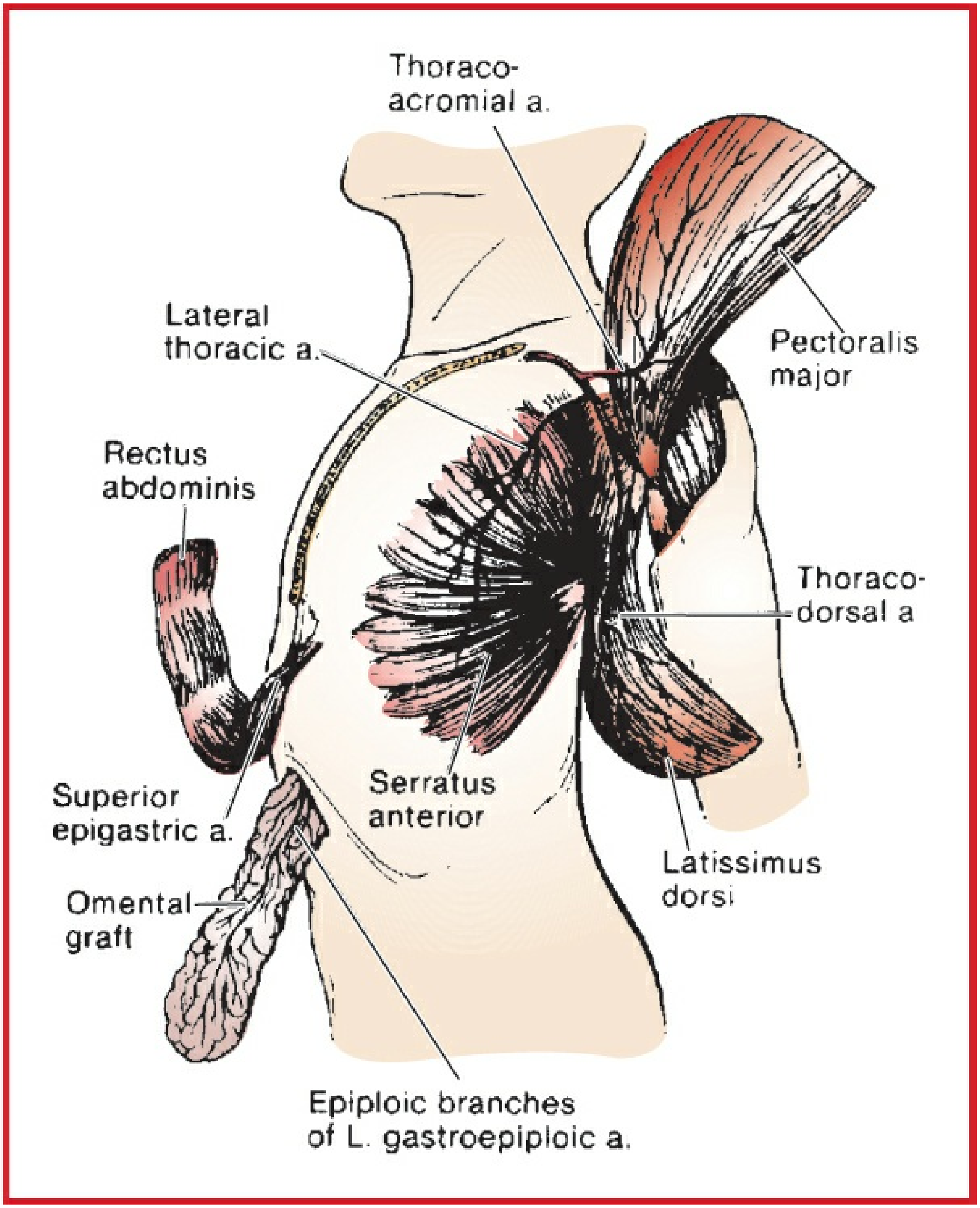


Figure 5.9 Extrathoracic muscle flaps that may be used to obliterate a postpneumonectomy empyema cavity. (Reproduced with permission from Miller JI Jr: Postsurgical empyema. In: Shields TW, LoCicero J III, Ponn RB, eds.General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

SUMMARY OF PROCEDURES

Position	Usually lateral
Incision	Along rib line

Special instrumentation	Bone instruments
Unique considerations	TB or fungal infection may be present
Antibiotics	As indicated by culture and sensitivity
Surgical time	2–3 h
EBL	500 mL or more
Postop care	ICU
Mortality	Minimal
Morbidity	Paradoxical chest wall motion → atelectasis → hypoxemia: 10% Pneumothorax: rare
Pain score	7–8

PATIENT POPULATION CHARACTERISTICS

Age range	Middle-aged or older adults
Male:Female	1:1
Incidence	Rare
Etiology	TB; pneumococcal infection; neoplasm; complication of pneumonectomy
Associated conditions	Immunosuppression

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Drainage of Empyema.

Suggested Reading

1. Seify H, Mansour K, Miller J, et al: Single-stage muscle flap reconstruction of the postpneumonectomy empyema space: the Emory experience. Plast Reconstr Surg 2007; 120(7):1886–91.

DRAINAGE OF EMPYEMA

SURGICAL CONSIDERATIONS

Description

Empyema is infection within the pleural space, and the primary treatment for it is drainage. Patients may be acutely ill or they may have a history of prolonged infirmity. The most common cause of empyema is extraparenchymal extension of a pneumonia, although other causes include trauma, BPF, and esophageal perforation. The three phases of an empyema are the exudative, fibrinopurulent, and organized phases. The early, or exudative, phase of an empyema is usually associated with fever, dyspnea, and a pleural effusion, with the diagnosis generally being made by thoracentesis. In more established infections, patients may complain of chronic symptoms, such as pain, dyspnea, and chest heaviness, and their medical history may include several previous courses of antibiotics.

Treatment of empyemas is based on the stage and the underlying cause of the infection. In the exudative phase, chest tube drainage alone may provide adequate therapy. Intrapleural tPA administration through the chest tube may assist in resolution. In the fibrinopurulent stage, thoracoscopic drainage with disruption of loculations and decortication (removal of the fine peel on the lung) is enough to drain the infected fluid and allow the underlying lung to expand. This procedure typically is done with the patient in the lateral position and involves three thoracoscopy ports. Blood loss is generally small, although large volumes of irrigation fluid may be necessary to thoroughly débride the thoracic cavity.

As the empyema becomes more established, the peel becomes thicker and more difficult to remove thoracoscopically. In such cases, an **open thoracotomy** is necessary. Due to the extensive intrapleural inflammation, blood and fluid losses may be substantial. To identify the correct plane between the lung and the thickened pleura, the lung may need to be reexpanded frequently throughout the procedure. Satisfactory drainage is accomplished when the infected fluid is removed and the lung expands freely. Because the peel often intimately adheres to the underlying lung, there may be a moderate postop air leak.

In patients too ill to undergo thoracotomy, **rib resection** with subsequent open drainage tube will permit the underlying lung to expand over a period of several weeks. Although this may be done under local anesthesia, a brief general anesthetic is often easier on the patient. In this operation, the patient is placed in the lateral position, and an incision is made over the rib corresponding to the most dependent portion of the empyema cavity. A 6-cm length of the rib is excised and a large-diameter (≥50 Fr) tube is inserted into the empyema cavity. More permanent open drainage is obtained by fashioning an **Eloesser flap.** In this procedure, a U-shaped flap of the skin is rotated into the empyema cavity after rib resection. This creates a long-term, skin-lined tube that will last indefinitely. A variant of this procedure—the **Clagett procedure—**is carried out for empyema (with or without BPF) following pneumonectomy because closed drainage rarely suffices in that situation. The principle is the same: drainage is accomplished through an epithelial-lined permanent opening. The opening is made anterolaterally and dependently so that drainage is effective and the patient can handle dressing changes without assistance. Segments of 2–3 ribs are removed, and the skin is sutured to the parietal pleura, leaving a permanent opening for drainage and irrigation. Without an underlying lung and with a relatively fixed mediastinum, this procedure is well tolerated physiologically.

Usual preop diagnosis

Nontuberculosis empyema (typically pneumococcal)

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

	Eloesser or Clagett
Position	Usually lateral
Incision	Over empyema pocket for Eloesser
Special instrumentation	None
Unique considerations	Patient may have BPF
Antibiotics	As indicated by culture and sensitivity
Surgical time	2 h
Closing considerations	Wound left open

EBL	100 mL
Postop care	PACU → room
Mortality	Minimal
Morbidity	Fluid drainage Bleeding: rare
Pain score	3–4

PATIENT POPULATION CHARACTERISTICS

Age range	Usually adults
Male:Female	1:1
Incidence	Decreasing
Etiology	Pneumonia; esophageal or bronchial leak; lymphatic or hematogenous spread of infection; posttrauma or thoracic surgery
Associated conditions	Bronchopleural fistula (BPF); sepsis; malnutrition

ANESTHETIC CONSIDERATIONS

(Procedures covered: thoracoplasty, drainage of empyema)

PREOPERATIVE

The guiding principle in the anesthetic management of empyema is to protect the unaffected lung from soiling by the affected side. These patients are often chronically ill with sepsis and cachexia, and there is usually an underlying BPF (which may require awake intubation).

Respiratory	Patients usually have preexisting pulmonary disease. Preop pulmonary findings may include collapse of the ipsilateral lung, impaired hypoxic pulmonary vasoconstriction 2° infection, and mediastinal shift to the ipsilateral side. Procedure often is performed for empyema in the presence of BPF following lung resection (particularly pneumonectomy), penetrating

	injury to the chest, or rupture of a cyst or bulla. When possible, surgeon should drain empyema under local anesthesia before induction, with patient sitting upright. If empyema is loculated, complete drainage may not be possible. **Tests:** Consider PFTs; ABG; obtain CXR to determine efficacy of preop chest drainage; if chest CT is available, look for airway obstruction that could interfere with DLT placement.
Cardiovascular	There may be ECG changes because of mediastinal shift to the affected side. **Tests:** As indicated from H&P
Neurologic	✓ Hx of back surgery and peripheral neuropathy. Examine the lumbar area for skin lesions, infection, and deformities. Avoid placement of epidural catheter in patient with neurologic problems or if obviously bacteremic or septic.
Musculoskeletal	Patients with lung cancer may have myasthenic (Eaton-Lambert) syndrome with resistance to depolarizing muscle relaxant and ↑ sensitivity to NDMRs. Monitor relaxation with peripheral nerve stimulator.
Hematologic	Transfuse patients with preop Hct < 25% or Hct < 30% in patients with CAD. (Hb level necessary to maintain adequate O_2 content.) Obtain autologous blood during the month before surgery or consider preop erythropoietin therapy in anemic patients.
Laboratory	Other tests as indicated from H&P
Premedication	Midazolam 0.5–2 mg iv if patient is anxious. When epidural opioids are planned, avoid systemic opioid or sedative premedication, which can potentiate postop respiratory effects of central neuraxial opioids.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or

	paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA; combined with epidural anesthesia/analgesia if thoracotomy is indicated and patient is not bacteremic or septic.

Induction	Consider inhalational induction (e.g., sevoflurane) if patient has significant BPF. Semi-Fowler position with a lateral tilt to make the infected side dependent, along with rapid control of the airway followed by isolation, makes contralateral contamination less likely. Rapid-sequence induction with cricoid pressure is an alternative. Intubate with DLT; isolate lungs to protect from aspiration and tension pneumothorax. Consider using DLT with the bronchial lumen to side opposite BPF. Contamination of the healthy lung from aspiration of pus is a major concern; thus, proper tube position should be verified by FOB, and adequate cuff inflation should be checked. Large DLT provides snug fit in the bronchus and limits aspiration. Pus may appear in the tracheal lumen (the lumen to the diseased lung); suction frequently to avoid soiling good lung.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	O_2 and isoflurane (1.0–1.5%); less required if epidural local anesthetics are used. No N_2O. Use FiO_2 = 0.5–1.0. A local anesthetic (e.g., 2% lidocaine with 1:200,000 epinephrine or 0.25% bupivacaine) can be infused or injected hourly into a thoracic (3–5 mL) or

	lumbar (5–10 mL) epidural catheter. Continuous infusion of local anesthetic generally provides better hemodynamic stability than hourly bolus injection. To enhance the effect of epidural analgesia, a loading dose of opiate (e.g., hydromorphone 0.4–1 mg [thoracic] or 1–1.5 mg [lumbar]) can be administered early in the surgery and at least 1 h before conclusion. Following intubation, isolate lung with DLT or BB. The chest tube is then removed, while the chest is prepped for the operation. Ventilate only the healthy lung. Because BPF is an abnormal communication between the bronchial tree and pleural cavity, if no chest tube is present, conventional intubation with IPPV can produce tension pneumothorax. Keep unclamped and do not remove a functioning chest tube until the lung is isolated and ventilation to the diseased lung is stopped. After the chest is opened, there is no chance of pneumothorax, but the large air leak through BPF may prevent satisfactory ventilation of that lung. High-frequency ventilation (HFV) is recommended by some, but studies show no benefit; in some patients, the BPF is actually increased with HFV.
Emergence	Before closing the chest, lungs are inflated gradually to 30 cm H_2O pressure to reinflate atelectatic areas and to check for significant air leaks. The surgeon will insert chest tubes to drain pleural cavity and aid lung reexpansion. Patient is extubated while still in OR. If postop ventilation is required (rare), the DLT is exchanged for an ETT. If BPF is still open, consider selective ventilation postop through DLT. It may be necessary to ventilate each lung separately; use smaller TVs to the lung with BPF. Alternatively, pressure-controlled ventilation may be used to avoid major air leaks through BPF. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if

	initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1; BSS Avoid hypervolemia	An overhydrated patient is at increased risk of right heart failure and pulmonary edema. Replace blood loss with crystalloid (1:3) or colloid (1:1). Third-space losses are negligible and do not need to be replaced. See Restrictive Fluid Management, Appendix K, p. K-4.
	Use vasopressor (ephedrine 5–10 mg iv bolus or phenylephrine 50–100 mcg iv bolus) if hypotensive	The use of epidural local anesthetics can cause ↓ BP in a volume-restricted patient; vasopressor is often needed.
Monitoring	Standard monitors ± CVP and/or PA line ± Arterial line	
Positioning	Axillary roll, "airplane" for the upper arm Avoid hyperextending arms ✓ and pad pressure points ✓ eyes, ears, and genitals	Placing the oximeter probe on the downside may help detect inadequate perfusion from compression.

POSTOPERATIVE

Complications	Tension pneumothorax Aspiration pneumonia	Functioning chest tube necessary to prevent tension pneumothorax.

	("down" lung)	
Multimodal pain management	Analgesic requirements minimal Parenteral opioids (iv, im, PCA), epidural, NSAID	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Alifano M, Gaucher S, Rabbat A, et al: Alternatives to resectional surgery for infectious disease of the lung: from embolization to thoracoplasty. Thorac Surg Clin 2012; 22(3):413–29.
2. Shiraishi Y: Surgical treatment of chronic empyema. Gen Thorac Cardiovasc Surg 2010; 58(7):311–6.

TRACHEAL RESECTION

SURGICAL CONSIDERATIONS

Description

The primary indications for **tracheal resection** are benign stricture and primary tracheal neoplasm. Benign strictures often are related to previous intubation or tracheostomy, whereas the most common malignant tumors include squamous cell carcinoma and adenoid cystic carcinoma. Proximal tracheal resections may also be required for trauma or idiopathic laryngotracheal stenosis. The preop assessment of patients with tracheal disease generally involves imaging with either CT or MRI, as well as bronchoscopy. Important considerations include the length and location of the lesion and the caliber of the airway. Although up to 50% of the trachea can be resected with a successful primary anastomosis, shorter segment resections are technically simpler and do not require special techniques to maximize tracheal mobility.

When operating on the upper trachea, a **cervical approach** is used. The patient is positioned supine with the neck extended. A transverse collar incision is used and subplatysmal planes are developed. The trachea is then extensively mobilized anteriorly and posteriorly. To minimize the risk of devascularizing the trachea, only the region to be removed should be circumferentially dissected. During this portion of the operation, care is taken to avoid injury of the recurrent laryngeal nerves. The trachea is then opened, the oral ETT is withdrawn into the proximal trachea, and a sterile armored ETT is passed across the operative field for cross-table ventilation. Fine, interrupted, absorbable sutures are placed, but not tied. Once all sutures are in place, the armored tube is removed and the oral ETT is positioned across the anastomosis. The ends of the trachea are approximated with minimal tension and the sutures are tied (Fig. 5.10). The neck is flexed for this portion of the procedure. A guardian suture may be placed from the chin to the chest wall to maintain neck flexion for several days. At the end of the procedure, the patient should be extubated to minimize airway irritation and disruption of the anastomosis.

When operating on the lower trachea, a **right thoracotomy** is used. The patient is positioned in the left lateral decubitus position. To facilitate exposure of the distal trachea, OLV using either a DLT or a SLT advanced into the left main bronchus is helpful.

As is apparent from the preceding discussion, all tracheal procedures require frequent communication between the surgeon and anesthesiologist. Occasionally, special techniques, such as jet ventilation or cardiopulmonary bypass (CPB), may be necessary for tracheal surgery.

Usual preop diagnosis

Tracheal stenosis or tumor (adenoid cystic carcinoma or squamous cell carcinoma most common)

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. See Appendix K, p. K-1.**

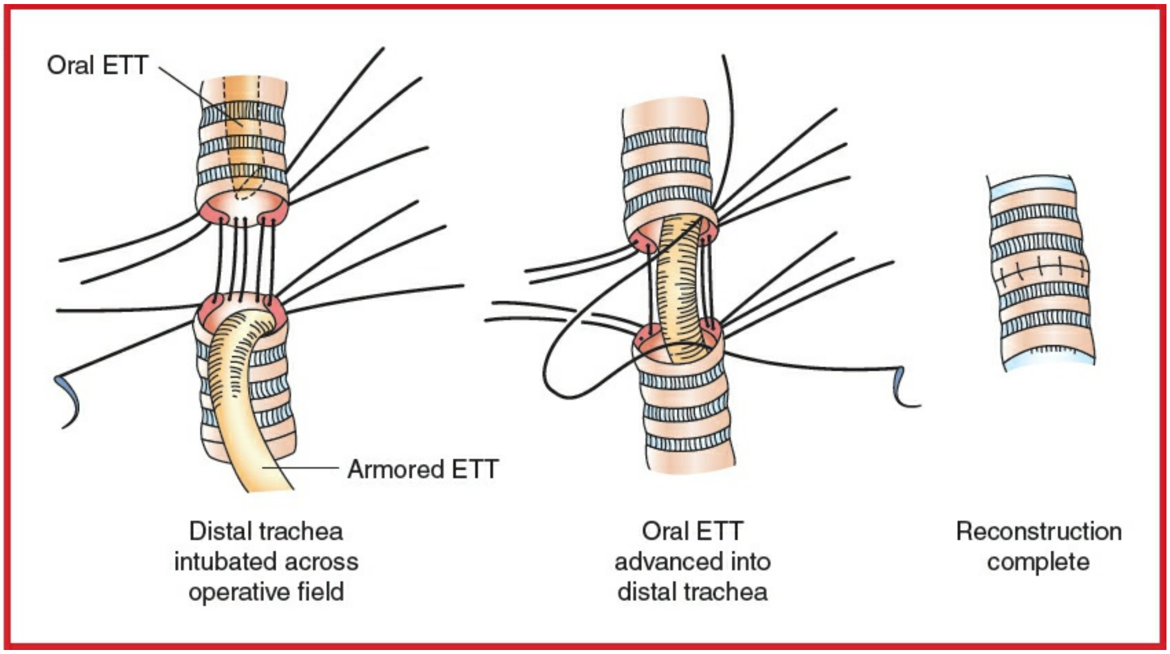


Figure 5.10 Stages of tracheal reconstruction. Note ETT in the distal trachea. (Reprinted from Grillo HC. Surgery of the trachea. Curr Probl Surg 1970; 7(7):3–59. Copyright © 1970 Elsevier. With permission.)

SUMMARY OF PROCEDURES

	Cervical Approach	Right Thoracotomy
Position	Supine	Left lateral decubitus
Incision	Transverse low cervical	Right thoracotomy
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	4 h	4 h
Closing considerations	Neck flexion (chin stitch)	⇐
EBL	<200 mL	<500 mL
Postop care	ICU	⇐
Mortality	<5%	⇐
Morbidity	Retained secretions Dehiscence Recurrent stenosis Recurrent/superior laryngeal nerve injury	⇐

	Granuloma	
Pain score	3–4	4–6

PATIENT POPULATION CHARACTERISTICS

Age range	Wide variation
Male:Female	1:1
Incidence	Rare
Etiology	Stenosis due to intubation or injury; tumor, either primary (e.g., smoking) or secondary (e.g., thyroid cancer)
Associated conditions	Carcinoid syndrome; cardiopulmonary disease; tracheoesophageal fistula (TEF)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Respiratory	Initial presentation may involve Sx of airway obstruction (stridor, cough, dyspnea), which may be misdiagnosed as asthma or pneumonitis. Patients presenting for tracheal resection almost exclusively have fixed obstruction but may have an associated dynamic component. A careful preop evaluation of the airway usually includes bronchoscopic delineation of lesion site and size. This, in combination with a review of CT images, will help to determine a plan including the type of induction, size of ETT, and location of ETT tip. **Tests:** PFT with flow/volume loops; bronchoscopy; CT scan to determine the extent of tracheal obstruction
Laboratory	Other tests as indicated from H&P
Premedication	Patients with stridor or critical airway lesions should not receive preop sedation. It is probably best to avoid sedation in all patients. Patients may be

	unable to lie flat 2° respiratory distress. Low-density helium-oxygen (heliox) mixtures have the clinical advantage of reducing airway resistance to flow past the obstruction, which may be beneficial in optimizing the patient prior to the procedure.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management for thoracotomy approach. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, combined with epidural or paravertebral catheter if sternotomy or thoracotomy approach is used. Surgery is divided into five phases: induction, dissection, open trachea, closure, and emergence. Induction, open trachea, and emergence are the critical and potentially dangerous stages. Close communication between surgeon and anesthesiologist is required.

Induction	Be prepared for airway emergency. Surgeon must be present and prepared for emergency rigid bronchoscopy and/or to perform tracheostomy below lesion. Induction depends on the site and degree of airway narrowing. Proximal or narrow lesions usually require inhalation induction or awake FOI, whereas distal or mild lesions may be approached with a rapid-sequence iv induction. Sevoflurane/O_2 is preferred for smooth inhalation induction with depression of cough reflex; avoid N_2O. High concentrations of sevoflurane may be necessary. Heliox (79% helium in oxygen) has been recommended to decrease resistance to flow past the obstruction; however, helium is not widely

	available and limits the FiO_2 that can be administered. Have a variety of laryngeal blades and uncut ETTs of various sizes, including small 5-mm tubes available. If ETT passes beyond lesion, IPPV can be started. If ETT cannot be passed, spontaneous ventilation with 100% O_2 and sevoflurane will be required if there is a dynamic obstruction, while patients with fixed lesions will tolerate PPV through a proximal tube. Careful and gradual dilation of the stenotic lesion may be required, using different sizes of ETTs or rigid bronchoscopy to facilitate placement of an adequate-size ETT for ventilation. For carinal resections, use a sterile ETT, which can be placed by surgeon directly into each bronchus during resection for cross-field ventilation. An armored tube is preferable because it avoids kinking during repeated manipulations by the surgeon. Another option is to place jet ventilation catheters into the mainstem bronchi and jet-ventilate the patient during the open trachea stage.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Inhalational anesthesia can be unreliable (and may pollute the OR environment) due to frequent tube changes and intermittent apnea. TIVA is preferable: use propofol (25–100 mcg/kg/min) and remifentanil (0.05–0.2 mcg/kg/min) infusions. After the airway is secured, muscle relaxation should be given to avoid movement or coughing during the procedure. Standard maintenance. FiO_2 = 1.0 if using apneic oxygenation; continuous monitoring with pulse oximetry is mandatory. Consider HFJV through a small-diameter catheter if ETT interferes with operation. HFJV will require iv anesthesia because inhalational agents cannot be delivered predictably; CPB can be used (rare). Analgesia requirements are

	highly dependent on surgical approach. Standard parenteral opioids are sufficient for the cervical approach, whereas epidural or paravertebral analgesia (similar to lobectomy) is used for open intrathoracic procedures.	
Emergence	Early extubation; presence of ETT and IPPV can disrupt fresh suture line. Remove ETT as soon as patient is awake enough to protect airway and breathing spontaneously but before bucking and coughing occur. The presence of a "guardian suture" (chest wall to chin) will make reintubation more difficult. Assess integrity of recurrent laryngeal nerve after high tracheal resections. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16–18 ga × 1 (left arm) BSS @ 3 mL/kg/h	✓ innominate artery compression, which will ↓ right-arm perfusion. See Goal Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ± Arterial line	Left radial artery cannulation permits uninterrupted monitoring of BP during periods of innominate artery compression. Placement of iv in the left arm allows unimpeded infusion. Right-extremity pulse oximetry will help detect innominate artery occlusion (which otherwise could lead to stroke).
Positioning	✓ and pad pressure points	

	✓ eyes	
Airway management	ETT replaced with sterile ETT and circuit intraop	After the trachea is divided, the surgeon places a sterile ETT in the distal trachea. The original ETT is withdrawn above the surgical site. The surgeon attaches a sterile anesthesia circuit to distal ETT for ventilation. Then, the surgeon places a suture through the distal tip of the original ETT. Before reanastomosis of the trachea, the distal trachea is suctioned to remove accumulated blood and secretions. After a posterior suture line is completed, the original ETT is pulled through the trachea, and the distal tube (which is below the resection) is removed. Reattach and ventilate patient through original ETT.
Complications	Tracheal edema	Corticosteroids (dexamethasone 6–8 mg iv) to ↓ tracheal edema.
	Injury to the neck	Any structure in the neck can be damaged, including superior and recurrent laryngeal nerves, the trachea, and the thoracic duct.

POSTOPERATIVE

Complications	Tracheal disruption	Neck swelling, subcutaneous

		emphysema, and inability to ventilate indicate loss of airtight anastomosis. Immediate reexploration of the neck is essential. May be fatal.
	Recurrent laryngeal nerve injury	Bilateral (occasionally unilateral) laryngeal nerve damage may → airway obstruction, necessitating reintubation. Mask ventilation may be ineffective.
Position	Airway edema Keep the head flexed to reduce tension on tracheal suture line	Place patient in head-up, neck-flexed position. Treat with nebulized racemic epinephrine if airway compromise occurs. If reintubation is required, a small ETT should be placed under direct vision or fiberoptic guidance, to avoid disruption of the anastomosis.
Multimodal pain management	Acetaminophen, NSAIDs, opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Ashiku SK, Mathisen DJ: Idiopathic laryngotracheal stenosis.Chest Surg Clin North Am 2003; 13(2):257–69.
2. Auchincloss HG, Wright CD: Complications after tracheal resection and

reconstruction: prevention and treatment. J Thorac Dis 2016; 8(Suppl 2):S160–7.
3. Gaissert HA, Grillo HC, Shadmehr MB, et al: Long-term survival after resection of primary adenoid cystic and squamous cell carcinoma of the trachea and carina. Ann Thorac Surg 2004; 78(6):1889–96, discussion 1896–7.
4. Hobai IA, Chhangani SV, Alfille PH: Anesthesia for tracheal resection and reconstruction. Anesthesiol Clin 2012; 30(4):709–30.
5. McRae K: Anesthesia for airway surgery. Anesthesiol Clin North Am 2001; 19(3):497–541.

TRACHEOBRONCHOPLASTY

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

Posterior splinting tracheobronchoplasty is performed for tracheobronchomalacia, when the airway collapses on exhalation. Patients often present with shortness of breath, cough, and recurrent respiratory infections; many have undergone Y-stent trials to determine whether their symptoms improved with airway stabilization. The operation is performed through a **right posterolateral thoracotomy**. The posterior distal trachea, bilateral mainstem bronchi, and bronchus intermedius are exposed. A Y-shaped mesh is sewn onto the membranous wall using many rows of sutures, placating and stabilizing it. Left lung ventilation is used during the operation; often a wire-reinforced SLT in the left mainstem bronchus is used so that the trachea is not distorted. When the mesh is sewn onto the left mainstem bronchus, intermittent ETT cuff deflation and apnea are used, so the cuff is not punctured. Flexible bronchoscopy is performed at the end of the procedure, withdrawing the ETT into the proximal trachea.

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES	
Position	Left lateral decubitus
Incision	Right posterolateral thoracotomy

Special instrumentation	Mesh
Antibiotics	Cefazolin 2 g iv
Surgical time	4–6 h
EBL	<500 mL
Postop care	ICU
Mortality	Minimal
Morbidity	Respiratory complications are common
Pain score	5–7

PATIENT POPULATION CHARACTERISTICS

Age range	Older adults
Male:Female	3:1
Incidence	Not uncommon
Etiology	Unknown
Associated conditions	Smoking history, asthma, bronchitis, emphysema, GERD

ANESTHETIC CONSIDERATIONS

Tracheobronchoplasty is a procedure performed for tracheobronchomalacia (cartilaginous or membranous) that typically obstructs more than 80% of the lumen on dynamic CT. Prior to the procedure, a trial period with a Y-stent is often used to predict if the surgery may be beneficial. The procedure is done through a high right thoracotomy and is therefore similar to a tracheal resection although the airway is not usually entered. The inverted Y-shaped membranous tracheobronchial wall is plicated with a mesh or biologic membrane with sutures that do not pierce the entire thickness of the wall; however, the risk exists and puncture of the cuff is possible.

PREOPERATIVE

The preop considerations for tracheobronchoplasty are similar to those for tracheal resection.

Respiratory	H&P complemented by lab and imaging studies to focus on the underlying disease, evaluating for pulmonary problems. Constant coughing, inability to raise secretions, and recurrent respiratory infection are typical. Often there is a history of smoking, bronchitis, emphysema, asthma, and GERD. **Tests:** Consider PFTs, inspiratory-expiratory CT (dynamic CT), and awake functional bronchoscopy to assess the degree of luminal obstruction. When respiratory distress is present at rest or when exercise tolerance is poor, an ABG may be indicated to assess increased pulmonary risk ($PaO_2 < 70$ and $PCO_2 > 45$).
Cardiovascular	Many patients with tracheobronchomalacia may have concomitant cardiac disease, which may need cardiac consultation to optimize cardiovascular status. **Tests:** ECG and stress echo may be indicated.
Hematologic	Adequacy of oxygen-carrying capacity (Hb/Hct) is important to ascertain. A type and screen if the starting hematocrit is low. **Test:** Hct
Laboratory	Hb/Hct for oxygen-carrying capacity and BMP to ascertain that the K^+ and creatinine levels are in the normal range for safe use of muscle relaxants.
Premedication	Antisialagogue (e.g., glycopyrrolate 0.2 mg iv) to decrease secretions and mild sedation with midazolam (0.5–2 mg iv) along with a small dose of fentanyl (25–50 mcg) may be needed. Avoid respiratory depression and oversedation.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80

	mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA combined with thoracic epidural opioids (T5 or T6) since the procedure involves a high thoracotomy.

Induction	Following preoxygenation, a standard induction using propofol (1–2 mg/kg) or etomidate (0.3 mg/kg). A depolarizing relaxant (succinylcholine 1 mg/kg iv) or a short-acting nondepolarizing relaxant (e.g., rocuronium 0.3–0.6 mg/kg iv) can be used. A small dose of fentanyl (1–2 mcg/kg iv) or remifentanil (0.25–1 mcg/kg iv) can be used to facilitate intubation.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Intubation	Intubation is performed using either (a) a long wire-reinforced endobronchial tube (WRETT) placed over a fiberoptic scope so the end of the tube lies just above the secondary carina of the left main bronchus or (b) a left-sided 37F or 39F DLT, which has the tracheal component shaved off by the surgeon resulting in a long endobronchial tube, which is again placed with a FOB to lie just above the secondary carina.
Maintenance	Isoflurane or sevoflurane in oxygen (FiO_2 = 1); VT, 4–6 mL/kg; PEEP, 5–10 cm. Only the left lung is ventilated during the surgery. When the procedure involves the posterior wall of the trachea and right main bronchus, the cuff of the endobronchial tube is inflated. During surgery on the left main bronchus, the cuff is intermittently deflated (with intermittent

	left lung ventilation). At the end of the procedure, a FOB is used to assess the result of the surgery.	
Emergence	Muscle relaxant is reversed and extubation is accomplished with patient sitting up. To ICU for the first 24 h. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18 ga × 1 + 16 or 14 ga × 1 Limit BSS to and colloids	Blood should be available but is rarely needed. Vasopressors (e.g., phenylephrine, ephedrine) are used to maintain MAP. See Restrictive Fluid Management, Appendix K, p. K-4
Monitoring	Standard monitors + Arterial line Urinary catheter +/− CVP or PA line	Mandatory to follow oxygenation continuously during OLV, typically using pulse oximetry and intermittent ABGs
Positioning	Left lateral, axillary roll + airplane for the upper arm Avoid hyperextending the arm ✓ and pad pressure points ✓eyes, ears, and genitals	✓radial pulses to ensure correct placement of axillary roll. Placing the arterial line or the pulse oximeter on the down arm may assist in monitoring perfusion.
Complications	Hypoxia	Pulse oximetry monitored continuously. Interrupt the procedure to allow oxygenation to improve

		before proceeding if necessary.
	Hypercapnia	Ventilate to normalize the $ETCO_2$ before proceeding.
	Hemorrhage	Be prepared, but usually not a problem.

POSTOPERATIVE

Complications	Airway irritation	Airway irritation causing cough (Rx, lidocaine topical spray prior to or during the procedure) or bronchospasm (Rx, nebulized bronchodilators, e.g., racemic epinephrine or albuterol). Occasionally steroids may be necessary.
	Hypoxia	Humidified O_2 by mask.
	Hypoventilation	Avoided by adequate reversal of neuromuscular blockade and using minimal-dose short-acting opioids.
	Hemorrhage	Usually self-limiting.
Multimodal pain management	Epidural ± Opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6). Try limiting opioids to decrease respiratory depression and ineffective coughing.
Tests	CXR	A postop CXR is often performed to confirm the absence of pneumothorax or

ERAS	PONV protocol (see Appendix B, p. B-1)	mediastinal emphysema. Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Buitrago DH, Wilson JL, Parikh M, et al: Current concepts in severe adult tracheobronchomalacia: evaluation and treatment. J Thorac Dis 2017; 9(1):E57–66.
2. Damle SS, Mitchell JD: Surgery for tracheobronchomalacia. Semin Cardiothorac Vasc Anesth 2012; 16(4):203–8.
3. Gangadharan SP, Bakhos CT, Majid A, et al: Technical aspects and outcomes of tracheobronchoplasty for severe tracheobronchomalacia. Ann Thorac Surg 2011; 91:1574–81.
4. McLaurin S, Whitener GB, Steinburg T, et al: A unique strategy for lung isolation during tracheobronchoplasty. J Cardiothorac Vasc Anesth 2017; 31(2):731–7.
5. Wright CD: Tracheal surgery: posterior splinting tracheoplasty for tracheomalacia. Oper Tech Thorac Cardiovasc Surg 2015; 20:31–45.

EXCISION OF BLEBS OR BULLAE

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

Pulmonary blebs or bullae requiring surgical treatment may vary from small, apical blebs—most usually seen in young people with spontaneous pneumothorax—to expanding, giant bullae causing respiratory distress. Specific indications for bullectomy include large size (>30% of the lung), recurrent pneumothorax, dyspnea in conjunction with compressed adjacent parenchyma, and recurrent infection of the bullae. The small blebs can be excised through **VATS**, while giant bullae are generally removed by **thoracotomy**. In either case, the goal is to resect the nonfunctional bullae and allow the compressed yet relatively preserved lung tissue to reexpand and contribute to gas exchange. The surgical technique generally involves **stapling** across the base of the bulla with reinforcing strips being applied to the staple line to minimize air leak. Alternatively, a **clamp and suture** technique may be used. However, the most important point is that an airtight closure should be obtained as a prolonged air leak can be very debilitating. Patients undergoing operation for giant bullae frequently have limited pulmonary reserve and present formidable operative risks. Because the operation is planned to improve their pulmonary function, however, these patients frequently do well following operation. **Pleural abrasion** or, rarely, **pleurectomy** may accompany the excision of blebs or bullae. The blebs in young patients with recurrent spontaneous pneumothorax usually are located at the apex of the upper lobe. Bullae in patients with emphysema are usually in the upper lobe but may be anywhere in the lung. Preop localization by CT scan is usually sufficient. The patient is usually in the lateral position.

Variant procedure or approaches

Patients with more generalized emphysema may be candidates for lung volume reduction surgery.

Usual preop diagnosis

Spontaneous pneumothorax 2° ruptured blebs; giant bullae causing respiratory distress

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURES

Position	Usually lateral
Incision	Thoracotomy or VATS
Special instrumentation	Staplers
Antibiotics	Cefazolin 2 g iv
Surgical time	1–3 h
EBL	<500 mL
Postop care	PACU → room; ICU or IIC for giant bullae
Mortality	Minimal
Morbidity	Air leak: 20% or more in giant bullae
Pain score	5–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Young adults (blebs/small bullae), elderly (large bullae)
Male:Female	3:1
Incidence	Not uncommon
Etiology	Emphysema (usually 2° smoking), congenital, infectious
Associated conditions	Spontaneous pneumothorax, emphysema, long smoking Hx, α-antitrypsin deficiency

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients with apical blebs tend to be young with normal lung function and otherwise healthy. Bleb resection in these patients tends to be a routine thoracoscopic wedge resection procedure. Patients with bullous emphysema, on the other hand, may have end-stage lung disease, often with pulmonary hypertension (HTN) and RV dysfunction. The risk of rupture of a bulla/bleb on the nonoperated side, with resultant tension pneumothorax, must be considered throughout the procedure. The majority of considerations and concerns therefore relate to the patient with bilateral disease. Most procedures are done thoracoscopically.

Respiratory	Cysts may be bronchogenic, postinfective, infantile, or emphysematous. With blebs, elicit Hx of repeat pneumothoraces. Bullae usually result from destruction of alveolar tissue; they represent end-stage emphysematous disease associated with severe COPD. Patients may have incapacitating dyspnea and limited pulmonary reserve. CO_2 retention ± hypoxia may be present. Obtain PFTs and ABG for baseline. **Tests:** CXR; presence of pneumothorax; if chest CT is available, look for bilateral bullous disease and rule out airway anomalies that could interfere with DLT placement; ABG, as indicated from H&P.
Cardiovascular	**Test:** ECG
Neurologic	✓ Hx for previous back surgery and peripheral

	neuropathy. Examine thoracolumbar area for skin lesions, infection, and deformities.
Hematologic	Transfuse patient with preop Hct < 25% (Hct < 30% if patient has CAD). Crossmatch 2 U of blood or obtain 1–2 U of autologous blood during the month before surgery, or consider erythropoietin therapy in patients who are anemic. **Test:** Hct
Laboratory	Other tests as indicated from H&P.
Premedication	Midazolam 1–2 mg iv if patient is anxious and not in respiratory distress. Minimize sedation if planning epidural placement, as the combination of parenteral sedation and epidural opioid, may result in significant respiratory depression.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA—combined with regional technique if open thoracotomy.

Induction	**Minor bleb:** standard induction **Major bullae (particularly bilateral):** main issue is risk of pulmonary tamponade/tension pneumothorax secondary to bulla expansion by PPV. Best to establish lung isolation prior to instituting PPV (DLT essential in bilateral disease, to allow for ventilation/exclusion of either side; BB sufficient for unilateral disease). Strategies include an awake intubation (difficult with DLT), a spontaneous

	ventilation induction (inhalational or iv), or, in patients with easy airway anatomy, a true rapid-sequence intubation. Lung isolation should immediately be confirmed with FOB prior to providing PPV. Rapid placement of a chest tube is essential should a bulla rupture.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Patients with severe COPD may have significant auto-PEEP. To avoid worsening dynamic hyperinflation of the lung, treat bronchospasm aggressively and allow adequate expiratory time (↓ I/E ratio, low RR). Caution with applied PEEP: it may increase total PEEP (and therefore air trapping). Minimize risk of bulla rupture by reducing ventilatory pressures (low tidal volumes, permissive hypercapnia, and pressure-controlled ventilation @ <20 cm H_2O). Inhalational anesthesia supplemented with epidural and local anesthetics or iv opioids. Avoid N_2O at all times because bullae may be filled with air.	
Emergence	Reexpand the lung under direct vision to check for major air leaks. Extubate patient early. In postbullectomy, unlike other lung resections, patients have greater functional lung tissue than preop. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 16 ga × 1 BSS @ 1–2 mL/kg/h Use vasopressors if hypotensive	Excess fluid predisposes to right heart failure. Epidural local anesthetics can ↓ BP in a volume-restricted patient;

		vasopressor is often needed (e.g., ephedrine 5–10 mg iv bolus or phenylephrine 50–100 mcg iv bolus). See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line ± CVP, or PA line, ± TEE	For patients with coexisting severe cardiac disease
Positioning	✓ and pad pressure points ✓ eyes, ears, and genitals Axillary roll, "airplane" for the upper arm	See Positioning.
Ventilation	DLT or BB needed to separate the lungs	Allows PPV of the nonoperative lung. Use gentle PPV (pressure control) with low Vt and permissive hypercapnia ($PaCO_2$ 50–70 mm Hg). Inspiratory pressure should be as low as possible (~10 cm H_2O), to reduce likelihood of rupture of bullae in nonoperative lung. Treat intraop hypoxemia with CPAP to "up" lung. In extreme cases, consider CPB (rare).
Complications	Tension pneumothorax Bronchopleural cutaneous fistula Hypoxia Hypercarbia Dysrhythmias	Can occur on either side during induction, only on nonoperated side after the chest is open, and again on either side postop. Presents with ↑ ventilatory pressure, progressive tracheal

		deviation, wheezing, and cardiovascular collapse. CXR to r/o tension pneumothorax. Rx: insertion of chest tube. Placement of a chest tube can create a bronchopleural cutaneous fistula. Rx: low Vt, spontaneous ventilation; may require DLT for differential ventilation. ✓ position of DLT, suction DLT, and avoid hypoventilation.

POSTOPERATIVE

Complications	Hypoventilation Dental damage Airway trauma Pneumothorax Hemorrhage Risk of aspiration Airway obstruction (bronchospasm, bleeding, dislodged tumor, FB)	Ensure complete reversal of muscle relaxants and patient is wide awake with good respiratory effort prior to extubation. Obtain ABG if patient has difficulty breathing or is overly sedated. If nerve blocks are used to depress gag reflex for awake FOI, keep patient NPO for several hours postop.
Multimodal pain management	Regional analgesia Epidural opioids Parenteral opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6). Parenteral opioids + intrapleural local anesthetics (0.5% bupivacaine + 1:200,000 epinephrine, 0.5

		mL/kg) + single-shot paravertebral blocks are adequate for thoracoscopy.
Tests	CXR	ABG if indicated.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Brister NW, Barnette RE, Kim V, et al: Anesthetic considerations in candidates for lung volume reduction surgery. Proc Am Thorac Soc 2008; 5(4):432–7.
2. Gasparini S, Zuccatosta L, Bonifazi M, et al: Bronchoscopic treatment o emphysema: state of the art. Respiration 2012; 84(3):250–63.
3. Holbek BL, Horsleben Petersen R, et al: Fast-track video-assisted thoracoscopic surgery: future challenges. Scand Cardiovasc J 2016; 50(2):78–82.
4. Palla A, Desideri M, Rossi G, et al: Elective surgery for giant bullous emphysema: a 5-year clinical and functional follow-up. Chest 2005; 128(4):2043–50.
5. Pompeo E, Tacconi F, Mineo D, et al: The role of awake video-assisted thoracoscopic surgery in spontaneous pneumothorax. J Thorac Cardiovasc Surg 2007; 133(3):786–90.

LUNG VOLUME REDUCTION SURGERY

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

Lung volume reduction surgery was initially described by Brantigan in 1958 but reintroduced by Joel Cooper in 1995 for the treatment of severe emphysema. Typically, patients referred for LVRS are chronically ill, requiring steroids, bronchodilators, and supplemental oxygen. With appropriate preop selection and perioperative care, these patients survive surgery and demonstrate improved pulmonary function. Physiologically, reducing the volume of the lung by resecting diseased tissue improves elastic recoil and decreases airway resistance. The chest cavity also is reduced in size, thereby improving chest wall and diaphragmatic function.

The procedure can be carried out either through a median sternotomy or minimally invasively. The **open approach** begins with a median sternotomy. OLV is initiated following opening of the pleurae. Often the diseased portions of the lung remain inflated, whereas healthy areas develop absorption atelectasis. These diseased portions are resected with the aid of a linear stapler. The visceral pleura is very thin; the stapling is done with bovine pericardium to bolster the staple line; and high inspiratory pressures (>20 cm H_2O) must be avoided. From 15% to 30% of the lung volume may be removed. Following careful examination for air leaks, the pleurae and chest wall are closed.

The **minimally invasive approach** is carried out using standard VATS techniques and instrumentation. Diseased tissue will have been identified preop using V/Q and CT scans. VATS forceps are used to guide this diseased tissue into the jaws of the stapler. Again, 15–30% of lung tissue may be removed by this means. At some centers, the anesthesiologist may be asked to measure inspiratory and expiratory volumes. Any difference between these volumes may represent an air leak requiring further exploration. Following this, access ports and the thoracotomy are closed, and chest tubes are placed. The patient is turned over to the opposite side, reprepped, and redraped, and the surgery is repeated. With either approach, patients should be extubated in the OR so that no unnecessary ventilatory pressures are put on the lungs. A functional epidural catheter and early extubation are important to the success of this procedure, especially in the very ill patient. Pleural drainage consists of one or two chest tubes per side; in contrast with lobectomy, they are left to water seal so as not to exert excessive negative pressure on the lung and disrupt the staple lines. A small pneumothorax is acceptable if the patient is not in respiratory distress. Ideally, patients are extubated at the conclusion of the operation. Their respiratory status is often tenuous, so close monitoring, vigorous pulmonary toilet, and good pain control are essential in the postop period.

Bronchoscopic LVRS often involves the endoscopic placement of one-way endobronchial valves that allow air to escape from selected lung segments. This will induce volume reduction through lobar atelectasis but only in areas devoid of collateral ventilation (assessed by CT imaging). Other bronchoscopic techniques include intrabronchial valves (conscious sedation for placement) or coil insertion (requiring GA). Coils appear to function by restoring the elastic recoil in small airways, thereby preventing early airway collapse during expiration (Table 5.3).

Usual preop diagnosis

COPD (emphysema)

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan. See Appendix K, p. K-1.

SUMMARY OF PROCEDURES

	Open LVRS	VATS LVRS
Position	Supine	Lateral decubitus
Incision	Median sternotomy	Three small incisions
Special instrumentation	Stapling devices, DLT	⇐+ Endoscopic instrumentation
Unique considerations	Bovine pericardium to bolster staple line	⇐
Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	2 h	1 h per side
Closing considerations	Avoid high PIPs (>20 cm H_2O)	⇐
EBL	Minimal	⇐
Postop care	Extubated in OR; avoid chest tube suction	⇐
Mortality	≤5–10%	⇐
Morbidity	Pneumothorax Infection Tearing of suture line Wound healing problems	⇐ ⇐ ⇐ ⇐
Pain score	6–8	4–6

PATIENT POPULATION CHARACTERISTICS

Age range	>50 yr

Male:Female	Male > female
Incidence	Although the incidence of emphysema is high in the general population, only a fraction of these patients will be candidates for LVRS
Etiology	Smoking, genetic factors
Associated conditions	CAD; pulmonary HTN; PVD; cerebrovascular disease

Table 5.3 Suggested Selection Criteria for LVRS

Medical history	Severe COPD (emphysema rather than chronic bronchitis) Age < 75 yr No cigarette smoking for 6 mo Lowest effective prednisone dose No previous chest surgery
Pulmonary function	FEV_1 > 30–35% of predicted $PaCO_2$ < 50 mm Hg TLC > 100% of predicted (hyperinflation)
Cardiac function	Mean PAP < 35 mm Hg (if pulmonary HTN is suspected) No evidence of LV dysfunction on dobutamine stress testing (if Hx of angina or CHF is present)
Radiographic	Hyperinflation, flattened diaphragm (CXR) Decreased upper lobe perfusion (ventilation-perfusion scan) Emphysema, with upper lobe predominance (CT scan)
Relative exclusion criteria	Continued smoking Illness other than emphysema that may cause severe dyspnea (e.g., CAD, CHF, cancer, interstitial lung disease, bronchiectasis) Severe malnutrition Obliteration of pleural space (e.g., pleurodesis or pleurectomy) Previous thoracic surgery Morbid obesity Severe pulmonary HTN (mean PAP > 35) Chest wall deformity with restrictive lung disease (e.g.,

	kyphoscoliosis, severe pectus deformity, $PaCO_2$ > 55 mm Hg)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

LVRS involves either laser thermal contraction or surgical resection of emphysematous lung tissue. Patients have end-stage emphysema with associated pulmonary HTN and RV dysfunction. These patients are a great challenge to the anesthesiologist because it may be difficult to maintain relatively normal physiologic parameters intraop and to have an awake, comfortable, and spontaneously breathing patient at the completion of surgery. Carefully selected patients may undergo endoscopic lung volume reduction with placement of valves (usually requiring only sedation) or coils (usually requiring GA).

Respiratory	These patients have advanced pulmonary emphysema. Examine patient for cyanosis, clubbing, RR, and pattern. Listen to the chest—breath sounds are often very distant or absent. Hx should include the use of O_2 supplementation, recent infection, severe bronchospasm, prior surgery on the chest, and other associated diseases, such as CAD or CHF. **Test:** PFT (± bronchodilators). Hyperinflation usually is indicated by TLC and RV value >120%; V/Q scan; CXR; chest CT scan; noninvasive exercise test (6-min walk); preop ABG—check for hypoxemia and hypercarbia.
Cardiovascular	These patients often have coexisting cardiac disease with pulmonary HTN. **Tests:** Right heart function can be determined by echocardiography or selective right heart catheterization, if indicated by H&P.
Premedication	Avoid premedication with sedative or opioids—cannot have respiratory depressants—patients have severe COPD and often are CO_2 retainers.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

	Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop pain management in open surgery. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA (with DLT or BB) ± epidural anesthesia. Place and test epidural catheter before surgery. (See Lobectomy, Pneumonectomy.)

Induction	Standard induction. DLT or BB is absolutely necessary during surgery to selectively ventilate each lung. DLT preferable for patients with severe bilateral disease with increased risk of pneumothorax (see Lobectomy, Pneumonectomy).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks for open surgery.
Maintenance	Inhalational agent and/or propofol. IV opioids and sedative agents should not be used (exception, remifentanil infusion intraop). Anesthesia may be supplemented by continuous or bolus administration of epidural anesthetics. Mechanical ventilation with O_2 and inhalation agent only. During TLV: Vt = 6–10 mL/kg, limit PIP to <25 cm H_2O (best to use PCV). Respiratory rate and inspiratory flow should be adjusted to minimize air trapping (low RR, long E-time). Beware of overinflation and "breath stacking," which can → pulmonary tamponade with severe ↓ BP and ↑ airway resistance. ABG to determine $ETCO_2$-$PaCO_2$ gradient. During OLV, use PCV for Vt = 4–8 mL/kg, long E-time, RR 6–10, and permissive hypercapnia ($PaCO_2$: 50–70 mm Hg); monitor lung

	compliance closely because the ventilated lung also has emphysema/bullous disease. Avoid overdistension with hyperinflation (→ pneumothorax) on ventilated, nonoperated lung. Patients with severe COPD may have significant auto-PEEP. To avoid dynamic hyperinflation of the lung, treat bronchospasm aggressively, allow adequate expiratory time (↓ I:E ratio), limit PIP to 15–20 cm H_2O, and cautiously use applied PEEP, as it may increase total PEEP (best managed with in-line spirometry). Moderate hypercapnia is well tolerated, but higher levels may result in significant pulmonary HTN, RV dysfunction, and tachydysrhythmias. CPAP to nonventilated lung may be necessary to maintain oxygenation. Maintenance of hemodynamic stability may require pressor support (e.g., ephedrine, phenylephrine, dopamine). Dynamic hyperinflation of the lung (pulmonary tamponade) during mechanical ventilation should be suspected if ↓ BP with ↑ PIP occurs.
Emergence	Emergence is a critical time for these patients. All physiologic parameters need to be optimized. Air leaks are common and may be worsened by prolonged mechanical ventilation, coughing, or straining on the ETT. Different options are available to achieve those goals, including deep extubation or deep conversion to spontaneous pressure support ventilation. Either way, ventilation has to be assisted until patients are wide awake, comfortable, warm, and maintaining adequate respiration with minimal support. These patients are critically dependent on good analgesia, upright/sitting position, and suctioning to ↓ mucous plugging (can be catastrophic). Recovery from GA occurs in the OR and may take 1–2 h. Ventilatory assistance via face mask with supplemental O_2 is usually necessary. ABG and CXR may be useful. When spontaneous ventilation and analgesia are satisfactory, the patient is

	transported to ICU. If postop mechanical ventilation is required, consider using pressure support ventilation with low levels of CPAP. The CPAP may help minimize the inspiratory work of breathing caused by lung hyperinflation. The pressure support mode of ventilation will permit control of airway pressure, while allowing patient control of $PaCO_2$. If postop intubation is anticipated, changing from DLT to ETT is required. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14–16 ga × 1 BSS @ 1–2 mL/kg/h Autologous PRBC	Fluid management to restore preop deficit and provide maintenance fluid but restrict fluids similar to lung resection surgery. Replace minor blood loss with colloid. Transfuse autologous blood for Hct < 30. See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors Arterial line Urinary catheter CVP and/or PA line	See Lobectomy, Pneumonectomy ✓ ABGs: baseline (preop) during sternotomy, 15 min after initiation of OLV, 15 min after initiation of OLV on the second lung, during closure of sternotomy, prior to extubation Useful for postop fluid management Depending on the level of PHTN, RV dysfunction, and

		comorbid CAD
Positioning	✓ and pad pressure points ✓ eyes, ears, and genitals	Axillary roll and support for upper airway are necessary for the lateral decubitus position (see Lobectomy, Pneumonectomy)

POSTOPERATIVE

Complications	Hypercarbia Hypoxemia Air leak/pneumothorax Hemorrhage	
Multimodal pain management	Lumbar or thoracic epidural opioids ± Local anesthetics	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with peripheral local anesthetics (see Appendix K, p. K-6). Essential that patient be comfortable. Begin infusion of epidural opioids and local anesthetics (see Lobectomy, Pneumonectomy). Breakthrough pain treatment options include iv hydromorphone (cautious boluses of 0.2–0.5 mg) and/or ketorolac (30 mg).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Geddes D, Davies M, Koyama H, et al: Effect of lung-volume-reduction surgery in patients with severe emphysema. N Engl J Med 2000; 343(4):239–45.

2. Hartigan PM, Pedoto A: Anesthetic considerations for lung volume reduction surgery and lung transplantation. Thorac Surg Clin 2005; 15(1):143–57.
3. Irons JF, Martinez G: Anaesthetic considerations for non-intubated thoracic surgery. J Vis Surg 2016; 2:61.
4. Ramaswamy A, Puchalski J: Bronchoscopic lung volume reduction: recent updates. J Thorac Dis 2018; 10(4):2519–27.
5. van Agteren JE, Carson KV, Tiong LU, et al: Lung volume reduction surgery for diffuse emphysema. Cochrane Database Syst Rev 2016; 10:CD001001.

BRONCHOPULMONARY LAVAGE

SURGICAL CONSIDERATIONS

Description

Whole-lung bronchopulmonary lavage is occasionally required for patients with pulmonary alveolar proteinosis—a condition characterized by excessive (or abnormal) surfactant production with resultant flooding of the lungs with proteinaceous fluid. Although the underlying cause of the condition is unclear, treatment consists of periodic whole-lung lavage and, more recently, GM-CSF administration. It is our practice to perform **unilateral lavage** only, although single-session, bilateral lavage has been reported. After induction of GA, a DLT is placed, and the correct position is confirmed bronchoscopically. The lung is then lavaged in aliquots of 500–1000 mL normal saline to dilute and wash out excess alveolar surfactant, pus, or mucus and to obtain material for cytologic and histochemical examination. Care should be taken not to overdistend the lung, and a running tally of fluid instilled and withdrawn should be performed to avoid overhydrating the patient. Frequently, 9–12 L of fluid is used, with the initial effluent being very cloudy and the final effluent being clear. Techniques that may improve the distribution of the lavage fluid include external chest percussion and tilting the operating table (laterally as well as in the craniocaudal directions).

Usual preop diagnosis

Pulmonary alveolar proteinosis; refractory asthma; cystic fibrosis; bronchiectasis; lipoid pneumonitis; silicosis; alveolar microlithiasis; inhalation of radioactive dust

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, postop pain management options, and physiotherapy plan.** **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURES

Position	Supine or lateral decubitus
Special instrumentation	Bronchoscope, DLT, lavage fluids
Antibiotics	None
Surgical time	45 min/side
EBL	None
Postop care	PACU → home
Mortality	Rare
Morbidity	Aspiration of lavage fluid Pneumothorax/hydrothorax Atelectasis
Pain score	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	17–35 yr
Male:Female	1:1
Incidence	Uncommon
Etiology	Pulmonary alveolar proteinosis, cystic fibrosis, bronchiectasis, lipoid pneumonitis, silicosis
Associated conditions	Asthma and other abnormalities of lung function

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Although whole-lung lavage is primarily a treatment for pulmonary alveolar proteinosis, it is also used as a therapeutic modality for many other lung-related conditions (see Usual preop diagnosis, above). Respiratory dysfunction of variable symptomatology is expected in these patients. Indications for whole-lung lavage include dyspnea on exertion, resting room air $PaO_2 < 60$ mm Hg, and shunt fraction >10–12%. Because the procedure requires GA with OLV, it is recommended that preop V/Q scans be obtained so that unilateral lung irrigation can be performed first on the more severely affected lung. If both lungs are equally diseased, lavage should be performed on the left lung initially to allow the larger right lung to be used for ventilation to provide better gas exchange. Patients then return in subsequent days or weeks for therapeutic lavage of the contralateral lung. A nonoperative treatment option for pulmonary alveolar proteinosis using granulocyte-macrophage colony-stimulating factor (GM-CSF) has shown promise for some patients with the acquired form of pulmonary alveolar proteinosis.

Respiratory	Patients with pulmonary alveolar proteinosis generally present with cough (nonproductive > productive), dyspnea, and fatigue. Physical findings consist of diffuse rales ± clubbing or cyanosis. CXR typically reveals diffuse bilateral patchy airspace consolidation. Pulmonary compliance is reduced 2° the restrictive disease pattern. PFTs show ↓ TLC, ↓ RV, ↓ VC, and ↓ D_LCO. Though nonspecific, baseline ABGs classically demonstrate respiratory alkalosis and hypoxemia, with a calculated elevation in A-a DO_2 gradient. Some patients may be O_2-dependent or have concomitant COPD 2° Hx of smoking. Secondary infections, especially in the respiratory tract, are well-recognized risks in these patients. Cessation of smoking (>6–8 wk) and prompt treatment of any underlying infections before whole-lung lavage may be prudent. **Tests:** CXR; PFT; ABG; V/Q scans; ± CT
Cardiovascular	Directed at any underlying disease process.
Laboratory	Abnormal elevation of serum LDH has been reported in some patients and has been shown to correlate with the severity of A-a DO_2. **Tests:** LDH; other tests as indicated from H&P
Premedication	Standard premedication. Avoid heavy sedation that might impair adequate gas exchange before induction. O_2 supplementation may be required after premedication.
ERAS	Check compliance with presurgical ERAS

	recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

GETA is required with OLV using a DLT. In pediatric patients or small adults where appropriate-sized DLT is not available and in patients who cannot tolerate OLV, the use of partial venoarterial CPB, venovenous bypass, or even ECMO has been reported. In some cases, sequential lobar lavage with smaller volume of lavage fluid has been achieved under conscious sedation for those patients in whom OLV is not possible.

Induction	Standard induction with thorough preoxygenation (FeO_2 > 90%). The largest DLT should be used to facilitate infusion and drainage of the lavage fluid. (Placement of DLT is discussed on p. 320.) Due to the underlying impaired respiratory function in these patients, instituting OLV will further compromise gas exchange. It is imperative to avoid spillage of lavage fluid into the ventilated lung during the procedure. This is achieved by verifying the correct position of the DLT with FOB and confirming the competency of the cuff seal by testing it against pressures as high as 50 cm H_2O. Baseline individual lung compliance, airway pressures, and ABG should be ✓'d after DLT placement.
ERAS	Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	100% FiO_2 with inhalational agent (e.g., sevoflurane) or TIVA using propofol and remifentanil infusions. Muscle relaxation is necessary to avoid movement or coughing that can cause leakage of lavage fluid into the ventilated lung. Because chest physiotherapy, including vibration and percussion, is required after each cycle of lavage filling, any deleterious change from baseline lung compliance decrease or airway pressure increase of the ventilated lung should prompt the anesthesiologist to look for evidence of

	spillage. Filling the nonventilated lung with lavage fluid can → ↓ pulmonary shunting (→ improved O_2 sat) and ↑ PVR (→ ↓ CO). Draining the lavage fluid can → ↑ pulmonary shunting (→ ↓ O_2 sat) and ↓ PVR (→ ↑ CO). Accurate inflow and outflow volume of the lavage fluid should be recorded to ensure return volumes are 90%.	
Emergence	At the conclusion of surgery, the lavaged lung should be adequately suctioned to remove any residual fluid. Bilateral lung ventilation should be reinstituted. Because the compliance of the lavaged lung is greatly reduced, higher airway pressures are required to reexpand it, but at the risk of causing barotrauma to the nonlavaged lung. A brief period of OLV to the lavaged lung, using large TVs or ↑ airway pressure, may be necessary to recruit the collapsed alveoli (higher pressure, including PEEP, is needed to counter the increase in surface tension after the removal of significant amounts of surfactant). The patient's trachea should be extubated after adequate reversal of muscle relaxation; however, if prolonged postop ventilation is anticipated (e.g., in those who aspirated lavaged fluid to the ventilated lung), changing the DLT to single-lumen tube is indicated. DLT should be maintained for patients in whom differential lung ventilation is required postop.	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 1–2 mL/kg/h	See Restricted Fluid Management, Appendix K, p. K-4
Monitoring	Standard monitors Arterial line ± CVP line ± PA line	Although warmed lavage fluid is used, some patients may become hypothermic after several hours of lavage under GETA. CVP or PA line is only needed as indicated by comorbid

		conditions. Consider fluoroscopic or TEE-guided positioning of a PA catheter to the lavaged lung. In doing so, the therapeutic potential of the catheter may be realized. Selective occlusion of pulmonary arterial blood flow can improve matching of blood flow to ventilation.
Positioning	Supine ✓ and pad pressure points ✓ eyes	Some prefer the lavaged lung to be dependent to minimize the risk of contaminating the contralateral lung with lavage fluid. Others prefer the lavaged lung to be nondependent because, in this position, perfusion will more closely match ventilation in the dependent lung. As a compromise, we perform lavage with the patient supine. However, intermittent manipulation of the patient position may be required to facilitate the lavage and drainage of any retained fluid. Prevention of DLT displacement during repositioning is paramount.
Complications	Hypoxemia	Hypoxemia during OLV is most commonly 2° luminal obstruction (by blood or pulmonary secretions) of the DLT, worsening of shunting, or malposition of DLT. Rx: suction the DLT; use PEEP to ventilated lung (may ↑

		shunting); return to two-lung ventilation; and ✓ DLT position. In extreme cases, temporarily inflating the balloon of the PA catheter (if available) may be necessary to improve shunting and, thus, oxygenation. Inadequate positioning of the DLT demands stopping further instillation of fluid. Most authors recommend suspending the lavage fluid 30 cm above the midaxillary line. Instilling fluid from excessive height or under pressure may precipitate pulmonary edema.
	Hypercarbia	Ensure adequate TV and RR.
	Aspiration	✓ DLT position. Turn patient to lavaged side down and head down, suction the ventilated lung, and reinstitute bilateral lung ventilation with PEEP, after the lavaged lung is thoroughly drained and suctioned. Termination of procedure may be needed in severe cases. Patient may need to be kept intubated after the procedure.

POSTOPERATIVE

Complications	Pneumothorax Hydrothorax Atelectasis	Obtain CXR; conservative measures if small pneumothorax or

		hydrothorax. Otherwise, chest tube placement will be required. Decrease in surfactant after lavage will → airspace collapse. Most patients will experience moderate coughing, which will help reexpand the atelectatic lung units.
Multimodal pain management	Acetaminophen or NSAID (ketorolac 30 mg iv)	Some patients may experience chest pain due to vigorous intraop percussion. Emphasize multimodal approach combining nonopioid and bolus opioid analgesics (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Ben-Abraham R, Greenfeld A, Rozenman J, et al: Pulmonary alveolar proteinosis step-by-step perioperative care of the whole lung lavage procedure. Heart Lung 2002; 31(1):43–9.
2. Cohen E, Eisenkraft JB: Bronchopulmonary lavage: effects on oxygenation and hemodynamics. J Cardiothorac Anesth 1990; 4(5):609–15.
3. Tan Z, Tan KT, Poopalalingam R: Anesthetic management for whole lung lavage in patients with pulmonary alveolar proteinosis. A A Case Rep 2016; 6(8):234–7.

ENDOBRONCHIAL ULTRASOUND-GUIDED TRANSBRONCHIAL NEEDLE ASPIRATION

SURGICAL CONSIDERATIONS

Vivek Kulkarni, MD

Description

Endobronchial ultrasound-guided transbronchial needle aspiration (EBUS-TBNA) ha been introduced as a less-invasive alternative to mediastinoscopy for pathologically staging the mediastinal lymph nodes. The procedure is performed by both pulmonologists and thoracic surgeons. This technique has become widely adopted and will likely play an increasingly important role in cancer staging.

The EBUS bronchoscope (Olympus America Inc., Center Valley, PA) contains a curvilinear ultrasound probe, at the distal end of the bronchoscope, which provides a linear continuous B-mode ultrasound image and color Doppler capability to allow identification of vascular structures. There are a 30° fiberoptic lens and biopsy channel through which a 22-gauge biopsy needle (NA-202C Olympus) can be deployed. A latex balloon can be placed over the ultrasound probe and inflated with saline to provide a fluid interface between airway and probe that improves ultrasonic image transmitted. EBUS-TBNA can be performed under GA or monitored anesthesia care (MAC). Th use of a laryngeal mask airway (LMA) or iGEL permits evaluation of the upper paratracheal lymph nodes that may not be accessible when a standard ETT is placed. The mediastinal lymph node stations (see Fig. 5.11) that are accessible by EBUS-TBNA include 2R, 2L, 4R, 4L, 7, 10R, 10L, 11R, and 11L. With standard cervica mediastinoscopy, the 11R and 11L lymph node stations cannot be assessed for biopsy, although it is highly unusual that involvement at these stations would impact treatment decisions.

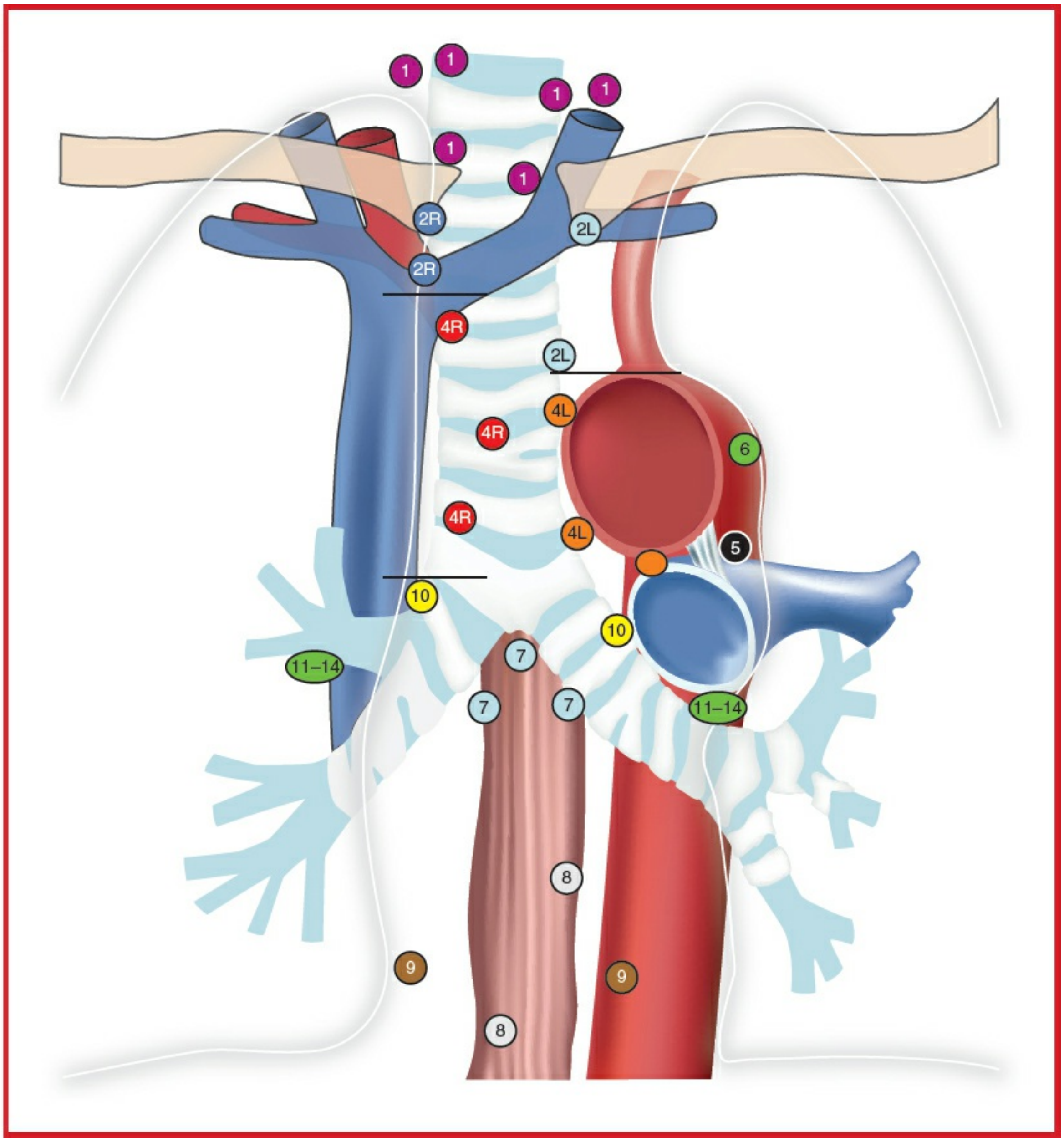


Figure 5.11 Regional lymph node stations for lung cancer staging.

At the beginning of the procedure, the EBUS bronchoscope is advanced into airway, and the ultrasound probe is used to scan the mediastinal lymph node stations of interest. Significant vascular structures, such as the pulmonary artery and aorta, appear hyperechoic and can be further identified with color Doppler. Once the lymph node is visualized, the 22-gauge biopsy needle is placed into the lymph node under direct ultrasound guidance. A suction syringe is applied to the biopsy needle, and the needle is passed into the lymph node for ~10 passes under ultrasound guidance. The aspirate is then placed on a slide with the assistance of a cytopathologist.

Rapid onsite examination by a cytopathologist in the OR is essential to confirm adequacy of diagnostic material, optimal specimen preparation, and preliminary diagnosis. The observation of moderate to abundant numbers of lymphocytes or pigmented histiocytes may serve as an indicator of adequate sampling of lymph nodes that are free of metastatic carcinoma.

Usual preoperative diagnosis

Carcinoma of the lung; sarcoidosis; lymphoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. **See Appendix K, p. K-1.**

■ SUMMARY OF PROCEDURES	
Position	Supine
Incision	None
Special instrumentation	Fiberoptic EBUS bronchoscope
Antibiotics	None
Surgical time	<1 h
EBL	Minimal
Postop care	PACU → home
Mortality	<0.1%
Morbidity	Pneumothorax: rare Airway bleeding: rare
Pain score	0–1

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Adults
Male:Female	Male > female
Incidence	Frequently part of lung cancer evaluation
Etiology	Lung cancer, lymphoma, sarcoidosis
Associated conditions	Lymphadenopathy, mediastinal mass

Suggested Readings

1. Adams K, Shah L, Edmonds L, et al: Test performance of endobronchial ultrasound and transbronchial needle aspiration biopsy for mediastinal staging in patients with lung cancer: systematic review and meta-analysis. Thorax 2009; 64(9):757–62.
2. Alsharif M, Andrade RS, Groth SS, et al: Endobronchial ultrasound-guided transbronchial fine-needle aspiration. Am J Clin Pathol 2008; 130(3):434–43.
3. Ernst A, Anantham D, Eberhardt R, et al: Diagnosis of mediastinal adenopathy-real-time endobronchial ultrasound guided needle aspiration versus mediastinoscopy. J Thorac Oncol 2008; 3(6):577–82.
4. Gomez M, Silvestri G: Endobronchial ultrasound for the diagnosis and staging of lung cancer. Proc Am Thorac Soc 2009; 6(2):180–6.
5. Sarkiss M, Kennedy M, Riedel B, et al: Anesthesia technique for endobronchial ultrasound-guided fine needle aspiration of mediastinal lymph node. J Cardiothorac Vasc Anesth 2007; 21(6):892–6.
6. Vaidya PJ, Munavvar M, Leuppi JD, et al. Endobronchial ultrasound-guided transbronchial needle aspiration: safe as it sounds. Respirology 2017; 22(6):1093–101.

ANESTHETIC CONSIDERATIONS

EBUS-TBNA is a procedure performed for biopsy of mediastinal nodes or mass lesions. The procedure is therefore similar to a mediastinoscopy but is less invasive. Pulmonologists may perform the procedure under sedation and local anesthesia, whereas thoracic surgeons often perform the procedure under GA to ensure patient immobility, which may be safer when sampling nodes close to vascular structures. Occasionally the procedure may be unsuccessful requiring conversion to a mediastinoscopy.

PREOPERATIVE

The preop considerations for EBUS-TBNA are similar to those for bronchoscopy and mediastinoscopy. If the mediastinal mass to be biopsied is compressing the airway or a vascular structure, proceed with caution as described under excision of anterior mediastinal tumor.

Respiratory	H&P complemented by lab and radiologic studies to focus on the underlying disease, evaluating for pulmonary problems. Consider PFTs with flow-

	volume loops for lesions that may be compressing the airways. When respiratory distress is present at rest or when exercise tolerance is poor, an ABG may be indicated to assess increased pulmonary risk (PaO_2 < 70 and PCO_2 > 45).
Cardiovascular	Many patients with tumor may have concomitant cardiac disease, which may need cardiac consultation to optimize cardiovascular status. **Tests:** ECG and stress echo may be indicated.
Musculoskeletal	Patients with lung cancer may have myasthenic syndrome with sensitivity to nondepolarizing relaxants and resistance to depolarizing relaxants. Monitoring neuromuscular blockade during the procedure is essential.
Hematologic	Adequacy of oxygen-carrying capacity (Hb/Hct) is important to ascertain. A type and screen may be indicated when the node or mass to be biopsied is abutting a vascular structure (check with surgeon). **Test:** Hct
Laboratory	Hb/Hct for oxygen-carrying capacity and BMP to ascertain that the K^+ and creatinine levels are in the normal range for safe use of muscle relaxants.
Premedication	Antisialagogue (e.g., glycopyrrolate 0.2 mg iv) to decrease secretions and mild sedation with midazolam (0.5–2 mg iv) along with a small dose of fentanyl (25–50 mcg) may be needed. Avoid respiratory depression and oversedation.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

IV sedation + topical anesthesia: if the procedure is to be done with sedation and topical anesthesia. Nerve blocks and topical anesthesia techniques are described under bronchoscopy. General anesthesia is the common technique requested by thoracic surgeons. For higher nodes, an LMA or iGEL is preferred, while lower nodes can be accessed via an ETT.

Induction	Following preoxygenation, a standard induction using propofol (1–2 mg/kg) or etomidate (0.3 mg/kg). A depolarizing relaxant (succinylcholine 1 mg/kg iv) or a short-acting nondepolarizing relaxant (e.g., rocuronium 0.3–0.6 mg/kg iv) can be used. A small dose of fentanyl (1–2 mcg/kg iv) or remifentanil (0.25–1 mcg/kg iv) can be used to facilitate intubation. Antibiotic is often not needed because only needle punctures are made.
ERAS	Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Intubation	Higher nodes are often easier to access with an LMA or iGEL (size no. 3 or larger), whereas lower nodes can be adequately accessed through an endotracheal tube. The ETT needs to be large enough to enable manipulation of the bronchoscope without damaging the balloon that surrounds the ultrasound probe. An 8.5-mm ID for women and 9.0-mm ID for men facilitate the procedure. The LMA/iGEL or ETT is linked to the anesthesia circuit through an angled bronchoscopy adaptor. The seal on the adaptor is often slit to allow easier to-and-fro passage of the bronchoscope → significant leak.
Maintenance	Because of the significant leak at the adaptor, TIVA is a good choice for anesthetic maintenance. Because of the significant leak, ventilation is performed with higher fresh gas flows either manually to overcome the leak and changing compliance or in a pressure-controlled mode with a FiO_2 sufficient to maintain adequate oxygen saturation. The adequacy of ventilation can only be assessed intermittently using $ETCO_2$ and manually closing the leak between biopsy attempts. Local

	anesthetic is often applied topically either prior to intubation or during surgery to decrease postop tracheal irritation caused by the procedure.	
Emergence	Muscle relaxant is reversed and the patient is extubated sitting upright with remifentanil infusion (0.03–0.05 mcg/kg/min) for a smooth wake-up. Supplemental oxygen postop.	
Blood and fluid requirements	IV: 18 ga minimum BSS @ 2 mL/kg/h	See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ± Arterial line Second pulse oximeter	Place the EKG electrodes away from the sternum in case an emergency sternotomy is necessary. Place a BP cuff or arterial line (when indicated) on the left side in the event of a possible mediastinoscopy (with intermittent pressure on the innominate artery). Place a pulse oximeter on both sides.
Positioning	Supine, arms tucked ✓ and pad pressure points	To facilitate surgery, the patient is usually turned 90° from the anesthesia provider with a jelly doughnut to stabilize the head.
Complications	Hypoxia	Pulse oximetry monitored continuously. Interrupt the procedure to allow oxygenation to improve before proceeding.
	Hypercapnia	$ETCO_2$ monitoring is inaccurate because of the leak at the adaptor. Intermittently pause the procedure and manually close

		the leak to confirm that $ETCO_2$ is not too high. Ventilate to normalize the $ETCO_2$ before proceeding.
	Hemorrhage	Often minor and local, which can be controlled with epinephrine solution irrigation. Rarely major, which may require an emergency sternotomy or thoracotomy.

POSTOPERATIVE

Complications	Airway irritation	Airway irritation causing coughing (decreased by lidocaine topical spray prior to or during the procedure) or bronchospasm, treated with nebulized bronchodilators, for example, racemic epinephrine or albuterol. Occasionally steroids may be necessary.
	Hypoxia	Humidified O_2 by mask.
	Hypoventilation	Avoided by adequate reversal of neuromuscular blockade and using minimal-dose short-acting opioids.
	Hemorrhage	Usually self-limiting.
Multimodal pain management	Usually minimal	
Tests	CXR	A postop CXR is often performed to confirm the absence of pneumothorax or mediastinal emphysema.
ERAS	PONV protocol (see	Encourage early enteral

Appendix B, p. B-1) nutrition and mobilization.

Suggested Readings

1. Lee HJ, Haas AR, Sterman DH, et al: Pilot randomized study comparing two techniques of airway anaesthesia during curvilinear probe endobronchial ultrasound bronchoscopy (CP-EBUS). Respirology 2011; 16:102–6.
2. Sarkiss M, Kennedy M, Riedel, et al: Anesthesia technique for endobronchial ultrasound-guided fine needle aspiration of mediastinal lymph node. J Cardiothorac Vasc Anesth 2007; 21(6):892–6.

BRONCHOSCOPY—FLEXIBLE AND RIGID

SURGICAL CONSIDERATIONS

Description

Bronchoscopy can be performed using either rigid or flexible instrumentation. **Flexible fiberoptic bronchoscopy (FOB)** is used for the diagnosis and evaluation of a variety of pulmonary conditions and can be accomplished using topical anesthesia and sedation without an anesthesiologist. Transbronchial biopsies can be performed in sedated patients, although more extensive interventions—such as laser ablation of a tumor, stent placement, and balloon dilation—require general anesthesia. When performed under GA, the bronchoscope should be passed through a size 8 or larger ETT. FOB is often performed through a single-lumen tube before lung resections.

Rigid bronchoscopy is more appropriate for evaluating hemoptysis and for intrabronchial procedures such as mechanical dilation of tracheal or bronchial strictures, laser or mechanical tumor debridement, and removal of foreign bodies that cannot be extracted with basket forceps through a flexible bronchoscope. Rigid bronchoscopy is performed under general anesthesia. With the patient's head and neck extended, the eyes, teeth, and gums must be protected, and the bronchoscope is inserted into the posterior pharynx until the epiglottis is visualized. The epiglottis is lifted anteriorly, with care being taken not to use the patient's teeth as a fulcrum. The bronchoscope is then advanced into the trachea (Fig. 5.12), and the diagnostic or therapeutic procedure is carried out. Ventilation is through the side-arm of the bronchoscope, and, as there is no cuff to prevent escape of anesthetic gases, high ventilatory volumes may be required. Because interventions (e.g., biopsy or tumor debridement) require removal of the bronchoscope viewing lens, the anesthesiologist must time ventilation appropriately. A Venturi ventilator may be useful when the viewing lens must be off for prolonged periods.

Laser bronchoscopy can be performed using either flexible or rigid bronchoscopes. The CO_2 laser is characterized by limited tissue penetration. As such, it is useful for superficial lesions of the upper airway. The Nd:YAG lasers use higher energies, can be directed by fiberoptic light guides, and can be used for tumor ablation. As both of these types of lasers rely on thermal damage to tissues, precautions—particularly $FiO_2 \leq 30\%$—must be taken to prevent the devastating complication of airway fire. **Photodynamic therapy** uses visible light to activate a photosensitive compound into a locally toxic drug. Because no thermal energy is involved, airway fires are not an issue.

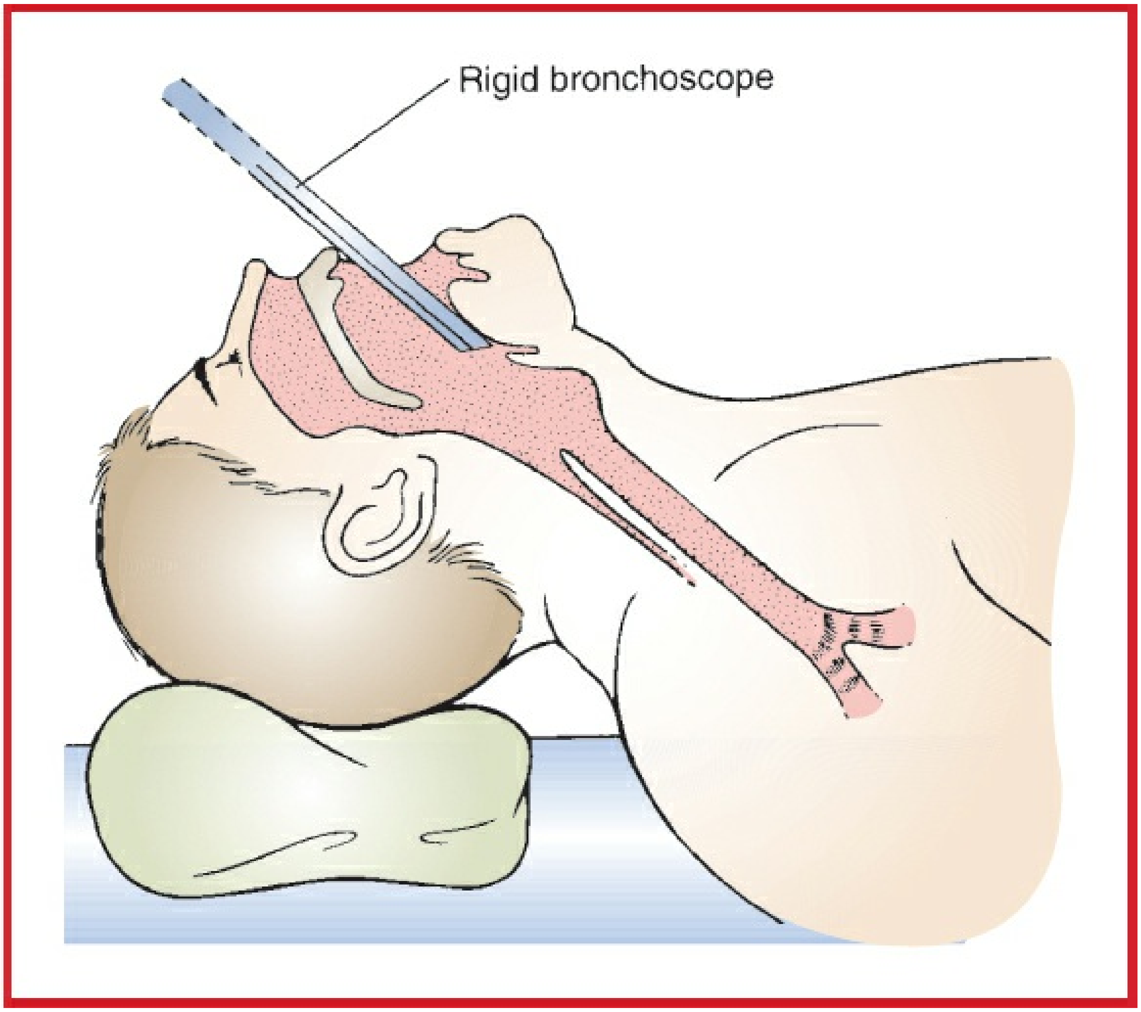


Figure 5.12 Patient positioning for rigid bronchoscopy.

Usual preop diagnosis

Carcinoma of the lung, primary or recurrent; hemoptysis; obstruction; foreign body; benign tumor; respiratory papillomatosis

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See Appendix K, p. K-1.

SUMMARY OF PROCEDURES

	Fiberoptic Bronchoscopy	Rigid Bronchoscopy	Laser Bronchoscopy
Position	Supine	⇐ (See Fig. 5.12)	⇐
Special instrumentation	FOB and instruments	Rigid bronchoscope and instruments	Nd:YAG laser and bronchoscope
Unique considerations	None	Shared airway	⇐ + Keep FiO_2 ≤0.4 during the use of laser
Antibiotics	Usually none	± Cefazolin 2g iv	⇐
Surgical time	<30 min	⇐	1 h
EBL	Minimal	⇐	⇐
Postop care	PACU then home	⇐	⇐
Mortality	Minimal	⇐	5%
Morbidity	Barotrauma Airway obstruction Pneumothorax	⇐ ⇐ ⇐ Tooth damage Tracheal laceration Pneumomediastinum Esophageal perforation	⇐ Airway fire Hemorrhage Perforation
Pain score	1	1	1

PATIENT POPULATION CHARACTERISTICS

Age range	Usually adults >50 yr
Male:Female	1:1

Incidence	Common
Etiology	Smoking, hemoptysis, aspiration of foreign body
Associated conditions	Lung cancer or metastatic spread to tracheobronchial tree, airway obstruction 2° tumor or FB, lung infiltrates

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for bronchoscopy range from asymptomatic to those in severe respiratory distress. FOB is performed most commonly as it is less invasive, is usually easily done under sedation, and allows for more distal airway examination. Rigid bronchoscopy is mostly reserved for specific interventional procedures such as hemorrhage control, foreign body removal, tumor debulking, and stent placement.

Respiratory	H&P to focus on underlying condition. Evaluate for acute and chronic pulmonary problems by H&P and lab and radiologic studies. **Tests:** Consider ABG in patient with respiratory distress, SOB at rest, or poor exercise tolerance. PaO_2 < 70 mm Hg and/or $PaCO_2$ > 45 mm Hg indicate significant respiratory impairment and predict increased risk; PFT with flow-volume loop (for airway lesions); CXR.
Cardiovascular	Many patients have Hx of cardiac disease. Cardiology consultation should be obtained for active/unstable cardiac issues or for patient with poorly controlled chronic disease. **Tests:** Consider ECG; others as indicated from H&P
Musculoskeletal	Patients with lung cancer may have myasthenic (Eaton-Lambert) syndrome with resistance to depolarizing muscle relaxants and ↑ sensitivity to NDMRs. Monitor relaxation with peripheral nerve stimulator.
Hematologic	Type and cross rarely required unless high risk of hemorrhage (✓ with surgeon). Adequate O_2 carrying capacity (Hct) is important. **Tests:** Hb/Hct
Laboratory	Other tests as indicated from H&P
Premedication	Antisialagogue (glycopyrrolate 0.2 mg iv, avoids central anticholinergic effects). Light sedation with midazolam 1–2 mg iv and/or fentanyl, 50–100 mcg

	iv. Avoid heavy sedation that might impair postop ventilation.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Flexible bronchoscopy

Anesthetic technique for flexible FOB requires sedation or GA. Anxious patients and those with respiratory compromise may not tolerate sedation for awake FOB, and patients with Hx of gastric reflux or aspiration are not candidates for awake FOB. Local anesthetic toxicity is a distinct possibility; ensure availability of resuscitative equipment.

Sedation with topical anesthesia	Sedate patient as necessary to ensure comfort and cooperation (midazolam 1–2 mg iv and/or fentanyl 50–100 mcg iv). Bronchoscopy is often better tolerated if using nasal route. Airway anesthesia can be provided with direct nerve blocks or topical anesthesia. **Nerve blocks:** Transtracheal local anesthesia—pass needle through the cricothyroid membrane, aspirate air into syringe, and then inject lidocaine (2%) 2–4 mL. Remove needle quickly because injection causes cough (spreads the anesthetic). Perform superior laryngeal nerve blocks: insert needle anterior to superior cornu of the thyroid cartilage. After resistance is felt, aspirate gently; then inject lidocaine (2%) 2 mL; repeat on the other side. **Topical anesthesia:** Spray the palate, the pharynx, the larynx, vocal cords, and the trachea with lidocaine (4%), using nebulizer, or have patient gargle viscous lidocaine (4%). Hold the base of the tongue forward and, using Krause forceps, place pledgets soaked in local anesthetic in each pyriform fossa to block the internal branch of the superior laryngeal nerve. Laryngeal structures are well topicalized with having the patient gargle 4%

	lidocaine. The trachea can be topicalized by administering 1% lidocaine through the working channel of the fiberoptic scope. Patient can hold a suction catheter in the mouth to remove oral secretions. A special face mask (Patil-Syracuse) incorporates a diaphragm through which the FOB can pass while patient breathes 100% O_2. Use a special oral airway (Ovassapian) to guide FOB over the back of the tongue into the trachea to prevent damage to FOB by teeth. Limit amount of suctioning by surgeon because suctioning through FOB decreases FiO_2 and FRC, → ↓ PaO_2.

General anesthesia

Almost any anesthetic technique is acceptable, including the use of an LMA. A large ETT has less resistance to air flow; minimum size is 8 mm (ID) for adult FOB. If patien requires ETT < 8 mm, use a pediatric FOB. Placing an LMA allows for examination o the proximal airway, and a size 2.5 or larger ProSeal LMA has an airway caliber equivalent to or larger than a size 8.0 ETT. The presence of a FOB in the ETT or LM increases airway resistance and may result in intrinsic PEEP and dynamic hyperinflation/pulmonary tamponade (hypotension).

Rigid bronchoscopy

This requires relatively deep GA, and usually paralysis for scope insertion.

Induction	Preoxygenate well. Standard induction. Consider short-acting paralytics (e.g., succinylcholine 1 mg/kg or rocuronium 0.3–0.6 mg/kg). Use only small amount of iv opioids because postop analgesic requirements are minimal; consider remifentanil (1 mcg/kg) to avoid postop respiratory depression.
ERAS	Verify preprocedure antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Commonly inhalation anesthesia with isoflurane or sevoflurane and 100% O_2. The adequacy of inhaled agent delivery may be hampered by ongoing

	suctioning. TIVA is an alternative: propofol (50–150 mcg/kg/min) and remifentanil (0.1–0.3 mcg/kg/min). Paralysis is usually used for ETT placement and is required during rigid bronchoscopy. May be provided with short-acting, nondepolarizing agents (atracurium, vecuronium, or rocuronium) or alternatively a succinylcholine drip (1 g/250 mL NS, titrated to effect; be aware of onset of phase II block at doses >5–6 mg/kg). Manual IPPV through side-arm of rigid bronchoscope. High flow (up to 20 L/min) to compensate for leak. Hyperventilate patient in preparation for periods of apnea. Ventilation must be interrupted whenever surgeon removes eyepiece to suction or biopsy. Manually ventilate to compensate for compliance changes that occur when bronchoscope is in the trachea (ventilating both lungs) and when it is in the bronchus (ventilating one lung). O_2 flush is used to compensate for leak and bypasses anesthetic vaporizer. Frequent flushing lowers anesthetic concentration. Sanders jet ventilator using Venturi effect and the Monsoon III high-frequency ventilator (Acutronic Medical Systems AG) are alternatives that allow for uninterrupted ventilation and may shorten the length of the procedure. They require TIVA. Entrainment of air results in variable FiO_2. Adequacy of ventilation is hard to determine as no $ETCO_2$ monitoring is available; prolonged procedures may require intermittent blood gas analysis or transcutaneous CO_2 monitoring.
Emergence	Place ETT, LMA, or iGEL at conclusion of rigid bronchoscopy. Patient must be fully awake before extubation with no residual neuromuscular blockade. Emergence can be “stormy.” Patient may cough violently to clear secretions and blood. Wake-up from remifentanil infusion tends to be smoother. Other considerations include early suctioning of the airway and lidocaine (1 mg/kg iv) to decrease airway

	reactivity. Provide postop O_2 supplementation (preferably humidified).	
Blood and fluid requirements	Blood usually not required IV: 18 ga × 1 BSS @ 2 mL/kg/h	Transfusion unnecessary unless complicated by massive hemorrhage; may require emergency thoracotomy. Usually restrict iv fluids to avoid fluid overload. See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors	**NB:** $ETCO_2$ not accurate during rigid bronchoscopy because of dilution effect at sample port.
Positioning	✓ and pad pressure points ✓ eyes Shoulder roll for rigid bronchoscopy	
Complications	Hypoxemia	Monitor pulse oximetry continuously. If patient is hypoxemic, surgeon must withdraw bronchoscope into the trachea. If problem persists, remove bronchoscope and ventilate by mask or ETT.
	Hypercapnia	Common, due to hypoventilation. Mild hypercarbia is well tolerated except for the setting of severe pulmonary HTN. CO_2 levels above 70 mm Hg may be associated with

		tachycardia, dysrhythmias, and cardiac depression. Easily treated with higher minute ventilation/hyperventilation. IV lidocaine for dysrhythmias.
	Bleeding Tracheobronchial injury Aspiration of debris	Requires frequent suctioning. For major hemorrhage, place uncut ETT down the healthy bronchus and ventilate the good lung. Consider DLT. May require thoracotomy using DLT or BB to isolate and/or tamponade bleeding site.
	Hypoxemia	Rx: supplemental O_2. Nebulized racemic epinephrine and steroids may ↓ airway edema. Humidified O_2 may ↓ airway irritation.
	Hypoventilation Dental damage Airway trauma Pneumothorax	Incomplete reversal of muscle relaxants or opioid overdosage can cause hypoventilation. Obtain ABG if patient has difficulty breathing or is oversedated. Be prepared to reintubate.

POSTOPERATIVE

Complications	Hemorrhage Risk of aspiration Airway obstruction Bronchospasm, bleeding, dislodged tumor, FB	If nerve blocks are used to depress gag reflex, no eating or drinking for several hours postbronchoscopy.
Multimodal pain management	Minimal pain	Emphasize multimodal approach combining nonopioid and occasionally po

		opioid analgesics (see Appendix K, p. K-6).
Tests	CXR	Obtain CXR in the recovery room to ✓ for atelectasis, pneumothorax, and mediastinal emphysema.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Bolliger CT, Sutedja TG, Strausz J, et al: Therapeutic bronchoscopy with immediat effect: laser, electrocautery, argon plasma coagulation and stents. Eur Respir J 2006; 27(6):1258–71.
2. Conacher ID: Anaesthesia and tracheobronchial stenting for central airway obstruction in adults. Br J Anaesth 2003; 90(3):367–74.
3. Goudra BG, Singh PM, Borle A, et al: Anesthesia for advanced bronchoscopi procedures: state-of-the-art review. Lung 2015; 193(4):453–65.
4. Hobai IA, Chhangani SV, Alfille PH: Anesthesia for tracheal resection anc reconstruction. Anesthesiol Clin 2012; 30(4):709–30.
5. Lohser J, Brodsky JB: Bronchial stenting through a ProSeal laryngeal mask airway.J Cardiothorac Vasc Anesth 2006; 20(2):227–8.
6. Moghissi K, Dixon K, Thorpe JA, et al: Photodynamic therapy (PDT) in early centra lung cancer: a treatment option for patients ineligible for surgical resection. Thorax 2007; 62(5):374–5.
7. Putz L, Mayné A, Dincq AS. Jet ventilation during rigid bronchoscopy in adults: a focused review. Biomed Res Int 2016; 2016:4234861.

AIRWAY LASER SURGERY

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients usually present with complications related to an airway mass (endobronchial, carinal, or tracheal). Resection tends to be a misnomer, as the laser is most commonly used for debulking of an unresectable lesion. Central airway obstruction is a primary concern when providing anesthesia.

Respiratory	It is imperative to define the exact location and magnitude of any central airway obstruction. This helps to estimate an appropriate ETT size and quantify the potential for obstruction during induction. CXR and CT scan must be studied. PFTs and flow-volume loops may also characterize the lesion. **Tests:** PFTs; ABGs; CXR; CT scan (to delineate airway anatomy, including site and severity of airway lesion)
Musculoskeletal	Although there may be no clinically detectable muscle weakness, some of these patients will have Eaton-Lambert syndrome → ↓ sensitivity to NDMRs and ↑ resistance to depolarizing muscle relaxants.
Hematologic	Transfuse patients with preop Hct <25% (or <30%, if CAD is present). **Test:** Hct
Laboratory	Other tests as indicated from H&P
Premedication	Minimal; avoid respiratory depressants; consider antisialagogue (e.g., glycopyrrolate).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

Usually GA. Unexpected patient movement may be disastrous, which is why sedation and local anesthesia are rarely practical. Nd:YAG laser can be transmitted through a flexible quartz monofilament passed through either a rigid bronchoscope or FOB. Rigid bronchoscope provides improved visibility and better debris retrieval. It also maintains the airway with less chance of fire, although it can reflect the laser beam, causing tissue damage. Manual ventilation through the side-arm may be difficult. FOB is used with local anesthesia or through ETT under GA. Laser-safe ETTs (required for surgery in the proximal trachea) include regular ETTs wrapped in metallic tape or commercially available "laser" tubes (usually some combination of aluminum, stainless steel, Teflon, and/or silicon). Fill cuff with saline (+ methylene blue to facilitate leak detection) and avoid any petroleum-based lubricants on ETT. Steps also must be taken to protect the OR staff from laser injury. These include safety glasses to avoid ocular damage and specially designed and properly fitted filter face masks to protect from inhalation of vaporized viral particles. Photodynamic therapy is simpler in its considerations and risks: routine GETA with large ETT, immobile patient, and no risk of fire.

Induction	**Without or with distal airway obstruction** (past carina) Standard induction. **With proximal airway obstruction:** Awake bronchoscopy may help to determine the feasibility of tracheal intubation. Awake fiberoptic intubation is the safest route. In patients with less severe obstruction, an inhalation induction with spontaneous ventilation may be appropriate. Avoid muscle relaxants until the airway has been secured. Several special laser ETTs are available though none guarantee against airway fire. The risk is minimized with Nd:YAG lasering of central airway lesions as the beam is directed distal to the tip of the ETT. **With rigid bronchoscopy:** GA with standard or rapid-sequence induction.
ERAS	Verify preprocedure antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Use isoflurane/desflurane/sevoflurane, air-O_2 mixture. Keep FiO_2 < 0.4. Avoid N_2O, which supports combustion. TIVA with propofol (50–150 mcg/kg/min) and remifentanil (0.1–0.3 mcg/kg/min) infusion is preferable during rigid bronchoscopy as it ensures anesthetic delivery to the patient and avoids anesthetic contamination of the OR. Frequent suctioning during bronchoscopic examination may make inhalation techniques unreliable. Short-acting, nondepolarizing relaxant or succinylcholine infusion should be used to provide an immobile patient during laser use. If the patient becomes hypoxemic, ventilate the lungs with higher FiO_2 and ask the surgeon to stop.
Emergence	Following rigid bronchoscopy, the patient's airway usually is maintained with an LMA/iGEL or ETT until awake and breathing well and protective airway reflexes have returned.

	Emergence can be "stormy," with bleeding and secretion clearance a problem. Patient should be recovered in the sitting position. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 14–16 ga × 1–2 BSS @ 1–2 mL/kg/h	There is a potential for massive blood loss following inadvertent perforation of a major blood vessel. See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors	Continuous pulse oximetry essential; monitor $ETCO_2$ to assess adequacy of ventilation ($ETCO_2$ value may be inaccurate; consider ABG during long cases). Keep alveolar PO_2 <40%.
Positioning	✓ and pad pressure points ✓ eyes and laser shields	
Complications	Airway obstruction	From tumor, blood, tissue debris, etc. May be catastrophic secondary to pneumothorax/pneumomediastinur and cardiac tamponade; lung isolation may be lifesaving.
	Hypoxemia/hypercarbia	From inadequate ventilation.
	Bleeding	Keep a BB handy to protect the healthy lung.
	Perforation of tracheobronchial tree	Can be massive from perforation of the blood vessel by laser. Apply topical epinephrine following laser photocoagulation to control bleeding.
	Airway fire	Rx: stop ventilation, remove O_2 source, and extubate trachea to decrease inhalation of toxic

		products. Douse fire with saline. Suction all debris from airway. Ventilate patient by mask and then reintubate. Prior to subsequent extubation, perform bronchoscopy to reevaluate airway damage and suction debris. Consider steroids.
	Venous air embolism	Rx: inform surgeon, d/c laser, and prevent further entrainment. If access is present, attempt to aspirate air. Immediately volume load and support circulation with vasopressors.

POSTOPERATIVE

Complications	Airway edema	Edema formation is difficult to appreciate simply based on epithelial changes during bronchoscopy. May progress dramatically after the procedure resulting in airway obstruction and need for CPAP or emergency reintubation. Steroids are routinely given to minimize edema formation (dexamethasone 6–8 mg iv). Nebulized racemic epinephrine is helpful. Complications such as hemorrhage or obstruction can be delayed up to 48 h.
Multimodal pain management	Minimal pain	Rarely requires analgesic.
Tests	Continuous pulse oximetry	

ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. McRae K: Anesthesia for airway surgery. Anesthesiol Clin North Am 2001; 19(3):497–541.
2. Ochroch EA: Laser endobronchial treatment does not need to occur in the operating room. J Cardiothorac Vasc Anesth 2005; 19(1):118–20.
3. Pawlowski J: Anesthetic considerations for interventional pulmonary procedures. Curr Opin Anesthesiol 2013; 26(1):6–12.
4. Sullivan EA: Anesthetic considerations for special thoracic procedures. Thorac Surg Clin 2005; 15(1):131–42.
5. Vaitkeviciute I, Ehrenwerth J: Bronchial stenting and laser airway surgery should not take place outside the operating room. J Cardiothorac Vasc Anesth 2005; 19(1):121–2.

THYMECTOMY

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

The two most common indications for thymectomy are **myasthenia gravis** and **thymoma.** The severity of myasthenia gravis can be classified using the Osserman scheme, which assigns Stage I to patients with ocular symptoms only, with Stages II–IV for progressive degrees of bulbar and systemic symptoms. Indications for surgery versus medical management remain controversial, with some neurologists referring nearly all patients with myasthenia gravis for surgery, whereas others refer only those with refractory symptoms. Patients referred for surgery often take a combination of pyridostigmine and immunosuppressants (steroids and azathioprine). In cases of severe myasthenia gravis, preop plasmapheresis may be helpful in minimizing periop muscle weakness. Patients with thymoma may be asymptomatic, although ~10–20% of them have a Hx of myasthenic symptoms.

Thymectomy can be performed through a complete median sternotomy, an upper sternal split (manubrium only), a cervical approach, or a minimally invasive approach (VATS or robot-assisted). The value of a complete **median sternotomy** is that it allows for removal of all anterior mediastinal tissue that may harbor small thymic rests. This is the most invasive approach, however, and the one associated with the greatest degree of intraop tissue injury. An **upper sternal split** is performed with the neck extended and a roll placed under the shoulder blades. Either a short vertical incision or a transverse incision at the level of the sternal angle may be used. Division of only the manubrium provides adequate exposure for identification, dissection, and removal of the thymus. Mobilization of the thymus can be accomplished without entering the pleural space. Care must be taken to avoid injuring the phrenic nerves. In contrast with the removal of anterior mediastinal tumors, thymectomy usually does not require OLV.

Transcervical thymectomy is performed through a collar incision similar to that used for thyroidectomy (Fig. 7.11-3). The cervical extensions of the thymus are identified and the dissection is advanced progressively into the neck. Attachments of the gland are cauterized, and a clip is placed on the thymic vein (which drains directly into the innominate vein). Exposure is aided by a special retractor that elevates the sternum anteriorly and exposes the anterior mediastinum. **Minimally invasive** thymectomy is performed through three or four small incisions with the patient in the supine position with a bump under the side of operation and arm abducted.

At the conclusion of the operation—whether it is done through the chest or the neck—the thymic bed is drained with a small suction drain. Preop medications should be resumed as soon as possible.

Usual preop diagnosis

Myasthenia gravis; thymoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include medication management, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, and physiotherapy plan. See Appendix K, p. K-1.

SUMMARY OF PROCEDURES

	Sternotomy	Cervical Approach
Position	Supine	⇐
Incision	Median sternotomy or minimally invasive	Suprasternal
Special instrumentation	None	Special sternal retractor

Antibiotics	Cefazolin 2 g iv	⇐
Surgical time	1–2 h	⇐
EBL	<500 mL	⇐
Postop care	ICU—special attention to muscle strength, related to respiratory function	⇐
Mortality	<5%	⇐
Morbidity	Phrenic nerve injury Infection Pneumothorax Hemothorax	⇐
Pain score	5–7	2

PATIENT POPULATION CHARACTERISTICS

Age range	Usually young adults
Male:Female	Females > males
Incidence	Infrequent
Etiology	Unknown
Associated conditions	Myasthenia gravis, benign or malignant thymoma, other autoimmune diseases (e.g., rheumatoid arthritis)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients presenting for thymectomy may have myasthenia gravis, an autoimmune disease of the neuromuscular junction characterized by muscle weakness and easy fatigability. Thymomas (benign or malignant) also may be associated with myasthenia gravis. The anesthesiologist needs to be aware of the possible compression effects of the tumor (see Excision of Other Mediastinal Tumors), the potential for respiratory failure, and anticipate complications of various treatment modalities.

Respiratory	Patient may have marked reduction in VC 2° muscle weakness; establish baseline spirometry values.

	Classic (Leventhal) criteria predicting the need for postop ventilation include duration of disease >6 yr, chronic comorbid pulmonary disease, pyridostigmine dose >750 mg/d, and VC < 2.9 L. Others include preop use of steroids and previous episode of respiratory failure. These predictors have not been widely validated. Inform the patient of the potential requirement for prolonged ventilation. **Tests:** PFTs, others as indicated from H&P
Cardiovascular	There is a rare association between myasthenia gravis and cardiomyopathy. Consider ECG and cardiac consult if indicated from H&P.
Neurologic	Review neurologic assessment. Patients often exhibit diplopia, ptosis, and easy fatigability of muscles. Difficulties with swallowing and speech are common. Review tests (EMGs, Tensilon® test) done by neurologist to evaluate the adequacy of drug therapy (steroids, anticholinesterases, azathioprine, and cyclosporin A). Azathioprine (Imuran) may actually antagonize neuromuscular blockade by inhibiting phosphodiesterase. Cyclosporin A is reserved for severe disease because of the side effects of renal insufficiency and HTN. It may prolong neuromuscular blockade. Up to 50% of the patients with thymomas will have myasthenia gravis, and >85% of myasthenics will have thymus abnormalities.
Musculoskeletal	Determine adequacy of anticholinesterase medication. Evaluate hand strength, inspiratory efforts, and PFTs. Note that an excess of anticholinesterase agents can cause weakness and respiratory failure (cholinergic crisis). A deleterious response (weakness) to Tensilon®, administration (10 mg), and the presence of cholinergic side effects (e.g., pupil constriction) characterize cholinergic crises. Plasmapheresis may offer temporary (days to weeks) improvement in symptoms. Typically, patients with worsening symptoms receive 4–8

	treatments prior to surgery. There should be a 24-h delay between the last plasmapheresis and surgery to restore clotting factors and immunoglobulins. Plasmapheresis also may transiently decrease plasma cholinesterase, which could prolong the effects of succinylcholine, mivacurium, and ester-type local anesthetics.
Endocrine	Other autoimmune phenomena occur in 10–15% of myasthenic patients. Check for symptoms of thyroid or adrenal disease. **Tests:** TSH; evaluate screening test if indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	Avoid premedication; for the anxious patient, give a small dose of midazolam (0.5–1 mg); avoid opioids or any other sedatives that may depress ventilation. Current recommendations for anticholinesterases suggest that, in mild disease without physiologic dependence, the morning dose can be held or halved. Patients with severe disease or with marked anticholinesterase dependence should receive their regular morning dose. For patients on steroids—depending on dosage and duration of steroid therapy—hydrocortisone (up to 100 mg iv bolus) before induction, then 100 mg q8h × 24 h, may be helpful. Note that acute steroid exposure in naïve patients may paradoxically exacerbate neuromuscular weakness. Consider aspiration prophylaxis.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of epidural or paravertebral nerve blocks for postop thoracotomy pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40–80 mg po with sip of water for patients with h/o

	severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA, combined with thoracic epidural opioids if transsternal approach will be used. Lung isolation may be requested by surgeon and can be accomplished with an EZ-Blocker™ or a DLT.

Induction	Inhalational induction with sevoflurane to avoid muscle relaxants entirely; however, patients with profound muscle weakness are at risk for aspiration 2° inadequate airway protective mechanism. Alternatively, iv propofol induction with remifentanil 1–2 mcg/kg (without muscle relaxants). If absolutely necessary, succinylcholine may be used to facilitate intubation, although myasthenic patients are considered resistant and require a dose of 1.5–2.0 mg/kg to obtain adequate conditions, which then will result in prolonged muscle weakness. Anticholinesterase therapy may prolong the duration of action of succinylcholine.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard maintenance. Patients with myasthenia gravis have ↑ sensitivity to nondepolarizing NMBs, which are rarely ever needed, as maintenance with inhalational agents provides adequate neuromuscular relaxation when combined with the existing baseline weakness. If relaxants are needed, titrate small amounts (1/10 ED_{95}) of drug, using a peripheral nerve stimulator to maintain a single twitch. Cisatracurium (20–30 mcg/kg q15–20min or infusion 1–3 mcg/kg/min) is useful because it is rapidly eliminated. With the advent of sugammadex, the judicious use of rocuronium is often possible.

	Alternatively, potent inhalation agents may provide adequate muscle relaxation and avoid the need for any muscle relaxant. Avoid drugs with neuromuscular blocking effects (e.g., antidysrhythmics, calcium channel blockers, diuretics, aminoglycosides, Mg^{2+}, iodinated contrast agents). Extremes of temperature, hypokalemia, and hypophosphatemia may also aggravate neuromuscular weakness and should be prevented. If the patient has normal ventilatory function, then spontaneous ventilation during cervical thymectomy may be appropriate. Patients undergoing sternotomy, and any patient with ↓ pulmonary reserve, require mechanical ventilation.	
Emergence	Criteria for extubation include head lift (5 sec), MIF > −25 cm H_2O, TV > 5 mL/kg, and full reversal evidenced by twitch monitor. Extubate when fully awake; usually immediate postop ventilation is not necessary. However, because of the variable response to muscle relaxation and delayed benefits of the surgery, some patients may require postop ventilation. Whether intubated or not, monitor patient for pulmonary function by measuring MIF and spirometry (Vt). Avoid residual pharmacologic neuromuscular blockade, which will → hypoventilation and ↑ risk of gastric aspiration if protective airway reflexes are inadequate. Discuss with surgeon local wound infiltration or nerve block (e.g., intercostal block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18 ga × 1 BSS @ 1–2 mL/kg/h	Consider lower extremity iv access when hemorrhage is anticipated. See Goal Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors	Avoid muscle relaxants if

		possible; use nerve stimulator.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hemorrhage Dysrhythmia Compression of mediastinal structures Pneumothorax	

POSTOPERATIVE

Complications	Pneumothorax Respiratory failure Phrenic nerve damage Myasthenic or cholinergic crises	Pleura can be entered—usually at the right side—if so, chest tube is needed (for Dx and Rx, see VATS).
Multimodal pain management	Acetaminophen NSAIDs Parenteral opioids Epidural opioids	Emphasize multimodal approach combining nonopioid analgesics with peripheral local anesthetics and limited opioid analgesics (see Appendix K, p. K-6). Avoid respiratory depression; for breakthrough pain, consider parenteral opioids (cervical incision) and epidural opioids for median sternotomy. Of note, anticholinesterases are reported to potentiate the effect of morphine.
Drug management	Usually ↓ anticholinesterase requirement in the immediate postop period	Reduce daily anticholinesterase by 20%. Beware of "cholinergic" crisis. Sx include ↑ salivation, sweating, lacrimation, abdominal cramps, urinary

		frequency, fasciculations, and weakness 2° anticholinesterase overdose. Rx: anticholinergic agent (e.g., atropine) ± intubation and mechanical ventilation.
Tests	Tensilon® (edrophonium) Muscle strength Pupil exam	Tensilon® test may differentiate between myasthenic (↑ strength) and cholinergic (↓ strength) crises. Determine muscle strength postop (grip strength and sustained head lift). Pupil dilation (myasthenic crisis), pupil constriction (cholinergic crisis).
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Blichfeldt-Lauridsen L, Hansen BD: Anesthesia and myasthenia gravis.Acta Anaesthesiol Scand 2012; 56(1):17–22.
2. Dillon F: Anesthesia issues in the perioperative management of myasthenia gravis. Sem Neurol 2004; 24(1):83–94.
3. Limmer KK, Kernstine KH: Minimally invasive and robotic-assisted thymu resection. Thorac Surg Clin 2011; 21(1):69–83, vii.
4. O'Neill GN: Acquired disorders of the neuromuscular junction.Int Anesthesiol Clin 2006; 42(2):107–21.
5. Shrager JB, Nathan D, Brinster CJ, et al: Outcomes after 151 extended transcervica thymectomies for myasthenia gravis. Ann Thorac Surg 2006; 82(5):1863–9.
6. Sungur Z, Sentürk M. Anaesthesia for thymectomy in adult and juvenile myasthenic patients. Curr Opin Anaesthesiol 2016; 29(1):14–9.

EXCISION OF OTHER MEDIASTINAL TUMORS

SURGICAL CONSIDERATIONS (ALSO SEE SURGEON'S PERSPECTIVE ABOVE)

Description

Mediastinal tumors are characterized by their location (anterior, middle, and posterior) and their size. Common anterior mediastinal tumors include thymic tumors (thymoma and thymic carcinoma), germ cell tumors, lymphoma, and substernal goiters. Typically, thymic and germ cell tumors are resected, whereas lymphomas are biopsied. Substernal goiters usually can be resected through the neck. Tumors in the anterior mediastinum usually are removed through a median sternotomy, a VATS, or a robot-assisted approach, whereas tumors in the middle and posterior mediastinum usually are removed through a lateral thoracotomy, a VATS, or a robot-assisted approach. Although middle and posterior mediastinal tumors usually do not present airway management problems, the issue of functioning neuroendocrine tissue must be considered. Mediastinal pheochromocytomas are uncommon middle mediastinal tumors. However, as with pheochromocytomas arising in other locations, appropriate preop adrenergic management is necessary.

Mediastinal tumors that are well encapsulated generally are removed in a straightforward fashion. If anterior mediastinal tumors are not well encapsulated and are attached to the pericardium or lung on either side, appropriate portions of these attached structures may be removed en bloc. If there is attachment to phrenic nerves on either side, one nerve may be sacrificed, if necessary, to remove the tumor completely. In patients with anterior mediastinal tumors, invasion of the major vascular structures, particularly the aorta and great vessels, presents an even greater problem.

Germ cell tumors of the anterior mediastinum are treated with chemotherapy initially. For nonseminomatous tumors, surgical resection is performed only if there is residual tumor. A common regimen for these patients consists of cisplatin, etoposide, and bleomycin, and because bleomycin is associated with pulmonary toxicity—particularly in conjunction with high concentrations of inhaled oxygen—care must be taken to keep the $FiO_2 < 40\%$ when conducting these operations.

Another common issue with patients with large anterior mediastinal masses is that of intrathoracic airway obstruction at the time of anesthetic induction. Although most mediastinal masses do not cause obstruction of the trachea or tracheobronchial tree, large mediastinal masses in the anterior mediastinum, in conjunction with muscle relaxation, can lead to complete obstruction of the airway with inability to ventilate the patient. Although rigid bronchoscopy may permit ventilation through the obstruction, it cannot be counted on to relieve the obstruction; therefore, only short-acting or no muscle relaxants (spontaneous ventilation) should be used in these patients.

Usual preop diagnosis

Thymoma; germ cell tumor; ganglioneuroma; lymphoma; schwannoma; substernal goiter; congenital bronchogenic or pericardial cysts

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb loading, postop pain management options, need for a chest tube, and physiotherapy plan.** **See Appendix K, p. K-1.**

SUMMARY OF PROCEDURES	
Position	Supine or lateral
Incision	Median sternotomy, lateral thoracotomy, or minimally invasive
Special instrumentation	Sternal or rib retractors
Antibiotics	Cefazolin 2 g iv
Surgical time	2 h
EBL	<500 mL
Postop care	Frequently ICU
Mortality	Minimal
Morbidity	Bleeding
Pain score	5–8

PATIENT POPULATION CHARACTERISTICS	
Age range	All ages
Male:Female	1:1
Etiology	**Anterior mediastinum:** thymoma, teratoma, pericardial cyst, lymphoma, parasternal (Morgagni) hernia, lipoma **Superior mediastinum:** goiter, aneurysm,

	parathyroid tumor, esophageal tumor, angiomatous tumor **Middle mediastinum:** lymphoma, lymph node inflammation, bronchogenic tumor, bronchogenic cyst **Posterior mediastinum:** neurogenic tumor, aneurysm (enteric cyst), esophageal tumor, bronchogenic tumor
Associated conditions	SVC syndrome, myasthenia gravis, recurrent laryngeal nerve damage, airway obstruction, dyspnea, Horner syndrome

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Mediastinoscopy.

Suggested Readings

1. Carter BW, Marom EM, Detterbeck FC: Approaching the patient with an anterio mediastinal mass: a guide for clinicians. J Thorac Oncol 2014; 9(9 Suppl 2):S102–9.
2. Marshall MB, DeMarchi L, Emerson DA, et al: Video-assisted thoracoscopic surgery for complex mediastinal mass resections. Ann Cardiothorac Surg 2015; 4(6):509–18.
3. Narang S, Harte HB, Body SC: Anesthesia for patients with a mediastinal mass Anesthesiol Clin North Am 2001; 19(3):559–79.
4. Viswanathans S, Campbell CE, Cork RC: Asymptomatic undetected mediastina mass: a death during ambulatory anesthesia. J Clin Anesth 1995; 7(2):151–5.

MEDIASTINOSCOPY

SURGICAL CONSIDERATIONS

Description

Mediastinoscopy is used for biopsy of mediastinal lymph nodes. The most commor indication for this procedure is bronchogenic carcinoma, although lymphadenopathy associated with lymphoma, sarcoidosis, and infectious granulomatous diseases is also indication for mediastinoscopy. **Cervical mediastinoscopy** provides access to the pretracheal, paratracheal, and anterior subcarinal nodes (Fig. 5.13), whereas **transthoracic mediastinoscopy** (also known as **anterior mediastinotomy** or **Chamberlain procedure**) provides access to the aortopulmonary lymph nodes. Previous mediastinoscopy and radiation are relative contraindications to this procedure. If a thoracic aneurysm is present or SVC is obstructed, mediastinoscopy is contraindicated because the anatomy is distorted and vessels can be punctured inadvertently by the mediastinoscope.

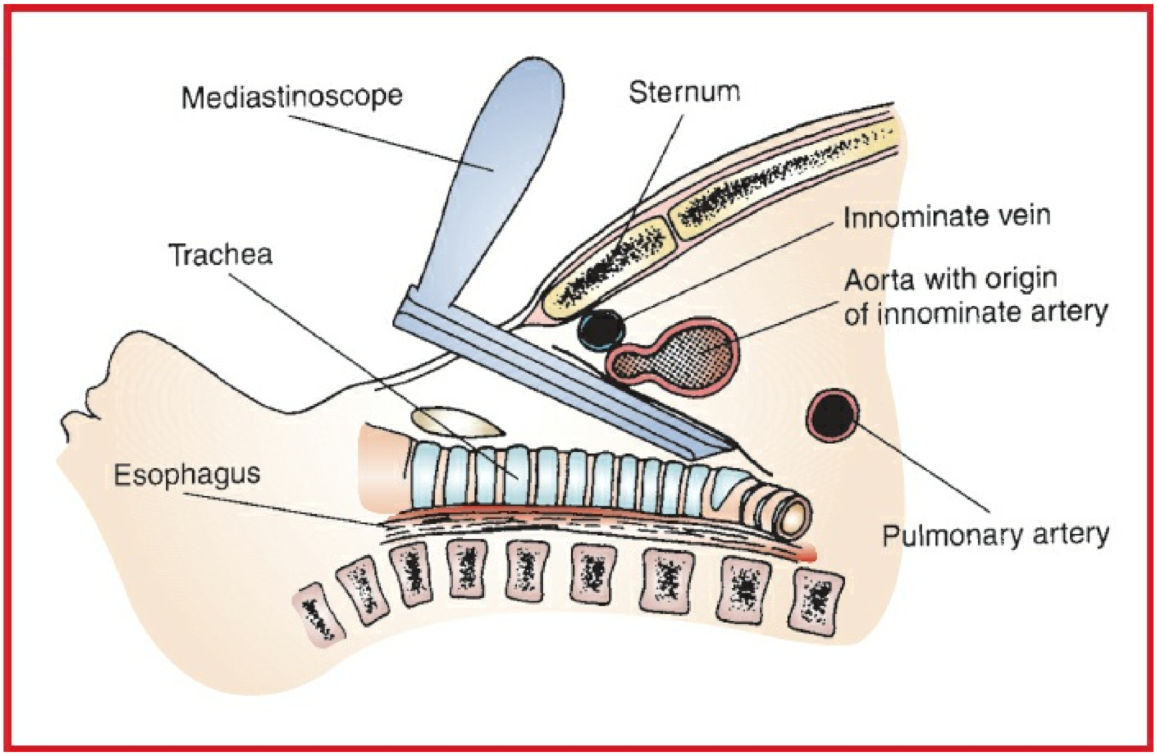


Figure 5.13 Mediastinoscope is inserted through a small cervical incision into the middle mediastinum, along the pretracheal plane. (Reproduced with permission from Baker RJ, Fischer JE, eds.Mastery of Surgery, 4th edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Variant procedure or approaches

For nodes on the left side of the mediastinum, **transthoracic mediastinoscopy** is performed through a limited anterior thoracotomy. In the classic **Chamberlain procedure,** the 3rd costal cartilage is resected, and the mediastinum is explored without entering the pleural space. As with cervical mediastinoscopy, visualization is often limited and lymph nodes should be aspirated before biopsy. If the pleural space is entered during the course of the procedure, either a chest tube can be placed postop or the pleural space can be aspirated immediately before wound closure. Patients should be extubated at the end of the operation. Cervical mediastinoscopy is usually an outpatient procedure, whereas patients undergoing transthoracic mediastinoscopy are usually hospitalized overnight.

Usual preop diagnosis

Carcinoma of the lung with enlarged mediastinal nodes; mediastinal node enlargement 2° lymphoma, thymoma, or others

ERAS: **Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, carb loading, and postop pain management options. See** **Appendix K, p. K-1.**

SUMMARY OF PROCEDURES

Position	Supine
Incision	For mediastinoscopy, low cervical; for anterior mediastinotomy, left or right parasternal in the 2nd interspace
Special instrumentation	Mediastinoscope
Antibiotics	Cefazolin 2 g iv
Surgical time	1 h
EBL	Minimal (but risk of significant blood loss if major vascular injury occurs)
Postop care	PACU and then same-day discharge
Mortality	<0.1%
Morbidity	Bleeding Pneumothorax: rare Vocal cord paralysis: rare Esophageal perforation: rare Pleural tear: rare Tracheal laceration: rare
Pain score	2 (mediastinoscopy), 2–3 (anterior mediastinotomy)

PATIENT POPULATION CHARACTERISTICS

Age range	Adults, usually >50 yr
Male:Female	Male > female
Incidence	Frequently part of evaluation for patients with lung cancer
Etiology	Lung cancer, lymphoma, thymoma, retrosternal goiter

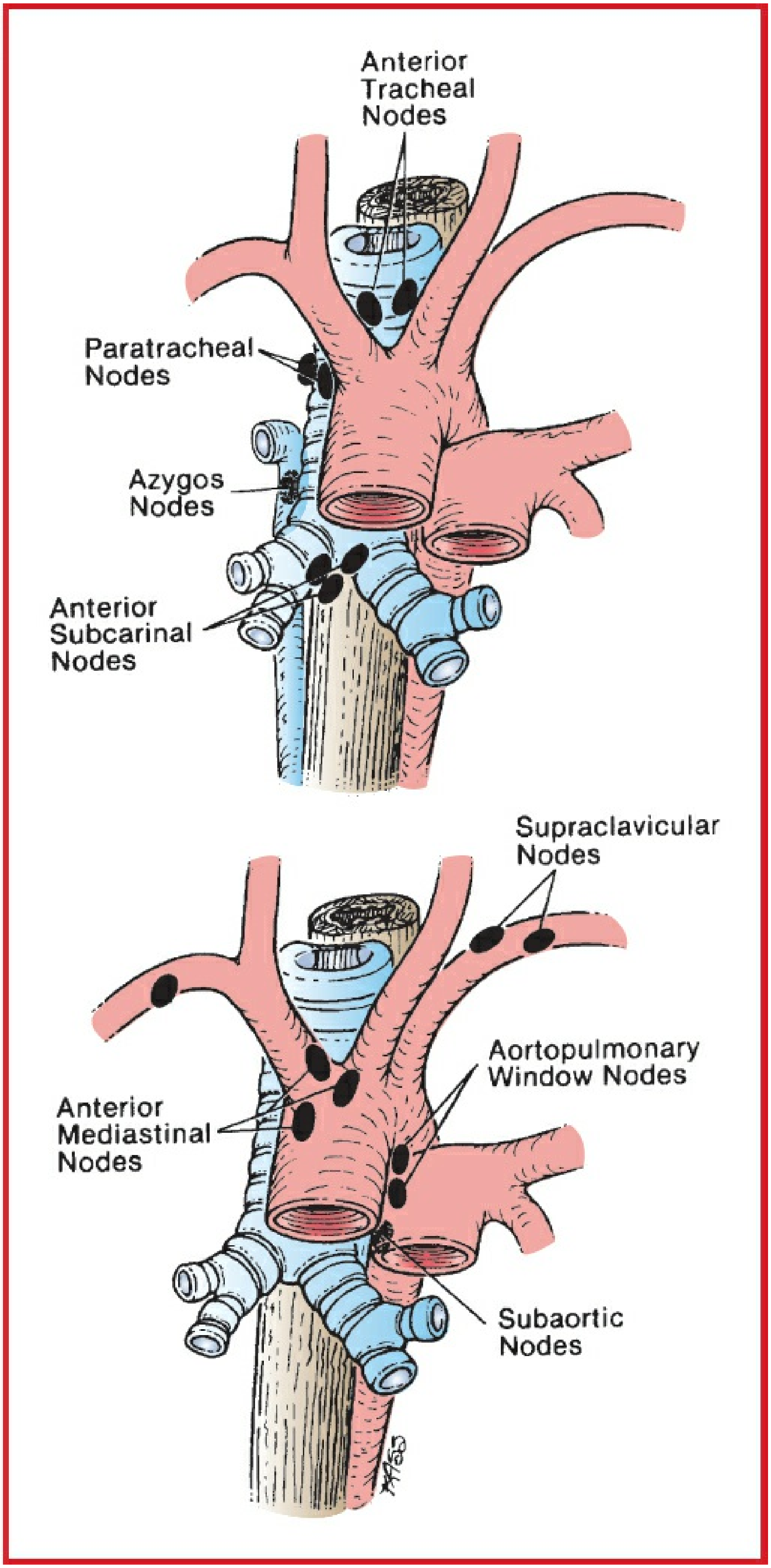
Associated conditions	Airway obstruction

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

(Procedures covered: excision of mediastinal tumor, mediastinoscopy)

Typically, these patients can be divided into two populations, depending on the presence or absence of a significant mediastinal mass (with the potential for catastrophic airway obstruction or cardiovascular collapse on induction of anesthesia). The preop assessment must focus on the differentiation of these two populations. Close consultation with the surgeon is essential in formulating the anesthetic plan. On occasion, patients with critical airway or cardiac compression may require a tissue biopsy for diagnostic purposes only. If general anesthesia poses a significant physiologic threat to the patient, search for an alternative, less-invasive biopsy site.



Anterior
Tracheal
Nodes
Paratracheal
Nodes
Azygos
Nodes
Anterior
Subcarinal
Nodes
Supraclavicular
Nodes
Aortopulmonary
Window Nodes
Anterior
Mediastinal
Nodes
Subaortic
Nodes

Figure 5.14 Lymph node sites accessible to mediastinoscope biopsy. Many of these can be reached by standard cervical mediastinoscopy. Anterior and aortopulmonary window nodes, however, require extended or anterior mediastinoscopy, VATS, or needle biopsy. (Reproduced with permission from Bocage J-P, Mackenzie JW, Nosher JL: Invasive diagnostic procedures. In: Shields TW, LoCicero J III, Ponn RB, eds. General Thoracic Surgery, 5th edition. Lippincott Williams & Wilkins, Philadelphia: 2000.)

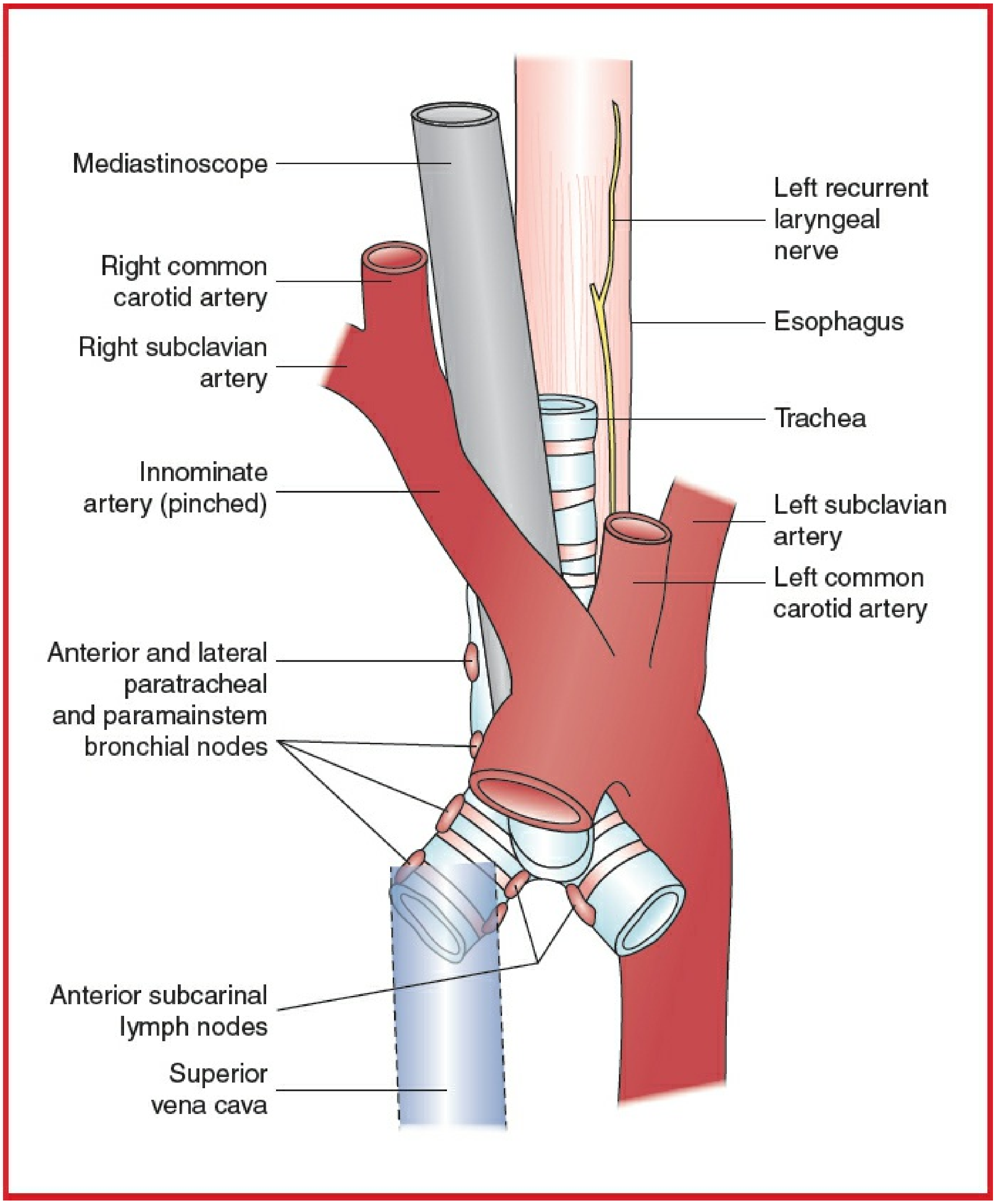


Figure 5.15 Relationship of mediastinoscope to the trachea and great vessels. (Reproduced with permission from Petty C: Right radial artery pressure during mediastinoscopy. Anesth Analg 1979; 58:428. Modified in Rogers MC:Principles & Practices of Anesthesiology. Mosby-Year Book, St. Louis: 1993.)

Respiratory	Question patient with anterior mediastinal mass about ability to lie supine and the presence of cough or dyspnea. Determine what position they prefer ("safe position"). Change in position may cause superior vena caval obstruction or cardiac and airway compression by mediastinal mass (which may be apparent only following induction or on emergence from anesthesia). On PE, ✓ for the presence of cyanosis, wheezing, or stridor in the upright and supine positions. If significant airway compression or SVC obstruction is present, the surgeon may delay surgery for radiation or chemotherapy. Patients with SVC syndrome (edema; venous engorgement of the head, neck, and upper body; supine dyspnea; ± headache; mental status change) may have significant airway edema. **Tests:** Classic teaching has recommended PFTs with flow-volume loops in upright and supine positions (Fig. 5.16 shows flow-volume loop) in patients with suspected central airway obstruction. These investigations demonstrate airflow obstruction during inspiration in patients with an extrathoracic mass and during expiration in patients with an intrathoracic mass. CT scans obtained with the patient supine delineate the location and extent of both airway and cardiac ± vascular compression. These images may predict patients at higher risk (>50% tracheal compression).
Cardiovascular	Intrathoracic vascular structures (e.g., right heart, PA, SVC) may be compressed → ↓ BP, hypoxia, and SVC syndrome. Involvement of the pericardium may → tamponade. **Tests:** Echo, CT/MRI if indicated by H&P
Musculoskeletal	Patients with lung cancer may have myasthenic (Eaton-Lambert) syndrome with resistance to depolarizing agents and ↑ sensitivity to NDMRs. Monitor relaxation with peripheral nerve stimulator.

Neurologic	Patient may have ↑ ICP if SVC is obstructed. Consider neurology consultation. Patients with preexisting carotid disease are at risk for stroke if there is significant compression of the innominate artery during the procedure.
Endocrine	Due to the diverse causes of an anterior mediastinal mass, consider comorbid thyroid disease and paraneoplastic syndromes.
Laboratory	Other tests as indicated from H&P
Premedication	Avoid sedation in patients with the potential for airway obstruction; otherwise, midazolam 1–2 mg iv may be appropriate. Antisialagogue may facilitate awake intubation. In patients with critical airway obstruction, heliox may prove useful.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40–80 mg po with sip of water for patients with h/o severe PONV.

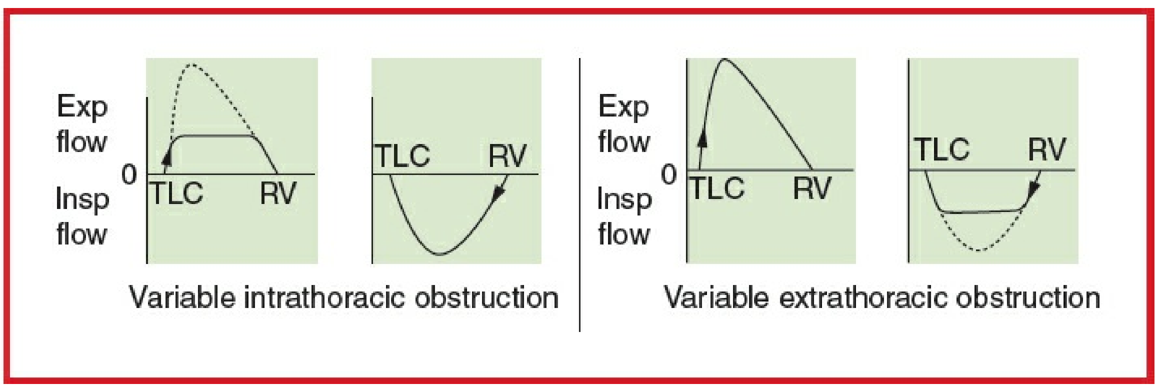


Figure 5.16 Flow-volume loop: with variable extrathoracic lesion, the alteration in the flow-volume loop is seen by flow limitation and a plateau on inspiration. The reverse occurs with variable intrathoracic lesions. (Reprinted from Acres J, Kryger MH: Clinical significance of pulmonary function tests.Chest 1981; 80:207–11. Copyright © 1981 The American College of Chest Physicians. With permission.)

INTRAOPERATIVE

Anesthetic technique

GETA, combined with epidural anesthesia if thoracotomy is planned.

Induction	Consider awake FOI (e.g., if symptomatic in supine

	position). Alternatively, a mask induction with sevoflurane/O_2 in a spontaneously breathing patient may be adequate in "the safe position." The use of neuromuscular blockers in high-risk patients may precipitate airway obstruction and CV collapse. Select a reinforced endotracheal tube when intubating patients with airway compression. Complete or partial airway obstruction by anterior mediastinal mass can also be due to changes in lung and chest wall mechanics associated with changes in the patient's position (sitting to supine during procedure) or to muscle relaxation. Consider placing the patient in the lateral or prone position in the event of central airway compression. A surgeon familiar with rigid bronchoscopy must be in the OR ready to bypass any obstruction. In rare instances, percutaneous cardiopulmonary bypass (CPB) may be necessary to complete the procedure safely. If this is indicated, it must be employed electively, not as a rescue measure. Salvage or emergent use of CPB in these patients is rarely successful.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	O_2 (100%) and isoflurane (1–1.5%) or sevoflurane (1.5–2.5%). Avoid N_2O, especially during OLV. Short-acting muscle relaxant and opioid as indicated.
Emergence	Extubation in OR
Blood and fluid requirements	IV: 14–16 ga × 1 BSS @ 1–2 mL/kg/h Blood in OR Have blood for transfusion available in OR prior to surgery. Patients with SVC syndrome may have impaired venous return from upper-limb veins. In these patients, a large-bore iv cannula should be placed in lower limb for fluid and blood

		transfusions. See Restrictive Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors ± Arterial line ± CVP/PA	Invasive monitors are appropriate in patients with large mediastinal masses. In the presence of SVC syndrome, CVP/PA catheters should be placed, using the femoral vein. BP cuff on the left arm, radial artery line (if used) and pulse oximeter on the right. Mass or mediastinoscope can compress the innominate artery, causing reduction in right radial pulse and right-arm BP. If only right-arm BP is measured, patients may be treated inappropriately for "hypotension" or cardiac arrest. Suspect great vessel compression if right-arm pressure is lower than left or if right-arm BP disappears in the presence of a normal ECG. Arterial compression can compromise cerebrovascular perfusion → cerebral ischemia → stroke. Place pulse oximeters on both hands so the right side can monitor compression of the innominate and warn the surgeon if prolonged.
Positioning	Head-up position ✓ and pad pressure	In patients with an anterior mediastinal mass, the head-

	points ✓ eyes	up position reduces mass compression effect on airway and vascular structures but may subject the patient to ↑ risk of VAE if venous bleeding occurs during the procedure. Patients with SVC obstruction, if placed in head-down position with IPPV (further impedes venous return of thoracic cavity), are at ↑ risk of airway edema and airway obstruction following extubation.
Complications	Bleeding	Surgical tamponade through mediastinoscope may be indicated. For major hemorrhage, emergency thoracotomy or median sternotomy may be required to stop bleeding. Can occur from laceration of mediastinal vein.
	Air embolism	Head elevation increases risk of embolism, particularly if patient breathes spontaneously. Monitor $ETCO_2$ and ETN_2.
	Airway rupture or obstruction	Requires immediate thoracotomy.
	Tracheal collapse	Acute obstruction may require rigid bronchoscope to reopen airway.
	Recurrent laryngeal nerve injury	If recurrent laryngeal nerve injury is suspected, the vocal cords should be examined during spontaneous breathing

at the time of extubation.

POSTOPERATIVE

Complications	Pneumothorax Phrenic/recurrent laryngeal nerve damage Tracheomalacia Bleeding	Bilateral laryngeal nerve damage may result in airway obstruction, necessitating reintubation. Mask ventilation may be ineffective. May occur in patients with long-standing mediastinal mass (e.g., retrosternal goiter).
Multimodal pain management	Acetaminophen Parenteral opioids ± Epidural	Emphasize multimodal approach combining nonopioid and bolus opioid analgesics (see Appendix K, p. K-6).
Tests	CXR on all patients to r/o pneumothorax	
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Barash PG, Tsai B, Kitahata LM: Acute tracheal collapse following mediastinoscopy. Anesthesiology 1976; 44(1):67–8.
2. Bechard P, Letourneau L, Lacasse Y, et al: Perioperative cardiorespiratory complications in adults with mediastinal mass. Anesthesiology 2004; 100(4):826–34.
3. Blank RS, de Souza DG: Anesthetic management of patients with an anterior mediastinal mass: continuing professional development. Can J Anesth 2011; 58:853–67.
4. Campos JH: An update on robotic thoracic surgery and anesthesia. Curr Opin Anaesthesiol 2012; 23(1):1–6.
5. Cata JP, Lasala J, Mena GE, et al: Anesthetic considerations for mediastinal staging procedures for lung cancer. J Cardiothorac Vasc Anesth 2018; 32(2):893–900.
6. Hnatiuk OW, Corcoran PC, Sierra A: Spirometry in surgery for mediastinal masses. Chest 2001; 120(4):1152–6.

7. Marshall MB, DeMarchi L, Emerson DA, et al: Video-assisted thoracoscopic surgery for complex mediastinal mass resections. Ann Cardiothorac Surg 2015; 4(6):509–18.
8. Petty C: Right radial artery pressure during mediastinoscopy. Anesth Analg 1979; 58(5):428–30.
9. Slinger P, Karsli C: Management of the patient with a large mediastinal mass: recurring myths. Curr Opin Anaesthesiol 2007; 20(1):1–3.

LUNG TRANSPLANT (ALSO SEE SURGERY FOR LUNG TRANSPLANTATION IN CHAPTER 6.4)

SURGICAL CONSIDERATIONS

Description

The most common indications for lung transplantation include emphysema, pulmonary fibrosis, cystic fibrosis, and pulmonary HTN. Patients with the first three of these diagnoses have marked abnormalities in mechanical pulmonary function, whereas patients with pulmonary HTN often have normal lung mechanics but very abnormal cardiac function. Patients with emphysema and pulmonary fibrosis often receive single-lung transplants, and those with cystic fibrosis require double-lung transplants. The best operation for patients with pulmonary HTN continues to be debated, with options including single-lung, double-lung, and heart-lung transplantation. Although candidates for lung transplantation, by definition, have end-stage lung disease, their overall state of health and functional abilities varies considerably. Furthermore, although these patients all have poor pulmonary function, many with multisystem disease are eliminated during the preop screening process. Thus, the remaining patients are generally well motivated and free of significant cardiac, renal, and vascular disease.

Single-lung transplants generally are carried out through a thoracotomy incision. Patients with emphysema as the underlying disease rarely require CPB, whereas those with pulmonary fibrosis (decreased pulmonary reserve and increased incidence of pulmonary HTN) more commonly require bypass. Although the need for bypass can be assessed at the outset of the operation by instituting OLV, a CPB circuit and perfusionist should always be available. The route for vascular access for bypass (transthoracic or through the groin vessels) must be considered by the surgeon at the start of the case.

Double-lung transplants generally are done through bilateral anterior thoracotomies or a clamshell incision (Fig. 5.2). Mobilization of the lung is facilitated by the use of a DLT. If the patient does not tolerate OLV, CPB can be established using aortic and atrial cannulation. Some surgeons routinely use a single-lumen ETT, with a defined plan for using CPB.

The procedure for both single- and double-lung transplantation is the same for each side. The native lung is mobilized, and the bronchovascular structures are divided. Mediastinal adenopathy and extensive pleural adhesions are the rule, rather than the exception, for patients with cystic fibrosis. Such pneumonectomies take significantly longer than those for emphysema. Once the native lung is removed, the donor lung is brought into the surgical field. The bronchial anastomosis is created first, and the lung is allowed to fall into the posterior costovertebral gutter. The venous anastomosis (atrial cuff anastomosis) is fashioned next. The final sutures are placed but left untied for later deairing. The arterial anastomosis is created last. Upon removal of the arterial clamp, the lung is perfused and deaired. The venous sutures are then tied and the atrial clamp is removed. After ensuring that the vascular anastomoses are hemostatic and that there is no air leak at the bronchial site, the incision is closed, or, in the case of double-lung transplant, attention is directed to the other side.

The anesthesiologist should be aware that hypotension is not uncommon on reperfusion of the donor lung but that it usually resolves spontaneously. Specific immunosuppression protocols vary from institution to institution, but intravenous administration of steroids immediately before lung reperfusion is a common practice. Lung transplant patients typically remain intubated following surgery; however, changing from a DLT to a single-lumen tube at the conclusion of the operation facilitates postop ET suctioning.

Usual preop diagnosis

COPD; pulmonary fibrosis; cystic fibrosis; pulmonary HTN; Eisenmenger syndrome

For Summary of Procedure and Anesthetic Management, see Surgery for Lung and Heart/Lung Transplantation, and Anesthetic Considerations for Lung Transplantation, p. 528.

Suggested Viewing

Links are available online to the following videos:

Part 1 of 3 VATS Thoracoscopic Lobectomy: https://www.youtube.com/watch?v=LPwsnQY3HkQ

Thoracotomy (On-Q catheter insertion for post-thoracotomy pain management): https://www.youtube.com/watch?v=XX_e4wTQ-90

Chest tube insertion: https://www.youtube.com/watch?v=EcIDkJHRq8c

Decompressing a Tension Pneumothorax: https://www.youtube.com/watch?v=0W-OmFBu93w

Thoracoscopic: Right Lower Lobectomy: https://www.youtube.com/watch?v=RgI4EcL0fDE

Sleeve Pneumonectomy for Lung Cancer: https://www.youtube.com/watch?v=GitI8FAU_vo